

Information Geometry and Dimensionality Reduction

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by

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Abstract of the Dissertation
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In the context of information geometry, the framework of differential geometry is applied to probability theory. It has a wide range of applications not limited to statistical inference, including image and signal processing, statistical learning, and hidden Markov models. Particularly in linear systems and time series analysis, distance between distributions are measured in statistical manifolds. Traditionally the information geometry of linear systems is studied for single-input-single output only. Moreover, unlike the widely used Euclidean metrics, information geometry based distances are usually difficult to compute and optimize. We go beyond previous work on statistics manifolds by showing how the theory can be generalized to multiple-input-multiple-output(MIMO) linear systems and be computationally efficient at the same time. We show the stability and usefulness of our approach with dimensionality reduction problems.

To summarize our main contributions: 1) We generalize information distance to MIMO LTI systems. When cepstrum are used as coordinates of the manifold, the geometry becomes Euclidean with nice theoretical and computational properties. 2) We developed a permutation invariant parameterization of linear system, which yields triangular input balanced(TIB) form. The TIB form is known to be numerically superior and a flexible framework for prior estimation. 3) We developed a new dimensionality reduction method for MIMO system which are better than the current state of the art methods. 4) We developed a numerical computation method for large Toeplitz matrix which is more efficient than existing methods. The method enables us to compute MIMO cepstrums and apply reduction algorithms on statistical manifolds. This work demonstrates the value of modeling linear systems on statistical manifolds. It also provides new tools for signal processing and time series analysis.

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1. Linear Time Invariant Systems

A causal discrete-time linear time invariant (LTI) system, which maps the input signal $\{x(t)\} \in \mathbb{R}^{q \times 1}$ to the output signal $\{y(t)\} \in \mathbb{R}^{p \times 1}$ as a function of t . We have the following equivalent representations of such a system:

1.1 The Impulse Response of LTI System

The mapping between input signal and output signal of an LTI system is described as follows

$$y(t) = h_0 x(t) + h_1 x(t-1) + h_2 x(t-2) + \dots \quad (1)$$

$$y(t) = \sum_{j=0}^{\infty} h_j x(t-j) \quad (2)$$

$$= \{h * x\}(t) \quad (3)$$

where $\{h_i\} \in \mathbb{R}^{p \times q}$ are called the impulse response of this LTI system. They characterize the mapping relationship as the convolution with input signal sequence. And a LTI system is called ‘Bounded Input Bounded Output’(BIBO) stable, if and only if the fourier transform of the impulse response is in the Wiener class, that is, absolute summable.

The (block) Toeplitz matrix Γ_T and (block) Hankel matrix Γ_H generated by the impulse response are also extremely useful in the context of LTI system, where

$$\Gamma_T = \begin{pmatrix} h_0 & h_1 & h_2 & \dots \\ & h_0 & h_1 & \dots \\ & & h_0 & \dots \\ & & & \dots \end{pmatrix} \quad (4)$$

$$\Gamma_H = \begin{pmatrix} h_0 & h_1 & h_2 & \dots \\ h_1 & h_2 & \dots & \\ h_2 & \dots & & \\ \dots & & & \end{pmatrix} \quad (5)$$

The Hankel matrix which admits a Krylov factorization is associated with the state space system. The Toeplitz matrix is closely connected with the covariance structure of LTI system output sequence as well as the information metric tensor of the system.

1.2 Z transform and Transfer Function

In signal processing, the Z-transform converts discrete-time signal into the frequency domain. The convolution of two sequences is taken to be the multiplication of polynomials as follows:

$$Y(z) = H(z) X(z) \quad (6)$$

where

$$Y(z) = \sum_{n=0}^{\infty} y_n z^n \quad (7)$$

$$X(z) = \sum_{n=0}^{\infty} x_n z^n \quad (8)$$

$$H(z) = \sum_{n=0}^{\infty} h_n z^n \quad (9)$$

Notice that the spectrum of the output signal is obtained by multiplying the z-transform of the unit impulse response sequence $\{h_n\}$ of the digital filter. An infinite impulse response digital filter with transfer function $H(z)$ given in (9) is stable if and only if all the poles of the function $H(z)$ lie outside the closed unit disk $|z| \leq 1$.

1.3 State Space Representation

The state-space representation is a mathematical model first introduced in control theory. The state-space is a space with axes being the state variables, and describes the LTI system in the time domain.

$$z(t+1) = A z(t) + B x(t) \quad (10)$$

$$y(t) = C z(t) + D x(t) \quad (11)$$

where A, B, C, D are $m \times m$, $m \times p$, $q \times m$ and $q \times p$ matrices independent of t . $z(t) \in \mathbb{R}^{m \times 1}$ is the state vector of the system. The matrix A is called the system matrix, B is called the control matrix. Matrix C , which is called observation matrix, describes how the state is measured to obtain the output of the system. The output sequence

$\{y(t)\}$ are $q \times 1$ vectors and input sequence $\{x(t)\}$ are $p \times 1$ vectors. The LTI system is called Multiple-Input-Multiple-Output(MIMO) if $p, q > 0$, and it is called Single-Input-Single-Output (SISO) if $p = q = 1$. Notice that even if the system is LTI (Linearly Time Invariant), which means the system is independent of time, the statistics of y is not independent of time, for example: $\text{Var}(y(t+1)) = \text{Var}(y(t)) + \text{Var}(x(t))$.

The state Covariance in this coordinate system is given by:

$$P \equiv E[z(t) z(t)^*] \quad (12)$$

$$= A E[z(t-1) z(t-1)^*] A^* + BB^* \quad (13)$$

$$= BB^* + APA^* \quad (14)$$

Notice that the equalities to the left of P follows from stationarity of the process. The equalities to the right of P follows from induction. **P is called the Controllability Gramian.** For completeness, we also give the definition of Observability Gramian:

$$Q = C^* C + A^* Q A = \sum_{k=0}^{\infty} A^{k*} C^* C A^k \quad (15)$$

The state space representation can be converted into the impulse response representation:

$$h_k = C A^{k-1} B \quad k \geq 1 \quad (16)$$

and $h_0 = I$.

2. LTI System Approximation

2.1 Problem Formulation

Our problem is, given a sequence of observations $\{y(t)\}$, $t_0 < t < t_1$, of a MIMO LTI system which has unknown transfer function $f(z)$, approximate $f(z)$ with rational function $g(z)$, such that the information distance between f and g , denoted as $I(f, g)$, is small.

2.2 Existing Methods ^[1]

Existing methods of model approximation and reduction can be classified into three categories: rational interpolation methods, balanced truncation and proper orthogonal decomposition (POD).

Over the last decade, numerous parametric model reduction methods based on a rational interpolation framework have been suggested. In the tangential interpolation setting, the model reduction task is posed as follows: Given a frequency interpolation point $\omega_i \in \mathbb{C}$, a parameter interpolation point $p \in \mathbb{R}^d$, and a nontrivial direction vector $r \in \mathbb{C}^m$, construct a reduced parametric model via projection such that $g(s, p)$ interpolates $f(s, p)$ r . For a discussion of projection based tangential interpolation for nonparametric systems, we refer the reader to [2]. The recent work by Baur et al. [3] provided a unifying projection-based framework for structure-preserving, tangential interpolation parametric model reduction, which also permitted a robust implementation. Recently Gugercin et al. [4] have introduced the Iterative Rational Krylov Algorithm (IRKA) to compute the frequency interpolation points optimally to minimize the H^2 error norm using the optimality conditions by Meier and Luenberger in [5].

In the systems and control theory community, balanced truncation (BT) [6-7] is one of the most common techniques for approximating linear dynamical systems without parametric dependency. In the parametric setting, BT can be employed to construct local reduced models at given parameter values.

Due to its broad applicability to linear and nonlinear systems, the proper orthogonal decomposition (POD) has become widely used in many different application domains as a method for computing the reduced basis. POD was introduced for the analysis of turbulent flows by Lumley [8], and is closely related to methods used in other fields such as Karhunen-L   ve expansions in stochastic process modeling [9-10], principal component analysis in statistical analysis [11], and empirical orthogonal eigenfunctions in atmospheric modeling [12].

3. Information Geometry and Information Distance

In approximation of LTI system, various distances between probability distributions are employed. The error of approximation introduced by the underlying approximation scheme is quantified in terms of those distances. There is an equivalency in measuring the error in the time domain and the frequency domain. In frequency domain, the most common error measures are H_∞ norm and H_2 norm.

3.1 Distribution Measures

The simplest way to create a signed measure is by taking a difference $\mu_1 - \mu_2$ of two nonnegative measures. In fact, one of the key properties of signed measures with finite total variation is that they can always be written as such a difference.

Definition 3.1.1. [Total Variation Distance]. The total variation distance between two probability distributions P_1 and P_2 is defined as

$$\|P_1 - P_0\|_1 = \int |p_1 - p_0| d\mu \quad (17)$$

independent of the dominating measure.

Definition 3.1.2.[Hellinger Distance]. The Hellinger distance $H(P_0, P_1)$ between two probability distributions P_0 and P_1 is defined as

$$H(P_0, P_1)^2 = \frac{1}{2} \|P_1 - P_0\|_2^2 = \frac{1}{2} \int |\sqrt{p_1} - \sqrt{p_2}|^2 d\mu \quad (18)$$

$$= 1 - \int \sqrt{p_1} \sqrt{p_2} d\mu \quad (19)$$

$$= 1 - \rho(P_0, P_1) \quad (20)$$

independent of the dominating measure, and $\rho(P_0, P_1)$ is known as the affinity between P_0 and P_1 .

Definition 3.1.3. [Fisher information matrix]. Given a statistical manifold with Coordinates $\theta = (\theta_1, \theta_2, \dots)$, one writes $p(x, \theta)$ for the probability distribution as a function of θ . Here x is drawn from the range R for a random variable X . The probability of the distribution is normalized by $\int_R p(x, \theta) dx = 1$; the Fisher information matrix takes the form:

$$F(\theta) = E((\partial_\theta \log p(x, \theta)) (\partial_\theta \log p(x, \theta))^T | \theta) \quad (21)$$

$$= \int_R p(x, \theta) (\partial_\theta \log p(x, \theta)) (\partial_\theta \log p(x, \theta))^T d\mu(x) \quad (22)$$

$$= \int_R \left(\frac{(\partial_\theta p(x, \theta))}{\sqrt{p(x, \theta)}} \right) \left(\frac{(\partial_\theta p(x, \theta))}{\sqrt{p(x, \theta)}} \right)^T d\mu(x) \quad (23)$$

Or equivalently,

$$F_{jk}(\theta) = \int_R \frac{\partial \log p(x, \theta)}{\partial \theta_j} \frac{\partial \log p(x, \theta)}{\partial \theta_k} p(x, \theta) dx \quad (24)$$

By taking the probability distributions for a statistical model as the points of a Riemannian manifold, we form a statistical manifold and the Fisher Information metric provides the Riemannian metric, with the variable θ being the coordinate axes on the manifold.

Definition 3.1.4:[Information Distance] Suppose that $\theta(t)$, $0 \leq t \leq 1$ is a smooth path connecting $\theta(0) = \theta_0$ and $\theta(1) = \theta_1$, the information distance between θ_0 and θ_1 , $I(\theta_0 \parallel \theta_1)$ is defined by

$$I(\theta_0 \parallel \theta_1) = \min \int_0^1 \sqrt{\dot{\theta}(t)^T F(\theta) \dot{\theta}(t)} dt \quad (25)$$

Where the information distance is independent with respect to smooth changes of coordinates on the manifold.

The reason why we are interested in approximation a MIMO LTI system in information distance stems from the following two lemmas.

Lemma 3.1.1[Lehmann, Romano][13]: The Hellinger distance and total variation distance satisfy

$$H(P_0, P_1)^2 \leq \frac{1}{2} \|P_1 - P_0\| \leq H(P_0, P_1) \sqrt{2 - H(P_0, P_1)^2}^{\frac{1}{2}} \quad (26)$$

which implies that the Hellinger distance and total variation distance generate the same metric topology.

Lemma 3.1.2[Mullhaupt[14]]: The information distance and Hellinger distance satisfy:

$$\sqrt{8} H(\theta_0, \theta_1) \leq I(\theta_0, \theta_1) \quad (27)$$

and that for any positive number $K < 1$, $F(\theta_0)$ non-singular, $\exists \theta_1 \neq \theta_0$ such that

$$KI(\theta_0, \theta_1) \leq \sqrt{8} H(\theta_0, \theta_1) \quad (28)$$

which implies that the upper bound in Lemma 1 is tightest possible.

One of the contexts in which the distances defined above arise naturally is hypothesis testing. Let's consider the simple problem of testing hypothesis P_0 against alternative P_1 , both are probability measures in the measurable space $(\Omega, \mathcal{F})$. For a test ϕ , denote the sum of the probability of rejecting P_0 when P_0 is true and the probability of rejecting P_1 when P_1 is true by

$$S_{P_0, P_1}(\phi) = \int_{\Omega} \phi(x) dP_0(x) + \int_{\Omega} (1 - \phi(x)) dP_1(x) \quad (29)$$

The following theorem gives the test ϕ^* that minimizes S_{P_0, P_1} over all possible tests.

Theorem 3.1.1. [Lehmann, Romano][13]: S_{P_0, P_1} is minimized by taking $\phi = \phi^*$ a.e. μ , where ϕ^* is any test satisfying $\phi^*(x) = 1$ if $p_1(x) > p_0(x)$ and $\phi^*(x) = 0$ if $p_1(x) < p_0(x)$. Further more,

$$S_{P_0, P_1}(\phi^*) = 1 - \frac{1}{2} \int |p_1 - p_0| d\mu \quad (30)$$

Definition 3.1.5.[Cepstrum] The cepstrum of a multivariate process y as the inverse Fourier transform of the matrix logarithm of the spectrum of y :

$$\frac{1}{2} \log \phi(e^{j\theta}) = \frac{1}{2} \sum_{k=-\infty}^{\infty} c(k) e^{-j k \theta} \quad (31)$$

where $c(k) \in \mathbb{R}^{p \times p}$ is the k th cepstral coefficient of y . For scalar processes this corresponds to the usual definition of the power cepstrum of a process. The sequence $\{c(k)\}_{k \in \mathbb{Z}}$ is real and even.

3.2 Information Distance of SISO LTI system

In the linear system case, the covariance matrix is positive definite and therefore Hermitian matrix with positive eigenvalues, the power spectral is real. Then the singular values are just the eigenvalues of the matrix. Let $F := \log(x)$ be a continuous function on $(0, \infty)$ with compact support. Given T_m the covariance matrix and that the Toeplitz matrix

$$T_m \equiv h_t h_t^* \quad (32)$$

is generated by

$$\phi(z) = |h(z)|^2 = h(z) h(z)^* \quad (33)$$

The following theorems will be needed in the derivation of Information Distance.

Theorem 3.2.1 (Szegö). Suppose $f : (-\pi, \pi) \rightarrow \mathbb{R}$ is a bounded function and let $\{T_m\}$ be the sequence of $m \times m$ Hermitian Toeplitz matrices generated by f ; then for any function F , continuous on the interval $\text{ess inf } f \leq \lambda \leq \text{ess sup } f$, one has:

$$\lim_{n \rightarrow \infty} \frac{1}{m} \sum_{j=1}^m F(\lambda_j(T_m)) = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(f(x)) dx \quad (34)$$

Where $\lambda_j(T_m), j = 1, \dots, n$ are the eigenvalues of T_n .

There has been generalizations of the classical Grenander-Szego's Theorem[15] in many directions. In the Tyrtshnikov result, the generating function was extended from L^∞ to L^2 , provided that the function F has compact support $\mathbb{R}$.

Theorem 3.2.2 .[Tyrtshnikov] [16]. If $f : (-\pi, \pi) \rightarrow \mathbb{C}$ is L^2 and let $\{T_m\}$ be the sequence of Hermitian Toeplitz matrices generated by f ; then for any continuous function F with compact support in $\mathbb{R}$, one has

$$\lim_{n \rightarrow \infty} \frac{1}{m} \sum_{j=1}^n F(\sigma_j(T_n)) = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(|f(e^{i\theta})|) d\theta \quad (35)$$

Where $\sigma_j(T_m), j = 1, \dots, n$ are the singular values of T_n .

This result is one of the many generalizations of the classical Grenander-Szego's Theorem[16]. In the Tyrtshnikov result, the generating function was extended from L^∞ to L^2 . The above theorem by Tyrtshnikov covers the case where f is complex valued and the matrices $\{T_m\}$ it generates are $m \times m$ square matrices; on the other hand, the notion of singular value makes sense also when the matrices involved are nonsquare.

Theorem 3.2.3.[Jensen][17]: Consider the family of p -variate normal distributions with common means vector μ but differing variance-covariance matrices Σ . The distance between two members of the family with variance-covariance matrices Σ_1 and Σ_2 is given by

$$I^2(\Sigma_1, \Sigma_2) = \frac{1}{2} \sum_{i=1}^p (\ln \lambda_{0i})^2 \quad (36)$$

where $0 < \lambda_{01} \leq \lambda_{02} \leq \dots \leq \lambda_{0p}$ are the eigenvalues of the determinantal equation:

$$|\Sigma_1 - \lambda \Sigma_2| = 0 \quad (37)$$

Given that the linear system is stable, minimum phase and

$$\frac{1}{2\pi i} \oint_{|z|=1} |\log(h(z))|^2 \frac{dz}{z} < \infty \quad (38)$$

from [Jensen 1981] we have the calculation of Information Distance:

$$\lim_{m \rightarrow \infty} \frac{1}{m} I^2(T_m, I) \quad (39)$$

$$= \lim_{m \rightarrow \infty} \frac{1}{m} \frac{1}{2} \sum_{i=1}^m (\ln \lambda_i)^2 \quad (40)$$

$$= \lim_{m \rightarrow \infty} \frac{1}{m} \frac{1}{2} \sum_{i=1}^m |\ln \lambda_i|^2 \quad (41)$$

$$= \frac{1}{4\pi} \int_{-\pi}^{\pi} |\ln |h(e^{i\theta})|^2|^2 d\theta \quad (42)$$

$$= \frac{1}{4\pi i} \oint_{|z|=1} \ln(h(z)h(z)^*) \ln(h(z)h(z)^*)^* \frac{dz}{z} \quad (43)$$

$$= \frac{1}{2} \|\log \phi(z)\|_{L_2}^2 \quad (44)$$

Another derivation of information distance comes from Choi and Mullhaupt [15][18], the cepstrum of the transfer function is defined as log of the transfer function:

$$\log h(z) \equiv a(z) = \sum_{k=0}^{\infty} a_k z^k \quad (45)$$

$$h(z) = \sum_{k=0}^{\infty} h_k z^k \quad (46)$$

Taking $a_0, a_1, \dots$ which are the transfer function cepstrum coefficients as the coordinates of the statistical manifold, the Fisher Information matrix is the identity, in other words, the manifold is Euclidean. And when $a_0 = 0$, the information distance between h and white noise is calculated as

$$I^2(h \parallel I) = \left(\min \int_0^1 \sqrt{\dot{\theta}(t)^T F(\theta) \dot{\theta}(t)} dt \right)^2 \quad (47)$$

$$= \frac{1}{2\pi} \int_{|z|=1} |\log h(\theta)|^2 \frac{dz}{z} \quad (48)$$

$$= \sum_{i=1}^{\infty} |a_i|^2 \quad (49)$$

which is the H_2 norm of the cepstrum. These two expressions for information distance are equivalent when $a_0 = 0$, i.e. $h_0 = 1$. However this is not true when h_0 is not 1, i.e. it corresponds to a scaled process from the original one. Now let $h(z) = h_0 g(z)$ where $g_0 = 1$, and also

$$\log h(z) = \sum_{k=0}^{\infty} a_k z^k \quad (50)$$

$$\log g(z) = \sum_{k=1}^{\infty} b_k z^k \quad (51)$$

$$\log \phi_h(z) = \log h(z) + \log h(z)^* \quad (52)$$

$$= 2a_0 + \sum_{k=1}^{\infty} a_k (z^k + z^{*k}) \quad (53)$$

$$\log \phi_g(z) = \log g(z) + \log g(z)^* \quad (54)$$

$$= \sum_{k=1}^{\infty} b_k (z^k + z^{*k}) \quad (55)$$

hence we have

$$\lim_{m \rightarrow \infty} \frac{1}{m} I^2(T_m(hh^*), I) \quad (56)$$

$$= \lim_{m \rightarrow \infty} \frac{1}{m} I^2(|h_0|^2 T_m(gg^*), I) \quad (57)$$

$$= \lim_{m \rightarrow \infty} \frac{1}{m} \frac{1}{2} \sum_{i=1}^m (\ln \lambda_{g_i} + \ln |h_0|^2)^2 \quad (58)$$

$$= \lim_{m \rightarrow \infty} \frac{1}{m} \frac{1}{2} \left(\sum_{i=1}^m |\ln \lambda_i|^2 + \ln |h_0|^2 \ln \det(T_m(gg^*)) + (\ln |h_0|^2)^2 \right) \quad (59)$$

since $g_0 = 1$, we have $\ln \det(T_m(gg^*)) = 0$, hence

$$\lim_{m \rightarrow \infty} \frac{1}{m} I^2(T_m(hh^*), I) \quad (60)$$

$$= \lim_{m \rightarrow \infty} \frac{1}{m} I^2(T_m(gg^*), I) + 2(\ln |h_0|)^2 \quad (61)$$

$$= \|\log g(z)\|_{H_2}^2 + 2(\ln |h_0|)^2 \quad (62)$$

$$= \frac{1}{2} \|\log \phi_g(z)\|_{L_2}^2 \quad (63)$$

Since

$$\ln h(z) = \ln(h_0 g(z)) = \ln(h_0) + \ln g(z) \quad (64)$$

This result is different from

$$\|\log h(z)\|_{H_2}^2 \quad (65)$$

$$= \|\log g(z)\|_{H_2}^2 + |\ln(h_0)|^2 \quad (66)$$

Notice that the latter derivation only requires the log of the transfer function to be in H^2 , while the first requires the existence of spectral density of the underlying LTI system. For example, the random walk, whose transfer function is given by:

$$h(z) = \frac{1}{1-z} \quad (67)$$

does not have a well defined spectral density. However, it has finite information length because the cepstrum of $h(z)$ is in H^2 . Now we claim that, with any large but finite number of samples N , threshold $\alpha \in (0, 1)$, there exists two processes, one stationary and the other not, such that any hypothesis test of one against the other will have sum of error probabilities no less than α . Here is an example. Notice that the following infinite sum converges:

$$\|\log \frac{1}{1-z}\|_{H_2}^2 \quad (68)$$

$$= \lim_{m \rightarrow \infty} \sum_{k=1}^m \frac{1}{k^2} \quad (69)$$

$$:= \lim_{m \rightarrow \infty} S_m \quad (70)$$

$$= \frac{\pi^2}{6} \quad (71)$$

Then by the definition of monotonic increasing converging infinite sum, for $\epsilon = 8 \left(1 - (2\alpha - \alpha^2)^{\frac{1}{2N}}\right)$, there

exist a number $m > 0$, s.t.

$$\frac{\pi^2}{6} - S_m < \epsilon \quad (72)$$

Now let P_0 be a stochastic process with transfer function

$$h_0(z) = \frac{1}{1-z} \quad (73)$$

P_1 be another process with transfer function

$$h_1(z) = \exp\left(\sum_{k=1}^{m+1} \frac{z^k}{k}\right) \quad (74)$$

where the information distance satisfy:

$$I^2(P_0, P_1) \quad (75)$$

$$= \|\log h_0 - \log h_1\|_{H_2}^2 \quad (76)$$

$$< \epsilon = 8 \left(1 - (2\alpha - \alpha^2)^{\frac{1}{2N}}\right) \quad (77)$$

It is obvious that P_0 is a random walk and hence non-stationary, and that since the transfer function of $h_1(z)$ does not have pole in the closed unit disc, P_1 is stationary. So we have:

$$I^2(P_0, P_1) < 8 \left(1 - (2\alpha - \alpha^2)^{\frac{1}{2N}}\right) \quad (78)$$

$$\Rightarrow H^2(P_0, P_1) < 1 - (2\alpha - \alpha^2)^{\frac{1}{2N}} \quad (79)$$

$$\Rightarrow 1 - H^2(P_0, P_1) > (2\alpha - \alpha^2)^{\frac{1}{2N}} \quad (80)$$

$$\Rightarrow 1 - (1 - H^2(P_0, P_1))^{2N} < 1 - (2\alpha - \alpha^2) \quad (81)$$

$$\Rightarrow 1 - (1 - (1 - H^2(P_0, P_1))^{2N})^{\frac{1}{2}} > \alpha \quad (82)$$

$$\Rightarrow 1 - (1 - \rho(P_0, P_1)^{2N})^{\frac{1}{2}} > \alpha \quad (83)$$

$$\Rightarrow 1 - (1 + \rho(P_0^N, P_1^N))^{\frac{1}{2}} (1 - \rho(P_0^N, P_1^N))^{\frac{1}{2}} > \alpha \quad (84)$$

$$\Rightarrow 1 - (2 - H^2(P_0^N, P_1^N))^{\frac{1}{2}} H(P_0^N, P_1^N) > \alpha \quad (85)$$

$$\Rightarrow 1 - \frac{1}{2} \|P_0^N - P_1^N\|_1 > \alpha \quad (86)$$

$$\Rightarrow S_{P_0, P_1}(\phi) > \alpha \text{ for all } \phi \quad (87)$$

where S_{P_0, P_1} is defined as in Theorem 3.1.1. Now we have proved our claim: Given an ideal level of sum of error probability, and a large but finite amount of data, there always exist processes that no hypothesis test on its stationary would satisfy that level of error probability and tell us its stationarity. The existence of such processes is irrelevant with the number of samples, as long as there are only finitely many. Since stationary processes are dense subsets of finite information processes, so even though we cannot tell if the real process is stationary or not, we still want to find a short memory process which is not distinguishable from the real one, so that we can make good approximations and good predictions. Information distance makes this possible.

3.3 Information Distance of MIMO LTI system

In a m input n output MIMO linear system, the input signals $x(t)$ is mapped to the output signals $y(t)$ as a function of t :

$$\begin{pmatrix} y_n(t) \\ y_{n-1}(t) \\ \dots \\ y_1(t) \end{pmatrix} = H_0 \begin{pmatrix} x_m(t) \\ x_{m-1}(t) \\ \dots \\ x_1(t) \end{pmatrix} + H_1 \begin{pmatrix} x_m(t-1) \\ x_{m-1}(t-1) \\ \dots \\ x_1(t-1) \end{pmatrix} + H_2 \begin{pmatrix} x_m(t-2) \\ x_{m-1}(t-2) \\ \dots \\ x_1(t-2) \end{pmatrix} + \dots \quad (88)$$

Where H_i are $n \times m$ constant matrices. And we can form a Toeplitz matrix

$$\begin{pmatrix} Y(t) \\ Y(t-1) \\ \dots \\ \dots \\ \dots \end{pmatrix} = \begin{pmatrix} H_0 & H_1 & H_2 & \dots & \dots \\ & H_0 & H_1 & H_2 & \dots \\ & & H_0 & H_1 & \dots \\ & & & H_0 & \dots \\ & & & & \dots \end{pmatrix} \begin{pmatrix} X(t) \\ X(t-1) \\ \dots \\ \dots \\ \dots \end{pmatrix} \quad (89)$$

where

$$Y(t) = \begin{pmatrix} y_n(t) \\ y_{n-1}(t) \\ \dots \\ y_1(t) \end{pmatrix} \quad (90)$$

$$X(t) = \begin{pmatrix} x_m(t) \\ x_{m-1}(t) \\ \dots \\ x_1(t) \end{pmatrix} \quad (91)$$

Which we denote as

$$Y(t) = H_t X(t) \quad (92)$$

Assuming the distribution of $\{x(t)\}$ to be i.i.d Gaussian(0,1) , then the covariance structure of $Y(t)$ is given by:

$$\begin{aligned} & E(Y(t) Y(t)^*) \\ &= E(H_t X(t) (H_t X(t))^*) \\ &= H_t E(X(t) X(t)^*) H_t^* \\ &= H_t H_t^* \end{aligned} \quad (93)$$

Where $H_t H_t^*$ is a block toeplitz matrix, and therein $Y(t)$ is a multivariate normal random vector. According to S.T. Jensen (1976)[14], the information distance based on Fisher information metric is directly related to the covariance structure of random variable, with respect to multivariate normal distribution.

In the Z domain, the same mapping is called a transfer function. Let $Y(z)$ and $X(z)$ be the Z transform of $Y(t)$ and $X(t)$ and we have:

$$\begin{aligned} Y(z) &= H(z) X(z) \\ H(z) &= \sum_{r=-\infty}^{\infty} H_r z^{-r} \end{aligned} \quad (94)$$

We specifically focus on the causal response case, $H_r = 0$ for all $r < 0$. Then we denote

$$\Phi(z) = H(z) H(z)^* \quad (95)$$

$$\log \Phi(z) = \sum_{k=0}^{\infty} \Psi_k (z^k + z^{-k}) \quad (96)$$

where

$$(\log \Phi(z))_0 = 2 \Psi_0 = \log \Phi(0) = \log (H(0) H(0)^*) = \log (H_0 H_0^*) \quad (97)$$

Now we define the inner product between two complex functions F and G in the Hilbert space:

$$\langle F, G \rangle = \frac{1}{2\pi i} \text{Tr} \oint F(z) G(z)^* \frac{dz}{z} \quad (98)$$

The following notions of matrix valued Hardy Spaces will be needed.

Definition 3.3.1 [H^p space] Given that $H^p(\mathbb{D})$ denotes the space of scalar valued Hardy functions of class p over $\mathbb{D}$, $1 \leq p \leq \infty$. And $H_{k \times j}^p(\mathbb{D})$ denotes the space of $k \times j$ matrix valued functions each entry of which belongs to H^p .

In particular, a matrix valued function $F(z)$ is in $H_{k \times j}^2(\mathbb{D})$ if and only if

$$\|F(z)\|_{H_{k \times j}^2} = \frac{1}{2\pi i} \oint_{|z|=1} \text{Tr}(F(z)F(z)^*) \frac{dz}{z} < \infty \quad (99)$$

Definition 3.3.2. [Essential Numerical Range]: If $f \in L^1(Q, \mathbb{C}^{n \times n})$, we set

$$\mathbf{ER}(f) \equiv \bigcap_{H \in \Theta} H \quad (100)$$

where Θ is the family of all closed half planes $H \subset \mathbb{C}$ such that $\mathbf{R}(f) \subset H$ holds for almost every $x \in Q$. If $\Theta = \emptyset$, then we agree to set $\mathbf{ER}(f) = \mathbb{C}$.

Theorem 3.3.1. [Essential numerical range]: Suppose $f \in L^1(Q, \mathbb{C}^{n \times n})$, and let $\{T_m\}$ be the set of $mn \times mn$ block Toeplitz matrices generated by f . Then for any function F holomorphic in an open disk containing $\mathbf{ER}(f)$, that is, the numerical range of T_m is contained in the essential numerical range of f . In particular, if λ is an eigenvalue of T_m , then $\lambda \in \mathbf{ER}(f)$.

And then we can proceed to introduce the generalized Grenander Szegő Theorem for block Toeplitz Matrices. We will be dealing with matrix valued functions integrable with respect to Lebesgue measure on the interval $Q = (-\pi, \pi)$.

Theorem 3.3.2. (Block Toeplitz I)[19]: Suppose $f \in L^\infty(Q, \mathbb{C}^{n \times n})$, and let $\{T_m\}$ be the set of $mn \times mn$ block Toeplitz matrices generated by f . Then for any function F holomorphic in an open disk containing $\mathbf{ER}(f)$,

$$\lim_{n \rightarrow \infty} \frac{1}{mn} \sum_{j=1}^{mn} F(\lambda_j(T_m)) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1}{n} \sum_{j=1}^n F(\lambda_j(f(e^{i\theta}))) d\theta \quad (101)$$

Notice that the above theorem is only valid for square matrices where we can compute the eigenvalues of the matrix valued function. The next theorem is more generalized and is valid even involving nonsquare matrices, in which case the singular values, instead of the eigenvalues are concerned.

Theorem 3.3.3. (Block Toeplitz II)[20]: Suppose $f \in L^1(Q, \mathbb{C}^{n \times k})$, where Q is $[-\pi, \pi]$ and let $\{T_m\}$ be the set of $mn \times mk$ block Toeplitz matrices generated by f . Then, for any uniformly continuous bounded function F , we

find the following asymptotic formula.;

$$\lim_{n \rightarrow \infty} \frac{1}{m \min(n, k)} \sum_{j=1}^{n k} F(\sigma_j(T_m)) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1}{\min(n, k)} \sum_{j=1}^{\min(n, k)} F(\sigma_j(f(e^{i\theta}))) d\theta \quad (102)$$

With generalized Szegő's theorem, the Information distance of a $n \times k$ MIMO system is therefore:

$$\lim_{m \rightarrow \infty} \frac{1}{m} \frac{1}{\min(n, k)} I^2(T_m, I) \quad (103)$$

$$= \lim_{m \rightarrow \infty} \frac{1}{m} \frac{1}{\min(n, k)} \frac{1}{2} \sum_{i=1}^{\min(n, k)} (\ln \lambda_i)^2 \quad (104)$$

$$= \frac{1}{4\pi} \frac{1}{\min(n, k)} \int_{-\pi}^{\pi} \sum_{j=1}^{\min(n, k)} (\ln \sigma_j(\Phi(e^{i\theta})))^2 d\theta \quad (105)$$

Specifically, for a square system where $n = k$ and the $mn \times mn$ block Toeplitz matrix T_m is positive definite, we have

$$\lim_{m \rightarrow \infty} \frac{1}{m n} I^2(T_m, I) \quad (106)$$

$$= \frac{1}{4\pi} \frac{1}{n} \int_{-\pi}^{\pi} \sum_{j=1}^n (\ln \lambda_j(\Phi(e^{i\theta})))^2 d\theta \quad (107)$$

$$= \frac{1}{4\pi} \frac{1}{n} \int_{-\pi}^{\pi} \sum_{j=1}^n (\lambda_j(\ln(\Phi(e^{i\theta}))))^2 d\theta \quad (108)$$

$$= \frac{1}{4\pi} \frac{1}{n} \text{Tr} \int_{-\pi}^{\pi} \ln(\Phi(e^{i\theta})) \ln(\Phi(e^{i\theta})) d\theta \quad (109)$$

$$= \frac{1}{2} \frac{1}{n} \|\Psi(e^{i\theta})\|_{L_{n \times n}^2}^2 \quad (110)$$

$$= \frac{1}{n} \text{Tr} \left(\sum_{k=1}^{\infty} \Psi_k^2 + 2 \Psi_0^2 \right) \quad (111)$$

This result suggests that the information distance between the matrix valued transfer function and the white noise is determined by the log of the singular value of the transfer function. This result can be further generalized to include situations in analogy to random walk in SISO case. Even if the spectral density does not exist, it is entirely possible for the process to have finite information length, as long as $\ln \Phi(z)$ is in $L_{n \times n}^2$ space, i.e. quadratically integrable. This condition is actually weaker than the existence of spectral density.

In the SISO LTI system we have two assumptions: we can always fix the first coefficient of the transfer function to be 1, while assuming the innovations $x(t)$ to be $N(0,1)$, because in the scalar case, a linear relationship stays

the same while we scale $x(t)$ and $y(t)$ at the same time. However in the MIMO system, the analogous assumption is not trivial anymore. It corresponds to scaling the system with some constant matrix.

The Kullback-Leibler Divergence can also be deduced in a similar pattern. For a $n \times n$ MIMO linear filter when we map white noise to $y(z)$

$$y(z) = H(z)x(z) \quad (112)$$

with spectral density

$$\Phi(z) = H(z)H(z)^* \quad (113)$$

Denote the $mn \times mn$ block Toeplitz matrix generated by $\phi(z)$ as T_m . With the generalized Szegő theorem, the per sample KL divergence for $y(z)$ and white noise is given by:

$$\lim_{m \rightarrow \infty} \frac{1}{m n} D_{\text{KL}}(\Phi \| I) \quad (114)$$

$$= \lim_{m \rightarrow \infty} \frac{1}{m n} \left(\frac{1}{2} \log \det(T_m^{-1}) - \frac{1}{2} \text{tr} I + \frac{1}{2} \text{tr}(T_n) \right) \quad (115)$$

$$= \lim_{m \rightarrow \infty} \frac{1}{2 m n} \sum_{i=1}^{mn} (-\log \lambda_i + \lambda_i - 1) \quad (116)$$

$$= \frac{1}{4 \pi} \frac{1}{n} \int_{-\pi}^{\pi} \sum_{j=1}^n (-\log(\lambda_j(\Phi(e^{i\theta}))) + \lambda_j(\Phi(e^{i\theta})) - 1) d\theta \quad (117)$$

$$= \frac{1}{4 \pi} \frac{1}{n} \int_{-\pi}^{\pi} (-\log(\det(\Phi(e^{i\theta}))) + \text{Tr}(\Phi(e^{i\theta})) - 1) d\theta \quad (118)$$

$$= \frac{1}{4 \pi} \frac{1}{n} \int_{-\pi}^{\pi} (-\text{Tr}(\ln(\Phi(e^{i\theta}))) + \text{Tr}(\Phi(e^{i\theta})) - 1) d\theta \quad (119)$$

$$= \frac{1}{4 \pi} \frac{1}{n} \text{Tr} \int_{-\pi}^{\pi} (\Phi(e^{i\theta}) - \Psi(e^{i\theta}) - I_n) d\theta \quad (120)$$

The two way Kullback-Leibler divergence is given by

$$\lim_{m \rightarrow \infty} \frac{1}{m n} (D_{\text{KL}}(\Phi \| I) + D_{\text{KL}}(I \| \Phi)) \quad (121)$$

$$= \frac{1}{4 \pi} \frac{1}{n} \int_{-\pi}^{\pi} (\text{Tr}(\Phi(e^{i\theta})) + \text{Tr}(\Phi(e^{i\theta})^{-1}) - 2) d\theta \quad (122)$$

where Tr denotes the trace of a square matrix.

3.4 Information Distance of Finite LTI System

Formulae (52) describes the asymptotic relationship between the transfer function cepstrum coefficients and information distance. While asymptotic considerations are justified by the law of convergence and increasing amount of data, it is almost equally important to emphasize the finiteness of data. The adequacy of sample size to constrain error when using asymptotic results can be exposed by statistical inference of large but finite sample size. The section introduces a lemma which provides an upper bound for finite sample size, a bridge that links large sample to asymptotic statistics. We would introduce two theorems and a corollary to proceed to the final result.

Corollary 3.4.1[21] (Corollary 3.3.10 in reference): Let $\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_n$ be $2n$ given nonnegative real numbers such that $\alpha_1 \geq \dots \geq \alpha_n \geq 0$ and $\beta_1 \geq \dots \geq \beta_n \geq 0$. If

$$\prod_{i=1}^k \alpha_i \leq \prod_{i=1}^k \beta_i, \quad k = 1, \dots, n \quad (123)$$

then

$$\sum_{i=1}^k \alpha_i \leq \sum_{i=1}^k \beta_i, \quad k = 1, \dots, n \quad (124)$$

More generally suppose $f(\cdot)$ is a given real-valued function such that $f(e^t)$ is increasing and convex on the interval $[\min(\alpha_n, \beta_n), \beta_1]$. Then

$$\sum_{i=1}^k f(\alpha_i) \leq \sum_{i=1}^k f(\beta_i), \quad k = 1, \dots, n \quad (125)$$

Theorem 3.4.1[23] (Theorem 3.3.14 in reference): Let $A, B \in M_n$, order the singular values of A, B, AB by $\sigma_1(A) \geq \dots \geq \sigma_n(A) \geq 0$, $\sigma_1(B) \geq \dots \geq \sigma_n(B) \geq 0$, $\sigma_1(AB) \geq \dots \geq \sigma_n(AB) \geq 0$. Then

$$\sum_{i=1}^k \sigma_i(AB) \leq \sum_{i=1}^k \sigma_i(A) \sigma_i(B), \quad k = 1, \dots, n \quad (126)$$

Lemma 3.4.1 : Let $F_m = \sum_{i=1}^m f_i z^i$ be the truncated transfer function of an LTI system, with corresponding Toeplitz Matrix

$$T_{F_m} = \begin{pmatrix} f_0 & 0 & 0 & 0 \\ f_1 & f_0 & 0 & 0 \\ \dots & \dots & \dots & 0 \\ f_m & f_{m-1} & \dots & f_0 \end{pmatrix} \quad (127)$$

Now let

$$T_{A_m} = \log T_{F_m} = \begin{pmatrix} a_0 & 0 & 0 & 0 \\ a_1 & a_0 & 0 & 0 \\ \dots & \dots & \dots & 0 \\ a_m & a_{m-1} & \dots & a_0 \end{pmatrix} \quad (128)$$

and $A_m = \sum_{i=1}^m a_i z^i$. Then the finite sample information distance between the m dimensional estimated process and white noise has the inequality

$$\frac{1}{m} I_m^2(F, I) \leq (\operatorname{Re} a_0)^2 + 2 \sum_{i=1}^m \left(1 - \frac{i}{m}\right) |a_i|^2 \quad (129)$$

Proof:

We denote the $m \times m$ Toeplitz Matrix with symbol F_m as

$$T_{m,i,j} = \begin{cases} 0 & j > i \\ f_{i-j} & i \leq j \end{cases} \quad (130)$$

Since $T_m T_m^*$ and $(\log(T_m T_m^*))^2$ are normal,

$$I_m^2(T_m, I) \quad (131)$$

$$= \frac{1}{2} \sum_{i=1}^k \lambda_i (\log(T_m T_m^*))^2 \quad (132)$$

$$= \frac{1}{2} \sum_{i=1}^k \log(\sigma_i(T_m T_m^*))^2 \quad (133)$$

From Theorem 3.4.1 we have

$$\sum_{i=1}^k (\sigma_i(T_m T_m^*)) \leq \sum_{i=1}^k (\sigma_i(T_m) \sigma_i(T_m^*)) \quad (134)$$

Now in particular, since $f(z) = (\log z)^2$ is analytic in $\mathbb{C} \setminus (-\infty, 0]$ we have

$$\sum_{i=1}^k \log (\sigma_i(T_m T_m^*))^2 \leq \sum_{i=1}^k \log (\sigma_i(T_m) \sigma_i(T_m^*))^2 \quad (135)$$

Thus we have

$$\frac{1}{2} \sum_{i=1}^k \log(\sigma_i(T_m T_m^*))^2 \quad (136)$$

$$= \frac{1}{2} \| \log (T_m T_m^*) \|_F^2 \quad (137)$$

$$= \frac{1}{2} \| \Sigma (\log (T_m T_m^*)) \|_F^2 \quad (138)$$

$$= \frac{1}{2} \| \log(\Sigma (T_m T_m^*)) \|_F^2 \quad (139)$$

$$\leq \frac{1}{2} \| \log \Sigma(T_m) \log \Sigma(T_m) \|_F^2 \quad (140)$$

$$= \frac{1}{2} \| \log T_m + \log T_m^* \|_F^2 \quad (141)$$

i.e.

$$\frac{1}{m} I_m^2(F, I) \leq 2 (\operatorname{Re} a_0)^2 + \sum_{i=1}^m \left(1 - \frac{i}{m}\right) |a_i|^2 \quad (142)$$

In case $\log F$ is nilpotent, or more generally any case where $\| \log T_m - \log T_m^* \|_F^2 = \| \log T_m + \log T_m^* \|_F^2$, then we have

$$\| \log T_m + \log T_m^* \|_F^2 = 2 \| \log T_m \|_F^2 \quad (143)$$

in which case

$$I_m^2(T_m, I) \leq \| \log T_m \|_F^2 \quad (144)$$

$$\frac{1}{m} I_m^2(F, I) \leq \sum_{i=1}^m \left(1 - \frac{i}{m}\right) |a_i|^2 \quad (145)$$

Notice that in equation (143), the coefficient of a_i with finite k all converges to 1 as $m \rightarrow \infty$.

4. Triangular Input Balanced Form

4.1 Rational Functions and Approximation Theory

In a MIMO LTI system, the Z-transform reduces the problems of linear system approximation to problems of approximation in the complex plane. One species of methods for such problems goes by the name of H^∞ approximation, based on results by Adamjan, Arov and Krein [1971][22] and Glover [1984][23], which justifies our causes of using rational functions as approximants. Recall the Hankel matrix in (5), the following well posed theorem clarifies the close relationship between SISO LTI system and the Hankel matrix.

Theorem 4.4.1.[Kronecker's Theorem]. Given an analytic function:

$$H(z) = \sum_{n=0}^{\infty} h_n z^n \quad (146)$$

we consider the singular and analytic parts of $H(z)$, namely

$$H_a(z) = h_0 \quad (147)$$

$$H_s(z) = \sum_{n=1}^{\infty} h_n z^n \quad (148)$$

$$H(z) = H_s(z) + H_a(z) \quad (149)$$

The infinite Hankel matrix $\Gamma_H = \{h_{i+l-1}\}$, $i, l = 1, 2, \dots$ has finite rank if and only if the singular part $H_s(z)$ is a strictly proper rational function in z . Furthermore, in this situation, the rank of Γ_H agrees with the number of poles of $H_s(z)$, with multiplicities taken into consideration.

To introduce the classical AAK theorem of best approximation, we introduce the following two notations:

$$R_m^s = \left\{ r_n(z) \mid r_n(z) = \frac{p_1 z^{n-1} + \dots + p_n}{z^n + q_1 z^{n-1} + \dots + q_n}, \text{ all poles of } r_n(z) \text{ lie in } |z| < 1, n \leq m \right\} \quad (150)$$

$$H_m^\infty = \{r(z) + g(z) : r(z) \in R_m^s, g(z) \in H^\infty\} \quad (151)$$

Theorem 4.4.2[Adamjan, Arov, and Krein]: Let $h(z)$ be a given function in $L^\infty = L^\infty(|z| = 1)$ such that Γ_h is a compact operator with s -numbers

$$s_1 \geq s_2 \geq \dots \geq s_\infty = 0 \quad (152)$$

and let (ξ_m, η_m) be the Schmidt pair corresponding to s_m where

$$\xi_m = [u_1^{(m)} u_2^{(m)} \dots]^T \quad (153)$$

$$\eta_m = [v_1^{(m)} v_2^{(m)} \dots]^T \quad (154)$$

m be a non-negative integer. Then we have

$$\inf_{r \in R_m^s} \|h - r\|_\Gamma = \inf_{f \in H_m^\infty} \|h - f\|_\Gamma \quad (155)$$

Then a solution to the extremal problem:

$$\|h - \hat{r}_m\|_\Gamma = \inf_{r \in R_m^s} \|h - r\|_\Gamma \quad (156)$$

where $\|\cdot\|_\Gamma$, namely the Hankel norm, is the spectral radius of the operator, is given by the singular part $\hat{r}_m(z) = [\hat{f}(z)]_s$, and

$$\hat{f}(z) = h(z) - s_{m+1} \frac{\eta_-(z)}{\xi_+(z)} \quad (157)$$

with

$$\xi_+(z) = \sum_{i=1}^{\infty} u_i^{(m+1)} z^{i-1} \quad (158)$$

$$\eta_-(z) = \sum_{i=1}^{\infty} v_i^{(m+1)} z^{i-1} \quad (159)$$

Moreover,

$$\|h - \hat{r}_m\|_\Gamma = \inf_{r \in R_m^s} \|h - r\|_\Gamma = s_{m+1} \quad (160)$$

This novel AAK approach gives the solution to the extremal problem: **a best Hankel-norm approximant is given by the singular part of a best L^∞ norm approximant**. This is the best approximation in the Hilbert space operator norm. Notice that the situation where infinite rank Γ_h is not excluded in the theorem, no available computation and algorithm exist for best approximation of infinite rank Hankel matrix operator. Finite rank rational function implies finite computability.[24]

However, it should be noticed that for MIMO linear systems, there is no analog of the Kronecker theorem, so that the rank of the Hankel matrix associated with the transfer function is not simple in general. It turns out that

calculating the McMillan degree of the transfer function is not difficult and this rank is equal to its corresponding block-Hankel matrix and the rank of minimal realization. In this setting, finite rank matrix valued rational functions are still the best choice. See [Chui and Chen 1997][23] for example for details.

4.2 Triangular Input Balanced Form

The TIB representation of systems was introduced in [A.Mullhaupt and K.Riedel][25-26]. It is a well structured, computationally and analytically advanced LTI system. The state space representation of a MIMO LTI system is given by

$$z(t+1) = Az(t) + Bx(t) \quad (161)$$

$$y(t) = C \cdot z(t) + x(t) \quad (162)$$

From realization theory, any proper rational matrix valued function $G(z)$ can be written in the form

$$G(z) = D + C(I_m - Az)^{-1}B \quad (163)$$

where $A \in \mathbb{C}^{n \times n}$, Stability indicates $|\lambda_k(A)| < 1$ for all k . To such a realization, we associate the block partitioned matrix

$$R(z) = \begin{pmatrix} D & C \\ B & A \end{pmatrix} \quad (164)$$

denoted as the realization matrix.

4.2.1 Similarity Transformation

Recall that the state space representation can be converted into the impulse response representation:

$$h_k = CA^{k-1}B \quad k \geq 1 \quad (165)$$

The following coordinate change which preserves the impulse response, maps one state space representation to another, both of which characterize the same LTI system. Let T be any non-singular transformation, under the coordinate change:

$$z \rightarrow T z \quad (166)$$

and accordingly

$$A \rightarrow T A T^{-1} \quad (167)$$

$$B \rightarrow T B \quad (168)$$

$$C \rightarrow C T^{-1} \quad (169)$$

the state space representation reads

$$T z(t+1) = T A T^{-1} (T z(t)) + T B x(t) \quad (170)$$

$$y(t) = (C T^{-1}) (T z(t)) + x(t) \quad (171)$$

Now the same system is represented in a different coordinate system. Straight forward computation yields the following: the stein equation

$$A P A^* + B B^* = P \quad (172)$$

is valid under such coordinate change, but of course with different Controllability Gramian P . Now starting from any realization $(\tilde{A}, \tilde{B}, C, D)$, we perform Schur decomposition on $\tilde{A}$, which yields

$$\tilde{A} = Q \hat{A} Q^* \quad (173)$$

where $\hat{A}$ is lower triangular, Q is unitary. The next step is to perform Cholesky factorization on $Q^* P Q$, which yields

$$Q^* P Q = L L^* \quad (174)$$

where L is lower triangular. Now let the coordinate system be transformed as:

$$z \rightarrow (L^{-1} Q^*) z \quad (175)$$

$$\begin{aligned} \tilde{A} &\rightarrow (L^{-1} Q^*) \tilde{A} (L^{-1} Q^*)^{-1} \\ \tilde{B} &\rightarrow (L^{-1} Q^*) \tilde{B} \end{aligned} \quad (176)$$

Denote the new realization as (A, B, C, D) and the stein equation becomes.

$$A A^* + B B^* = I_m \quad (177)$$

where the Contrallability Gramian equals identity. This transformation suggests that, starting from any realization, it is always possible to finally arrive, via invertible transformations, at the form in which A is triangular and P is identity. Notice that since BB^* is positive semidefinite, A will have to be a contraction. The eigenvectors of AA^* is in the right nullspace of BB^* . BB^* would have k non-zero eigenvalues. (m-k) of the eigenvalues of AA^* will be one, others smaller than one.

Now let's discuss the following question: why would it be appealing to have the Contrallability Gramian equal to identity. As is well known, any realization of ARMA model always corresponds to the choice of T, the coordinate transformation, which can be computationally disastrous. The natural and intuitive way is to choose $P = I$, so that the LTI system would be less sensitive to perturbation. In the following example the choice of T makes noise x_2 ten times bigger:

$$T = \begin{pmatrix} \frac{1}{10} & 0 \\ 0 & 10 \end{pmatrix} \quad (178)$$

$$\begin{pmatrix} \frac{1}{10} & 0 \\ 0 & 10 \end{pmatrix} \begin{pmatrix} z_1(t+1) \\ z_2(t+1) \end{pmatrix} = T A T^{-1} \begin{pmatrix} \frac{1}{10} & 0 \\ 0 & 10 \end{pmatrix} \begin{pmatrix} z_1(t+1) \\ z_2(t+1) \end{pmatrix} + \begin{pmatrix} \frac{1}{10} & 0 \\ 0 & 10 \end{pmatrix} B x(t) \quad (179)$$

while the transformed system still depicts the same underlying LTI model. Realisticly, with a bad choice A,B and C, perturbation could be enormously amplified, resulting in a computationally unstable system and an extremely bad approximation result. What we can do is to pick A, B without any knowledge of C, to make sure $(B \ AB \ A^2 B \ \dots)$ does not turn the system into an ill conditioned one. This would lead to good performance for high dimensional models under the existence of noises.

5. Reproducing Kernel Hilbert Space and Schur Parametrization

5.1 Reproducing Kernel Hilbert Space[27]

Definition 5.1.1.[Reproducing Kernel Hilbert Space][28]: A reproducing kernel Hilbert Space is a Hilbert space H of C^p valued functions defined on some Ω open in C , with a $C^{p \times p}$ valued function $K(z, \omega)$ s.t. for every choice of $\omega \in \Omega$, $c \in C^p$ and $f \in H$

$$1) \quad K(z, \omega) c \in H \quad (180)$$

$$2) \quad \langle f, K(\cdot, \omega) c \rangle_H = c^* f(\omega) \quad (181)$$

The reproducing kernel has the following properties:

- 1) The reproducing kernel is unique.
- 2) It is a positive function in the sense

$$\sum_{i,j=1}^r c_j^* K(\omega_j, \omega_i) c_i \geq 0 \quad (182)$$

For $X = \text{span}(f_1, \dots, f_n)$. Let $F \equiv (f_1 \ f_2 \ \dots \ f_n)$ where f_i are column basis for the space, then the X is a Reproducing Kernel Hilbert Space, with

$$K_\omega(\cdot) = \sum_{i,j=1}^n f_i(\cdot) (P^{-1})_{ij} f_j(\omega)^* = F P^{-1} F^*(\omega) \text{ where } P = F^* F \quad (183)$$

Where P is the orthogonal projection onto F . To verify this is indeed the kernel of space X , if $f = F C$.

$$\langle f, K_\omega c \rangle \quad (184)$$

$$= \langle F C, F P^{-1} F^*(\omega) c \rangle \quad (185)$$

$$= c^* F(\omega) P^{-1} F^* F C \quad (186)$$

$$= c^* F(\omega) C \quad (187)$$

$$= c^* f(\omega) \quad (188)$$

For Hardy space, from Cauchy's Integral the kernel reads:

$$K_\omega(z) = \frac{I_m}{1 - \bar{\omega} z} \text{ where } \omega \in \mathbb{D}, z \in \mathbb{D} \quad (189)$$

A simple verification yields:

$$\langle f, K(\cdot, \omega) c \rangle_{H_2^p} = c^* \int_{\mathbb{T}} \frac{I_m f(z)}{1 - \omega \bar{z}} \frac{dz}{z} = c^* \int_{\mathbb{T}} \frac{I_m f(z)}{z - \omega} dz = c^* f(\omega) \quad (190)$$

For QH_2^p , from (128), the kernel is:

$$K_\omega(z) = \frac{Q(z) Q(\omega)^*}{1 - \bar{\omega} z} \text{ where } \omega \in \mathbb{D}, z \in \mathbb{D} \quad (191)$$

since for $f \in H_2^p$, $Qf \in QH_2^p$

$$\langle Qf, K(\cdot, \omega) c \rangle_{QH_2^p} \quad (192)$$

$$= c^* \int_{\mathbb{T}} \frac{Q(\omega) Q(z)^* Q(z) f(z)}{1 - \omega \bar{z}} \frac{dz}{z} \quad (193)$$

$$= c^* \int_{\mathbb{T}} \frac{Q(\omega) f(z)}{z - \omega} dz \quad (194)$$

$$= c^* Q(\omega) f(\omega) \quad (195)$$

For $\mathcal{H}(Q) = H_p^2 - QH_p^2$ the kernel is given by:

$$K_\omega(z) = \frac{I_p - Q(z) Q(\omega)^*}{1 - \bar{\omega} z} \text{ where } \omega \in \mathbb{D}, z \in \mathbb{D} \quad (196)$$

Verification: for $f \in \bar{H}_2^p$ where $Qf \in \mathcal{H}(Q) H_2^p$:

$$\langle Qf, K(\cdot, \omega) c \rangle_{QH_2^p} \quad (197)$$

$$= c^* \int_{\mathbb{T}} \frac{Q(z) f(z) - Q(\omega) Q(z)^* Q(z) f(z)}{1 - \omega \bar{z}} \frac{dz}{z} \quad (198)$$

$$= c^* \int_{\mathbb{T}} \frac{Q(z) f(z)}{1 - \omega \bar{z}} \frac{dz}{z} - c^* Q(\omega) \int_{\mathbb{T}} \frac{f(z)}{1 - \omega \bar{z}} \frac{dz}{z} \quad (199)$$

$$= c^* Q(\omega) f(\omega) \quad (200)$$

$$(201)$$

Since $Qf \in \mathcal{H}(Q) \subset H_2^p$, the first part of the third equation yields $c^* Q(\omega) f(\omega)$ and the second part is evaluated

outside the unit circle, where the domain $f \in \bar{H}_2^p$ is defined. The function is analytic outside the unit circle, so the second part equals zero. Which proves that this is the reproducing kernel of the RKHS.

A matrix $J \in \mathbb{C}^{m \times m}$ is said to be a signature matrix if $J = J^*$ and $J^* J = I$. And the space θH_2^p has orthogonal compliment denoted as $\mathcal{H}(\theta)$, where the inner product is defined as: $\langle f, g \rangle_J = \langle f, Jg \rangle$, its reproducing kernel is

$$K_\theta(z, \omega) = \frac{J - \theta(z) J(\theta(\omega))^*}{1 - \bar{\omega} z} \quad (202)$$

Let the function be $f(z) = \frac{\begin{pmatrix} u \\ v \end{pmatrix}}{1 - \bar{\omega} z}$ where $\omega \in \mathbb{D}$, $u \in \mathbb{C}^p$, $\|u\| = 1$, and $v \in \mathbb{C}^p$. And we have

$$\Theta(\omega, u, v)(z) = I_{2p} - (1 - z) \frac{1 - |\omega|^2}{1 - \|v\|^2} \frac{\begin{pmatrix} u \\ v \end{pmatrix} \begin{pmatrix} u^* & v^* \end{pmatrix}}{1 - \bar{\omega} z} J \quad (203)$$

The following theorems about reproducing kernel Hilbert space will be useful in later sections.

Theorem 5.1.1 .[RKHS theorem]²⁷: Let S be a $p \times q$ matrix valued function which is analytic and contractive in the unit disc, and be isometric. Then for every choice of α , $\alpha_1, \dots, \alpha_n$ in the unit disc, and $\eta_n, \eta_n, \dots, \eta_n$ in $\mathbb{C}^q$, $(R_\alpha f)(\lambda) = \frac{f(\lambda) - f(\alpha)}{\lambda - \alpha}$.

1) $R_\alpha S\eta \in \mathcal{H}(S)$

2) If $f \in \mathcal{H}(S)$, then $R_\alpha f \in \mathcal{H}(S)$

Where $\mathcal{H}(S) = H_p^2 - SH_p^2$, i.e. $\mathcal{H}(S)$ is the orthogonal compliment of SH_p^2 in H_p^2 .

Theorem 5.1.2. [Decomposition of J Inner function]: Let $\mathcal{M} = \text{span}(f_1, \dots, f_n)$ be an n dimensional subspace of H_m^2 which is R_α invariant for some point α in the unit disc and suppose that the Pick matrix $\mathbb{P}$ with i, j entry,

$$\mathbb{P}_{i,j} = \langle J f_j, f_i \rangle, \quad i, j = 1, \dots, n, \quad (204)$$

is positive definite. Then the space $\mathcal{M}$ endowed with the J inner product is an $\mathcal{H}(U)$ space based on a rational J inner function U which admits a factorization of the form:

$$U = U_1 \dots U_n \quad (205)$$

where each of the U_i , $i = 1, \dots, n$ is an elementary J inner factor (i.e. $\dim(\mathcal{H}(U)) = 1$) which is analytic in the closed unit disc. More over, in this instance,

$$\mathcal{M} = H_m^2 - UH_m^2 \quad (206)$$

in which $-$ indicates that the orthogonal complement of UH_m^2 in H_m^2 is taken with respect to the J inner product.

5.2 Linear Fractional Transformation

A Linear Fractional Transformation is a mapping between two matrices in $\mathbb{C}^{p \times p}$. Let

$$\Theta = \begin{pmatrix} \Theta_{11} & \Theta_{12} \\ \Theta_{21} & \Theta_{22} \end{pmatrix} \quad (207)$$

be a $(2p \times 2p)$ rational J -inner function analytic in the unit disc. And let A be a $p \times p$ rational inner function. Then we have $(\Theta_{21}A + \Theta_{22})$ is invertible in the unit disc and:

$$B \equiv T_\Theta(A) = (\Theta_{11}A + \Theta_{12})(\Theta_{21}A + \Theta_{22})^{-1} \quad (208)$$

$$\hat{T}_\Theta(A) = (A\Theta_{12} + \Theta_{22})^{-1}(A\Theta_{11} + \Theta_{13}) \quad (209)$$

is inner. The inverse transformation is:

$$\begin{aligned} B\Theta_{21}A + B\Theta_{22} &= \Theta_{11}A + \Theta_{12} \\ \Rightarrow B\Theta_{22} - \Theta_{12} &= (\Theta_{11} - B\Theta_{21})A \end{aligned} \quad (210)$$

$$\begin{aligned} &\Rightarrow T_\Theta^{-1}(B) = A \\ &= (\Theta_{11} - B\Theta_{21})^{-1}(B\Theta_{22} - \Theta_{12}) \end{aligned} \quad (211)$$

Notice that $T_\Phi \circ T_\Psi = T_{\Phi\Psi}$. Then we have the following result: for $B = T_{\theta(\omega, u, v)}(A)$,

$$\begin{pmatrix} B \\ I_p \end{pmatrix} = \Theta \begin{pmatrix} A \\ I_p \end{pmatrix} (\Theta_{21}A + \Theta_{22})^{-1} \quad (212)$$

where $\Theta(\omega, u, v)$ is given by (147), satisfies the interpolation condition

$$B(\omega)^* u = v \quad (213)$$

Proof:

$$(u^* \quad -v^*) \Theta(\omega, u, v)(\omega) \quad (214)$$

$$= (u^* \quad -v^*) \left(I_{2p} - (1 - \omega) \frac{1 - |\omega|^2}{1 - \|v\|^2} \frac{\begin{pmatrix} u \\ v \end{pmatrix}}{1 - \bar{\omega}\omega} \frac{(u^* \quad v^*)}{1 - \omega} J \right) \quad (215)$$

$$= (u^* \quad -v^*) - (1 - \omega) \frac{1 - |\omega|^2}{1 - \|v\|^2} \frac{1 - \|v\|^2}{1 - |\omega|^2} \frac{(u^* \quad -v^*)}{1 - \omega} = 0 \quad (216)$$

$$(u^* \quad -v^*) \begin{pmatrix} B(\omega) \\ I_p \end{pmatrix} = (u^* \quad -v^*) \Theta(\omega, u, v)(\omega) = 0 \quad (217)$$

$$\Rightarrow B(\omega)^* u = v \quad (218)$$

■

We have just showed the mapping from Θ to B : $B = T_\Theta(A)$, from $\mathcal{H}(Q)$ to $\mathcal{H}(B)$ is an isometry, Also associated with this mapping is the interpolation property (159). Now the following theorem would show the property of the inverse mapping from B , which has the interpolation condition, to A .

Theorem 5.2.1 [27]: Let B be a rational inner function, and let $\Theta(\omega, u, v)$ be the J -inner function. $\exists$ an inner function A s.t. $B = T_\Theta(A)$ if and only if the interpolation condition $B(\omega)^ u = v$ is satisfied. And we have $\deg B = \deg A + 1$; Notice that the mapping*

$$\tau : \begin{pmatrix} f_1 \\ f_2 \end{pmatrix} \rightarrow f_1 - B f_2 \quad (\mathcal{H}(\Theta) \text{ to } H_2^p) \quad (219)$$

Will map the generator of $\mathcal{H}(\Theta)$ as follows

$$\frac{\begin{pmatrix} u \\ v \end{pmatrix}}{1 - \bar{\omega}z} \rightarrow \frac{u - B(z)v}{1 - \bar{\omega}z} \quad (220)$$

Now that we know given $B, \Theta(\omega, u, v)$ which satisfy the interpolation condition

$$B(\omega)^* u = v \quad (221)$$

there exists an inner matrix valued function A , such that

$$B = T_{\Theta}(A) \quad (222)$$

Another important result is that if B is inner, and $\omega \in \mathbb{D}$, then exists $u \in \mathbb{C}^p$, $\|u\| = 1$ s.t. $\|B(\omega)^* u\| < 1$.

From the above two results we reach the following: given B, ω (the poles), we will be able to find interpolation directions u , such that $B(\omega)^* u = v$ and $\|v\| < 1$; From Θ , we will be able to obtain A whose degree is one smaller than B. By doing this recursively and we end up with a degree one matrix.

$$R(z) = \begin{pmatrix} I_p & & \\ & -z I_n & I_n \\ & & I_p \\ & I_n & \end{pmatrix} \quad (223)$$

$$\mathcal{R}_{p,n}(X(z)) \equiv \begin{pmatrix} I_p & 0 \end{pmatrix} T_{R(z)}(X(z)) \begin{pmatrix} I_p \\ 0 \end{pmatrix} \quad (224)$$

The above construction is associated with the LTI system in the following way: if

$$X(z) = \begin{pmatrix} D & C \\ B & A \end{pmatrix} \quad (225)$$

then we have

$$\mathcal{R}_{p,n} \left(\begin{pmatrix} D & C \\ B & A \end{pmatrix} \right) = D + C(z I_n - A)^{-1} B \quad (226)$$

which is the transfer function of the LTI system characterized by A, B, C and D. Details will be discussed in the next section.

5.3. Lossless System: Recursive Construction of Balanced Realizations

In this subsection, the recursive matrix factorization of [Olivi, Hanzon and Peeters][1] of all-pass transfer functions is presented in the frame work of balanced realization and linear fractional transformation. Given the rational transfer function $G(z)$, for each pair of matrices U and V, there exists a mapping denoted as $\mathcal{F}_{U,V}$ which is as follows :

$$F(z) = \begin{bmatrix} F_1(z) & F_2(z) \\ F_3(z) & F_4(z) \end{bmatrix} = V \begin{bmatrix} 1 & 0 \\ 0 & G(z) \end{bmatrix} U^* \quad (227)$$

where $F_1(z)$ is $p \times p$, $F_4(z)$ is scalar, $F_2(z)$ is $p \times 1$ and $F_3(z)$ is $1 \times p$.

$$\mathbb{F}_{U,V} : G(z) \rightarrow F_1(z) + \frac{F_2(z) F_3(z)}{z - F_4(z)} \equiv \tilde{G}(z) \quad (228)$$

$\tilde{G}(z)$ has the same size as $G(z)$, $p \times p$. The state space realization matrix of $G(z)$ is:

$$\begin{pmatrix} D & C \\ B & A \end{pmatrix} \quad (229)$$

where D is $p \times p$, A is $n \times n$ and we say this realization matrix has order n . The state space realization of $\tilde{G}(z)$ is given by:

$$\begin{pmatrix} \tilde{D} & \tilde{C} \\ \tilde{B} & \tilde{A} \end{pmatrix} = \begin{pmatrix} V & 0 \\ 0 & I_n \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & D & C \\ 0 & B & A \end{pmatrix} \begin{pmatrix} U^* & 0 \\ 0 & I_n \end{pmatrix} \quad (230)$$

where $\tilde{D}$ is $p \times p$, $\tilde{A}$ is $(n+1) \times (n+1)$.

$$\tilde{G}(z) = \tilde{D} + \tilde{C} (z I_n - \tilde{A})^{-1} \tilde{B} \quad (231)$$

Two properties of the transformation^[23]:

- i) Any minimal balanced realization of a $p \times p$ rational lossless function the observability and controllability Gramians are both equal to the identity matrix and the associated realization matrix is unitary.
- ii) Given a $p \times p$ rational function $G(z)$, with realization matrix has order n and is unitary, then $G(z)$ is lossless and has McMillan degree $\leq n$. The realization is minimal if and only if A is asymptotically stable. Then we say it's balanced.

5.4 Tangential Schur Algorithm

With the recursive construction of Balanced Realizations of lossless System, we have the tangential Schur Algorithm as follows:

- 1) $\omega_k \in \mathbb{C}$, $|\omega_k| < 1$
- 2) $u_k \in \mathbb{C}^{p \times 1}$, $\|u_k\| = 1$

- 3) $\xi_k \in \mathbb{C}$, $|\xi_k| = 1$
- 4) $H_k(u, v, \omega, \xi)$ is a $2p \times 2p$ constant J-unitary matrix
- 5) $G^{(n)}$ is a $p \times p$ lossless transfer matrix of McMillan degree n ,

Then there exists $v_k \in \mathbb{C}^{p \times 1}$, $\|v_k\| < 1$, and the tangential schur algorithm is:

$$v_k = G^{(k)}(\omega) u_k \quad (232)$$

$$\Theta_k = \Theta(u_k, v_k, \omega_k, \xi_k, H_k(u_k, v_k, \omega_k, \xi_k)) \quad (233)$$

$$G^{(k-1)} = T_{\Theta_k}^{-1}(G^{(k)}) \quad (234)$$

With each choice of unit vectors u_k ($k = n, \dots, 1$) yields of set of $G^{(k)}$ ($k = n, \dots, 1$), where each $G^{(k)}$ has lossless of McMillan degree k and we can write:

$$G^{(n)} = T_{\Theta_n}(T_{\Theta_{n-1}}(\dots(T_{\Theta_1}(G^{(0)}))) = T_{\Theta_n \Theta_{n-1} \dots \Theta_1}(G^{(0)}) \quad (235)$$

Where $G^{(0)}$ is a constant unitary matrix

Theorem 5.4.1[Tangential Schur][29]: Let $\Theta(z)$ be an elementary J-inner factor analytic inside the unit disc. If $T_\Theta(z)$ coincides with a mapping $F_{U,V}$ with U and V unitary, then $\Theta(z)$ can be written as

$$\Theta(z) = H(uv^*) S_{u,\omega}(z) H(\bar{\omega} uv^*) \begin{bmatrix} P & 0 \\ 0 & Q \end{bmatrix} \quad (236)$$

for some P, Q are some unitary $p \times p$ matrices. In that case one can take $U = \hat{U} \begin{bmatrix} 1 & 0 \\ 0 & P \end{bmatrix}$, $V = \hat{V} \begin{bmatrix} 1 & 0 \\ 0 & Q \end{bmatrix}$, where

$$\hat{U} = \begin{pmatrix} \frac{\sqrt{1-|\omega|^2}}{\sqrt{1-|\omega|^2\|v\|^2}} u & I_p - \left(1 + \frac{\omega \sqrt{1-\|v\|^2}}{\sqrt{1-|\omega|^2\|v\|^2}}\right) u u^* \\ \frac{\bar{\omega} \sqrt{1-\|v\|^2}}{\sqrt{1-|\omega|^2\|v\|^2}} & \frac{\sqrt{1-|\omega|^2}}{\sqrt{1-|\omega|^2\|v\|^2}} u^* \end{pmatrix} \quad (237)$$

$$\hat{V} = \begin{pmatrix} \frac{\sqrt{1-|\omega|^2}}{\sqrt{1-|\omega|^2\|v\|^2}} v & I_p - \left(1 - \frac{\sqrt{1-\|v\|^2}}{\sqrt{1-|\omega|^2\|v\|^2}}\right) \frac{v v^*}{\|v\|^2} \\ \frac{\sqrt{1-\|v\|^2}}{\sqrt{1-|\omega|^2\|v\|^2}} & -\frac{\sqrt{1-|\omega|^2}}{\sqrt{1-|\omega|^2\|v\|^2}} v^* \end{pmatrix}$$

And there is a mapping between (G, ω, u) and (v, G^0) . There are richness of possibilities to choose ξ , H , ω , and u ; For example we can employ $\xi = 1$, $H_k = I_p$. We can fix all interpolation points at zero and let u_k vary among a canonical basis of $\mathbb{C}^p$. And we have

$$\Phi = \begin{pmatrix} V_n & \\ & I_n \end{pmatrix} \begin{pmatrix} 1 & & \\ & V_{n-1} & \\ & & I_{n-1} \end{pmatrix} \dots \begin{pmatrix} I_{n-1} & & \\ & V_1 & \\ & & 1 \end{pmatrix} \quad (238)$$

$$\Psi = \begin{pmatrix} I_{n-1} & & \\ & U_1^* & \\ & & 1 \end{pmatrix} \begin{pmatrix} I_{n-2} & & \\ & U_2^* & \\ & & I_2 \end{pmatrix} \dots \begin{pmatrix} U_n & & \\ & I_n & \end{pmatrix} \quad (239)$$

$$R_n = \begin{pmatrix} D_n & C_n \\ B_n & A_n \end{pmatrix} = \Phi \begin{pmatrix} I_{n+1} & \\ & R_0 \end{pmatrix} \Psi \quad (240)$$

6. Null Basis Parametrization

6.1 Parametrizing MIMO TIB LTI System: What do we expect?

Parameterization is a mathematical process of deciding and defining the parameters necessary for a complete description and specification of a model, deemed to be relevant to some design goal. To be compatible with TIB, it is essential to construct parametrization in the Triangular Input Balanced context, fitting the goal of the TIB model.

Now let $R_n^{l \times k}$ denote the transfer function associated with $l \times k$ MIMO LTI system of degree n . In the TIB context we are concerned of the inner part of the transfer function, i.e. A and B . We denote the space of TIB $\{A, B\}$ as T_n^k where A is $n \times n$ and B is $n \times k$, as we work with the inner part. A parameterization of the TIB model is essentially a mapping from the parameter set Ω to the space $\{T_n^k\}$. This mapping is, by definition, a description of how the element of T_n^k are formed from the parameters in Ω .

Here we discuss our expectations of a parametrization of TIB realization based on the design goal of the model.

1) **Nested Mapping.** The existence of a TIB realization for every LTI system was shown in Section 4.2.1 with a constructive proof. However, it can be computationally dangerous to follow that routine in constructing TIB realization, since the triangular matrix A can be extremely ill conditioned. A parameterization of LTI system with good computational property is desired for the mapping from a random LTI realization to TIB LTI realization. This design goal would require us to have not only one mapping but a nested mapping: a mapping from any LTI system realization to the parameterization, and a mapping between the parameterization with TIB realization, denoted as

$$\mathbb{U} : R_n^{l \times k} \rightarrow \Omega \quad (241)$$

$$\mathbb{V} : \Omega \rightarrow T_n^k \quad (242)$$

For example, in SISO TIB, $\mathbb{U}$ maps any LTI realization to the poles of the system, and SISO TIB pairs are constructed from the poles and the poles only.

2) **Identifiability.** The requirement that mapping $\mathbb{V}$ being injective and has an inverse naturally leads to a parametrization with less redundancy. Generally speaking this is a crucial property since in LTI system reduction or a search algorithm, the size of the parameter space is decisive in the computation efficiency and uniqueness of solution.

3) **Similarity Invariance.** In the TIB realization context, the inner part A and B are prefixed. From a statistical inference point of view, this unique design allow us the freedom of choice in prior distributions of the system, which introduces variance-bias tradeoff to system estimation. The prior expresses the specific and definite information regarding the distribution of the system. A natural intuition is to have the mapping between parameter set Ω and the space of LTI system be bijective, since despite the variations of the representation, the information itself should be independent of the realization, and depend only on the LTI system. We already know that LTI system is invariant under similarity transformations to the LTI realization, it is appealing to also have this property in the parameterization.

In this section, we investigate the connection between MIMO orthonormal basis and Schur parameterization of MIMO LTI system, providing a new parametrization of MIMO TIB LTI system. In Chapter 4 the TIB parametrization in terms of $\{A, B\}$ as well as mapping between SISO TIB and system poles are introduced. In Chapter 5 the Schur parameterization is reviewed. In Section 6.2 we recall some properties of Blaschke Potapov Factors and its relations to MIMO LTI system. In Section 6.3 we introduce the method in [Xiao Yu 2015][30], which reveals the Schur parameterization of TIB system. In Section 6.4 we provide the construction of MIMO orthonormal basis from QR matrix decomposition point of view. The connection between the null basis and Schur parameterization is revealed by the Nevanlinna Pick problem in Section 6.5, and a mapping satisfying the aforementioned properties between our parameter set and MIMO TIB realization is presented as our main result for this chapter.

6.2 Blaschke Potapov Factors

This section focus on some useful properties of Blaschke Potapov Factors and the following theorem reveals its connection to the MIMO approximation context.

Theorem 6.2.1. Every rational J-inner matrix valued function $U(\lambda)$ can be represented as a finite product of elementary Blaschke Potapov factors. Moreover, if the McMillan degree of $U(\lambda)$ is equal to r , then $U(\lambda)$ may be expressed as the product of r primary Blaschke Potapov factors that are each normalized at a point α times a constant J-unitary matrix on either the left or the right.

In [Olivi 2010][31], the parameterization of Blaschke-Potapov factors where $J = I_n$ was provided, namely:

$$\beta_{\omega,u}(z) = I + \left(\frac{z - \omega}{1 - \omega^* z} - 1 \right) u u^* \quad (243)$$

where $\frac{z - \omega}{1 - \omega^* z}$ is scalar Blaschke factor, $|\omega| < 1$, $\|u\|_2 = 1$. The inverse function

$$\beta_{\omega,u}(z)^{-1} = I + \left(\left(\frac{z - \omega}{1 - \omega^* z} \right)^{-1} - 1 \right) u u^* \quad (244)$$

So the Blaschke Potapov factor is a Holomorphic function in the unit disk, and its inverse has a simple pole at the point ω . The determinant of a rational J-Inner function, i.e. the Blaschke Potapov product is a finite Blaschke product

$$\det \left(\prod_k \beta_{\omega_k, u_k}(z) \right) = \prod_k \frac{\omega_k^* - z}{1 - \omega_k z} \quad (245)$$

The algebra of Blaschke Potapov Product is very similar to the scalar case except for commutativity. The zeros of the product belongs to the open unit disk and the poles belong to the compliment of the closed unit disk, which means there is no pole-zero cancellation in multiplication and we have the following relation on McMillan degree

$$\deg \left(\prod_k \beta_{\omega_k, u_k}(z) \prod_j \beta_{\omega_j, u_j}(z) \right) = \deg \left(\prod_k \beta_{\omega_k, u_k}(z) \right) + \deg \left(\prod_j \beta_{\omega_j, u_j}(z) \right) \quad (246)$$

According to a result due to [V.P. Potapov][32-33], suppose that $\{\omega_k\}$ is a sequence in $\mathbb{D}$ and $\{u_k\}$ are unit vectors and that

$$\sum_k (1 - |\omega_k|) < \infty \quad (247)$$

Then the Blaschke Potapov product

$$\prod_k \beta_{\omega_k, u_k}(z) \quad (248)$$

converges for all $z \in \mathbb{D}$. It is easy to observe that

$$\beta_{\omega,u}(\omega) u = 0 \quad (249)$$

In this way, given poles $(\omega_1, \omega_2 \dots \omega_n)$ and null vectors $(u_1, u_2, \dots, u_n)$, the product of Blaschke-Potapov factor

$$G_k(z) = \beta_{\omega_1, u_1}(z) \beta_{\omega_2, u_2}(z) \dots \beta_{\omega_n, u_n}(z) \quad (250)$$

is lossless and satisfies the interpolation condition

$$\underline{G_k(\omega_k) u_k = 0} \quad (251)$$

And the other interpolation conditions can be obtained recursively by (148).

6.3 Schur parameterization of TIB

[Olivi, Marmorat, Hanzon, Peeters][34] pointed out that if the Schur parameter v in (219) is equal to zero, then one would have a Input Balanced System where A is lower triangular, which we call ‘TIB’; XiaoYu[36] combined tangential Schur algorithm and AAK theorem to produce a new MIMO reduction methods that is compatible with TIB.

The tangential Schur algorithm is interpreted as a unified framework for orthogonal filters, including the well-known lattice filters, TIB filters of both SISO and MIMO cases. Here we will focus on the TIB filters. Recall from section 5.4 that the interpolation condition of tangential Schur algorithm is given by $v_k = G^{(k)}(\omega) u_k$. Let $v_k = 0$, and $G^{(k)}(\omega_k) u_k = 0$, in this case, u_k are called null vectors, and corresponds to MIMO TIB case. ω_k are the poles and together with u_k form the atlas of MIMO TIB system A and B. In this case,

$$\hat{U} = \begin{pmatrix} \sqrt{1 - |\omega|^2} & u & I_p - (1 + \omega) u u^* \\ \bar{\omega} & & \sqrt{1 - |\omega|^2} u^* \end{pmatrix} \quad (252)$$

$$\hat{V} = \begin{pmatrix} 0 & I_p \\ 1 & 0 \end{pmatrix} \quad (253)$$

A nice little lemma gives:

$$\Phi = \begin{pmatrix} V & \\ & I_n \end{pmatrix} \begin{pmatrix} 1 & & \\ & V & \\ & & I_{n-1} \end{pmatrix} \dots \begin{pmatrix} I_{n-1} & & \\ & V & \\ & & 1 \end{pmatrix} = \begin{pmatrix} I_p & & \\ & I_n & \\ & & 1 \end{pmatrix} \quad (254)$$

where V is as in (176). Within this unified frame work, given n pairs of poles and null vectors $\{\omega_k, v_k\}_{k=1, \dots, n}$, where

$$|\omega_k| < 1 \quad (255)$$

$$\|v_k\|_2 = 1 \quad (256)$$

the parameterization of TIB system yields the updating formulae as follows

$$D_n = D_{n-1} - (1 + \bar{\omega}_n) D_{n-1} u_n u_n^* \quad (257)$$

$$C_n = \begin{bmatrix} \sqrt{1 - |\omega_n|^2} & D_{n-1} u_n & C_{n-1} \end{bmatrix} \quad (258)$$

$$B_n = \begin{bmatrix} \sqrt{1 - |\omega_n|^2} u_n^* \\ B_{n-1} (I_p - (1 + \bar{\omega}_n) u_n u_n^*) \end{bmatrix} \quad (259)$$

$$A_n = \begin{bmatrix} \omega_n & \\ \sqrt{1 - |\omega_n|^2} B_{n-1} u_n & A_{n-1} \end{bmatrix} \quad (260)$$

Where the matrix

$$\begin{bmatrix} B_n & A_n \\ D_n & C_n \end{bmatrix} \quad (261)$$

is unitary, and A_n will be lower triangular. So we have $A A^* + B B^* = I$, and (A, B) is a triangular input balanced pair. The pole-null vector pairs determines completely the inner part of the transfer function, as opposed to poles only in the SISO case.

6.4 Orthonormal Basis of MIMO System: A Linear Algebra Approach

The Nevanlinna Pick Problem: given a set of points $\omega_1, \dots, \omega_n$ in the open unit disc, and two sets of vectors: $\xi_1, \dots, \xi_n$ in $\mathbb{C}^p$ and $\eta_1, \dots, \eta_n$ in $\mathbb{C}^q$, the objective is to find necessary and sufficient conditions on the data to insure the existence of a $p \times q$ matrix valued function S which is analytic and contractive in the unit disc such that

$$\xi_j^* S(\omega_j) = \eta_j^* \quad (262)$$

for $j = 1, \dots, n$ and to describe all the existing solutions when these conditions are met. It is presumed that the functions

$$\frac{\xi_j}{1 - \omega_j^* z}, j = 1, \dots, n \quad (263)$$

are linearly independent. Independence will be true if $\omega_i \neq \omega_j$ and ξ_i is non-zero, or $\omega_i \neq \omega_j$ and ξ_i, ξ_j are linearly independent.

A fundamental role is played here by the Pick matrix $\mathbb{P}$ with entries

$$\mathbb{P}_{ij} = \frac{\xi_i^* \xi_j - \eta_i^* \eta_j}{1 - \omega_i^* \omega_j} \quad i, j = 1, \dots, n \quad (264)$$

Theorem 6.2.1[27]. The Nevanlinna Pick problem admits a solution S which is a $p \times q$ matrix valued function which is analytic and contractive in the open unit disc, if and only if the Pick matrix $\mathbb{P}$ with entries defined in (272) is positive semidefinite.

Suppose next that $\mathbb{P} > 0$, and let

$$v_i = \begin{bmatrix} \xi_i \\ \eta_i \end{bmatrix} \quad \text{and} \quad f_i = \frac{v_i}{1 - \omega_i^* z} \quad (265)$$

According to RKHS theorem 1, there exist an $m \times m$ rational J inner function U such that

$$\mathcal{M} = \text{span} \{f_1, f_2, \dots, f_n\} \quad (266)$$

endowed with the J inner product is an $\mathcal{H}(U)$ space.

Now in the previous section, we have shown the connection between TIB LTI systems and tangential Schur algorithm, where the TIB system corresponds to system with $\{\eta_j\}$ are all zero. So the span of the null space becomes:

$$f_i = \frac{v_i}{1 - \omega_i^* z} \quad (267)$$

where

$$v_i = \xi_i \quad (268)$$

$$J = I \quad (269)$$

Normalization yields

$$f_i = \frac{v_i \sqrt{1 - |\omega_i|^2}}{1 - \omega_i^* z} \quad (270)$$

The span of the left kernel of a matrix is the orthogonal compliment of the column space of the matrix, which in our case, is the inner part of the matrix valued transfer function, i.e. the basis of the transfer function. With this space available, we are able to generalize the results of [A.P. Mullhaupt][35] in the TIB SISO case. Let's consider a Walsh like approximation. We denote the basis and transfer function of a $l \times m$ MIMO LTI system with degree n as:

$$S_n = \text{span} \{f_1, f_2, \dots, f_n\} \quad (271)$$

$$\begin{aligned} T(z) &= t_0 + t_1 z + t_2 z^2 + \dots \\ &= [T_1(z) \ T_2(z) \ \dots \ T_m(z)] \end{aligned} \quad (272)$$

$$t_j = [t_j^1, \dots, t_j^m] \quad (273)$$

where $T_j(z)$ is the j th column of $T(z)$, a vector valued function. t_j are matrix valued coefficients of $T(z)$. In Walsh like approximation, the perpendicularity condition of H^2 approximation gives:

$$\left(T_j - \sum_{i=1}^n c_i f_i \right) \perp \text{span}(S_n) \quad (274)$$

thus for $k = 1, \dots, n$:

$$\left\langle \left(T_j - \sum_{i=1}^n c_i f_i \right), f_k \right\rangle \quad (275)$$

$$= \int_{|z|=1} \left(\frac{v_k}{1 - \omega_k^* z} \right)^* \left(T_j - \sum_{i=1}^n c_i \frac{v_i}{1 - \omega_i^* z} \right) \frac{dz}{z} \quad (276)$$

$$= v_k^* T_j(\omega_k^*) - \sum_{i=1}^n c_i \frac{v_k^* v_i}{1 - \omega_i^* \omega_k} = 0 \quad (277)$$

The matrix representation yields

$$\begin{pmatrix} v_1^* & v_1^* \omega_1^* & v_1^* \omega_1^{*2} & \dots \\ v_2^* & v_2^* \omega_2^* & v_2^* \omega_2^{*2} & \dots \\ \dots & \dots & \dots & \dots \\ v_n^* & v_n^* \omega_n^* & v_n^* \omega_n^{*2} & \dots \end{pmatrix} \begin{pmatrix} v_1 & v_2 & \dots & v_n \\ v_1 \omega_1 & v_2 \omega_2 & \dots & v_n \omega_n \\ v_1 \omega_1^2 & v_2 \omega_2^2 & \dots & v_n \omega_n^2 \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \dots \\ c_n \end{pmatrix} = \begin{pmatrix} v_1^* T_j(\omega_1^*) \\ v_2^* T_j(\omega_2^*) \\ \dots \\ v_n^* T_j(\omega_n^*) \end{pmatrix}$$

We recognize that

$$\begin{pmatrix} v_1^* & v_1^* \omega_1^* & v_1^* \omega_1^{*2} & \dots \\ v_2^* & v_2^* \omega_2^* & v_2^* \omega_2^{*2} & \dots \\ \dots & \dots & \dots & \dots \\ v_n^* & v_n^* \omega_n^* & v_n^* \omega_n^{*2} & \dots \end{pmatrix} \begin{pmatrix} t_0^j \\ t_1^j \\ t_2^j \\ \dots \\ \dots \end{pmatrix} = \begin{pmatrix} v_1^* T_j(\omega_1^*) \\ v_2^* T_j(\omega_2^*) \\ \dots \\ v_n^* T_j(\omega_n^*) \end{pmatrix} \quad (279)$$

So the interpolation system is just the normal equation of least square (l^2 norm) approximation:

$$\min_c \left\| \begin{pmatrix} v_1 & v_2 & \dots & v_n \\ v_1 \omega_1 & v_2 \omega_2 & \dots & v_n \omega_n \\ v_1 \omega_1^2 & v_2 \omega_2^2 & \dots & v_n \omega_n^2 \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \end{pmatrix} \begin{pmatrix} c_{11} & c_{12} & \dots & c_{1m} \\ c_{21} & c_{22} & \dots & c_{2m} \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ c_{n1} & c_{n2} & \dots & c_{nm} \end{pmatrix} - \begin{pmatrix} t_0 \\ t_1 \\ \dots \\ \dots \\ \dots \end{pmatrix} \right\|_{H_2} \quad (280)$$

Orthogonalizing the design matrix yields:

$$\begin{pmatrix} v_1 & v_2 & \dots & v_n \\ v_1 \omega_1 & v_2 \omega_2 & \dots & v_n \omega_n \\ v_1 \omega_1^2 & v_2 \omega_2^2 & \dots & v_n \omega_n^2 \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \end{pmatrix} = Q R \quad (281)$$

And denote:

$$C = Q^* \begin{pmatrix} t_0 \\ t_1 \\ \dots \\ \dots \\ \dots \end{pmatrix} \quad (282)$$

The least square problem is then equivalent to

$$\min_c \left\| U Q C - U \begin{pmatrix} t_0 \\ t_1 \\ \vdots \\ \vdots \\ \vdots \end{pmatrix} \right\|_{H_2} \quad (283)$$

where U is any unitary transformation, since it preserves the l_2 norm. The QR decomposition of the design matrix corresponds to finding the orthonormal basis of the space spanned by $\{f_1, f_2, \dots, f_n\}$. This QR decomposition will preserve the rationality of the basis. We choose

$$\Phi_1(z) = \frac{\sqrt{1 - |\omega_1|^2}}{1 - \omega_1 z} v_1 \quad (284)$$

and construct $\Phi_{n+1}(z)$. Observe that the normalized functions in the one dimensional space is given by

$$(\text{span}\{f_1, f_2, \dots, f_{n+1}\}) \cap (\text{span}\{f_1, f_2, \dots, f_n\}^\perp) \quad (285)$$

Before we provide proof by induction for the orthonormal basis of MIMO system, we denote the inner product in the H^2 space as $\langle \cdot, \cdot \rangle$, calculation yields the inner product between any two of the basis then can be derived as follows:

$$\left\langle \frac{v_m \sqrt{1 - |\omega_m|^2}}{1 - \omega_m z}, \frac{v_n \sqrt{1 - |\omega_n|^2}}{1 - \omega_n z} \right\rangle \quad (286)$$

$$= \frac{\sqrt{1 - |\omega_m|^2} \sqrt{1 - |\omega_n|^2} v_n^* v_m}{1 - \omega_m \omega_n^*} \quad (287)$$

Then we have the H^2 distance between any two of the basis as follows:

$$d\left(\frac{v_m}{1 - \omega_m z}, \frac{v_n}{1 - \omega_n z}\right) \quad (288)$$

$$= \left\| \frac{v_m}{1 - \omega_m z} - \left\langle \frac{v_m}{1 - \omega_m z}, \frac{v_n}{1 - \omega_n z} \right\rangle \frac{v_n}{1 - \omega_n z} \right\|_{H^2} \quad (289)$$

$$= 1 + \left(\frac{|\omega_m - \omega_n|}{|1 - \omega_m \omega_n^*|} - 1 \right) (v_2^* v_1)^2 \quad (290)$$

This result suggests how the distance between two basis is determined by the position of the pole as well as their direction. If v_2 and v_1 are parallel, the result is consistent with the SISO case as in proposed by Walsh[36].

Theorem 6.4.1: Given the reproducing kernel Hilbert space

$$\text{span} \{ f_1, f_2, \dots, f_n \} \quad (291)$$

where

$$f_i = \frac{v_i \sqrt{1 - |\omega_i|^2}}{1 - \omega_i z} \quad (292)$$

$$\|v_i\|_2 = 1, \quad |\omega_i| < 1 \quad (293)$$

and v_i is an $m \times 1$ vector. The orthonormal basis of the n dimensional space is given by:

$$\Phi_{k+1}(z) = \frac{\sqrt{1 - |\omega_{k+1}|^2}}{1 - \omega_{k+1} z} \prod_{j=1}^k \left(I + \left(\frac{\omega_j^* - z}{1 - \omega_j z} - 1 \right) x_j x_j^* \right) x_{k+1} \quad (294)$$

$$= \frac{\sqrt{1 - |\omega_{k+1}|^2}}{1 - \omega_{k+1} z} \prod_{j=1}^k \beta_{\omega_j, x_j}(z) x_{k+1} \quad (295)$$

$$x_j = \frac{\beta_{\omega_{j-1}, x_{j-1}}(\omega_j^*)^* \dots \beta_{\omega_1, x_1}(\omega_j^*)^* v_j}{\| \beta_{\omega_{j-1}, x_{j-1}}(\omega_j^*)^* \dots \beta_{\omega_1, x_1}(\omega_j^*)^* v_j \|_2} \quad (296)$$

Proof: See Appendix.

■

We can verify the perpendicular condition:

$$\langle \Phi_n(z), f_k \rangle \quad (297)$$

$$\begin{aligned}
&= \int_{|z|=1} \frac{\sqrt{1-|\omega_n|^2}}{1-\omega_n z} \left(\frac{\sqrt{1-|\omega_k|^2}}{1-\omega_k z} \right)^* v_k^* \prod_{j=1}^{n-1} \left(I + \left(\frac{\omega_j^* - z}{1-\omega_j z} - 1 \right) x_j x_j^* \right) x_n \frac{dz}{z} \\
&= \frac{\sqrt{1-|\omega_n|^2} \sqrt{1-|\omega_k|^2}}{1-\omega_n \omega_k^*} v_k^* \prod_{j=1}^{n-1} \beta_{\omega_j, x_j}(\omega_k^*) x_n
\end{aligned} \tag{299}$$

$$= \frac{\sqrt{1-|\omega_n|^2} \sqrt{1-|\omega_k|^2}}{1-\omega_n \omega_k^*} \begin{cases} 0 & k < n \\ v_k^* \prod_{j=1}^{n-1} \beta_{\omega_j, u_j}(\omega_k^*) x_n & k \geq n \end{cases} \tag{300}$$

Now let's assume, without loss of generality, $m \leq n$. Then we have

$$< \Phi_n(z) , \Phi_m(z) > \tag{301}$$

$$= \int_{|z|=1} \frac{\sqrt{1-|\omega_n|^2}}{1-\omega_n z} \left(\frac{\sqrt{1-|\omega_m|^2}}{1-\omega_m z} \right)^* u_m^* \prod_{j=m-1}^1 \beta_{\omega_j, u_j}(z)^{-1} \prod_{j=1}^{n-1} \beta_{\omega_j, u_j}(z) u_n \frac{dz}{z} \tag{302}$$

$$= \int_{|z|=1} \frac{\sqrt{1-|\omega_n|^2} \sqrt{1-|\omega_m|^2}}{1-\omega_n z} u_m^* \prod_{j=m}^n \beta_{\omega_j, u_j}(z) u_n \frac{dz}{z - \omega_m^*} \tag{303}$$

$$= \begin{cases} \frac{\sqrt{1-|\omega_n|^2} \sqrt{1-|\omega_m|^2}}{1-\omega_n \omega_m^*} u_m^* \prod_{j=m}^n \beta_{\omega_j, u_j}(\omega_m^*) u_n = 0 & m < n \\ 1 & m = n \end{cases} \tag{304}$$

These functions correspond to the columns of a QR decomposition:

$$\left(\frac{v_1}{1-\omega_1^* z} \quad \dots \quad \frac{v_n}{1-\omega_n^* z} \quad \dots \right) = QR \tag{305}$$

$$= (\Phi_1 \dots \Phi_n \dots) R \tag{306}$$

where Φ_1 are column vectors, R is the corresponding constant upper triangular matrix. We then have the expansions

$$\frac{v_q}{1 - \omega_q^* z} = \sum_{j=1}^{q-1} \Phi_j(z) R_{jq} + R_{qq} \Phi_q(z) \quad (307)$$

The error of approximating a unit norm basis in H_n^2 is then seen to be:

$$\min \|f_q - \sum_{j=1}^{q-1} c_j f_j\|_2 \quad (308)$$

$$= \|R_{qq}\| \quad (309)$$

$$= \left\langle \Phi_q(z), \frac{v_q}{1 - \omega_q z} \right\rangle \quad (310)$$

$$= \frac{\sqrt{1 - |\omega_q|^2}}{1 - |\omega_q|^2} v_q^* \prod_{j=1}^{q-1} \beta_{\omega_j, u_j}(\omega_q^*) x_q \quad (311)$$

$$= \frac{\sqrt{1 - |\omega_q|^2}}{1 - |\omega_q|^2} \left(\left\langle \beta_{\omega_{q-1}, x_{q-1}}(\omega_q^*)^* \dots \beta_{\omega_1, x_1}(\omega_q^*)^* v_q, \beta_{\omega_{q-1}, x_{q-1}}(\omega_q^*)^* \dots \beta_{\omega_1, x_1}(\omega_q^*)^* v_q \right\rangle \right) / \left(\|\beta_{\omega_{q-1}, x_{q-1}}(\omega_q^*)^* \dots \beta_{\omega_1, x_1}(\omega_q^*)^* v_q\|_2 \right) \quad (312)$$

where

$$x_q = \frac{\beta_{\omega_{q-1}, x_{q-1}}(\omega_q^*)^* \dots \beta_{\omega_1, x_1}(\omega_q^*)^* v_q}{\|\beta_{\omega_{q-1}, x_{q-1}}(\omega_q^*)^* \dots \beta_{\omega_1, x_1}(\omega_q^*)^* v_q\|_2} \quad (313)$$

and we also have

$$R_{jq} = \left\langle \Phi_j(z), \frac{v_q}{1 - \omega_q z} \right\rangle \quad (314)$$

$$= \frac{\sqrt{1 - |\omega_j|^2}}{1 - \omega_j \omega_q^*} v_q^* \prod_{n=1}^{j-1} \beta_{\omega_n, u_n}(\omega_q^*) x_j \quad (315)$$

6.5 Null Basis Parameterization of TIB

In the Tangential Schur Algorithm, when the interpolation points are the eigenvalues of the dynamic system and all the Schur parameters are all equal to the null vector, Linear fractional transformation admits a Potapov factorization and A of the dynamic system is lower triangular. The duality between being Schur parameter and interpolation of Blaschke Potapov factor forms an atlas on the embedded submanifold of $H_2^{p \times p}$, and provides the framework for a search algorithm for system degree reduction.

From a constructive point of view, we always hope the interpolation conditions will be static, all of which be solved by the final transfer function. However, this duality is dynamic, i.e. the interpolation condition only exists temporarily at its specific degree. Specifically in the TIB case, the interpolation relationship of Blaschke Potapov product of degree n in general will not be inherited by Blaschke Potapov product of $n+1$, the next search/construction step. This fact could be verified if we check the updating formulae of MIMO TIB realization of 2×2 case, where $\{\omega_2, u_2\}$ is no longer the interpolation basis of the system.

It is even more obvious if we check the transfer function itself. The lossless transfer function is a Blaschke Potapov product, where each update is equivalent to a left multiplication by the Blaschke Potapov factor. But that is on the same side of interpolation condition, and the interpolation direction and points of all previous steps changes unless the last two Blaschke Potapov factors commute. As a result of this evolutionary property, the tangential Schur algorithm on the atlas converges to local minimum rather than global minimum. And in basis construction, the relationship between interpolation point/direction to the system is also temporal, easily changed by their successor.

In this section, we introduce a new parameterization which establishes a nested mapping from any (A, B) pair to what we call Null Basis Pairs, and from Null Basis Pairs to triangular input balanced (A, B) pairs. First we make some denotations. Then we introduce the results of this section. The proofs will be at the end.

We name the $b_i = \{\omega_i, v_i\}$ pair in the orthonormal basis function as **Null Pair**, Consisting of a pole and a Null Basis. The name comes from its interpolation property and connection to LTI system: v_i is the null direction of the Blaschke Potapov factor B_{ω_i, v_i} at interpolation point ω_i , which characterizes the span of LTI system.

Let $b_i^m = (\omega_i, v_i)$ denote elements in the open subset Π_m of $\mathbb{C}^1 \times \mathbb{C}^m$ where $|\omega_i| < 1$ and $\|v_i\| = 1$. Let $f_i = \frac{v_i}{1 - \omega_i z}$ denote the corresponding vector valued function in the corresponding open subspace of H_2^m space.

Let Ω_0 denote the open subset of $\mathbb{C}^{n \times n} \times \mathbb{C}^{n \times m}$ consisting of all the controllable pairs of (A, B) . A pair (A, B) is called controllable if

$$\bigcap_{k=0}^{k=\infty} \text{Ker}(B^T (A^T)^k) = \{0\} \quad (316)$$



Tow pairs (A_1, B_1) and (A_2, B_2) are said to be equivalent if there exists a non-singular matrix $n \times n$ matrix T such that

$$A_1 = T A_2 T^{-1} \quad (317)$$

$$B_1 = T B_2 \quad (318)$$

Let $E\Omega$ denote the set of equivalent classes of all triangular input balanced controllable pairs.

Lemma 6.5.1 Given any set of equivalent class in $E\Omega$, every pair (A, B) in the equivalent class span the same Krylov row subspace.

Proof: See 6.6 Appendix

■

From the previous section we have the formula for orthonormal basis of the reproducing kernel Hilbert space,

$$\Phi_n(z) = \frac{\sqrt{1 - |\omega_n|^2}}{1 - \omega_n z} \prod_{j=1}^{n-1} \beta_{\omega_j, x_j}(z) x_n \quad (319)$$

$$x_n = \frac{\beta_{\omega_{n-1}, x_{n-1}}(\omega_n^*)^* \dots \beta_{\omega_1, x_1}(\omega_n^*)^* v_n}{\| \beta_{\omega_{n-1}, x_{n-1}}(\omega_n^*)^* \dots \beta_{\omega_1, x_1}(\omega_n^*)^* v_n \|_2} \quad (320)$$



Where $\beta_{\omega, x}$ is the Blaschke Potapov Factor associated with pole ω and null vector x . The basis span the $\mathcal{H}(U)$ space where U is the corresponding Blaschke Potapov factor. Formula (323) characterize the relationship between Null Basis and Null vector parameterizations. The following theorem gives representation of TIB LTI system using the null basis. Null basis also implicitly changes the direction of update, such that each basis is static with respect to adaptive update of parameter space.

Theorem 6.5.1 Let $b_k = \{\omega_k, v_k\}$ denote the Null basis pair that corresponds to $m \times 1$ vector valued function f_k . And let $\{b_1, \dots, b_n\}$ be a sequence of Null Basis pairs and $\{f_1, \dots, f_n\}$ be the corresponding $m \times 1$ vector valued function, the following recursive updating formulae at step $k, k \leq n$, generates $(A_k B_k)$ that are triangular input balanced.

$$L_k = [J_{k-1} L_{k-1} \quad t_{k-1} y_{k-1}] \quad (321)$$

$$t_k = \sqrt{1 - |\omega_k|^2} \quad (322)$$

$$y_k = \frac{\prod_{i=k-1}^1 \beta_{\omega_i, \frac{y_i}{\|y_i\|}}(\omega_k^*)^* v_k}{\| \prod_{i=k-1}^1 \beta_{\omega_i, \frac{y_i}{\|y_i\|}}(\omega_k^*)^* v_k \|_2} \quad (323)$$

$$J_k = I - (1 + \omega_k^*) y_k y_k^* \quad (324)$$

$$A_k = \begin{bmatrix} A_{k-1} & 0 \\ t_k y_k^* L_k & \omega_k \end{bmatrix}, \quad B_k = \begin{bmatrix} B_{k-1} \\ t_k y_k^* \prod_{i=n-1}^1 J_i \end{bmatrix} \quad (325)$$

Proof: See 6.6 Appendix

■

Lemma 6.5.2 will be one of the major results of this Chapter, which reveals the connection between the null basis parameterization and the null vector parameterization. Even though the two recursive updating formulae yields the same result A_n and B_n , the intermediate step results are totally different. Besides, the updating direction are reversed. This reverse relationship comes from updating to different sides of the Blaschke Potapov Product. It can also be verified from the updating formulae that the null basis recursion updates (A, B) pair to the bottom row, leaving everything else unchanged, while the tangential Schur algorithm updates makes change to the pair from last step, and updates to the top row. While the tangential Schur algorithm acts on the columns of the system. To the system (A, B, C, D) this difference seems trivial, however when it comes to the Krylov row space, with which we approximate the impulse response, there is a huge difference. The existing Krylov subspace is not preserved in the tangential Schur algorithm approach. This potentially leads to the permutation problem and some other issues, which makes it harder to manipulate and construct the orthogonal basis. The relationship between u_j and v_j is sufficient but not necessary condition for equivalence of Krylov row space, since Null Basis is invariant to permutation.

Lemma 6.5.2 Let $b_k = \{\omega_k, v_k\}$ denote the Null basis pair that corresponds to $m \times 1$ vector valued function f_k , $c_k = \{\omega_k, u_k\}$ denote the Null vector pair. Let $\{b_1, b_2, \dots, b_n\}$ be a sequence of Null Basis pairs and $\{c_n, c_{n-1}, \dots, c_1\}$ be a sequence of null vectors both in Π_m . Then the Tangential Schur algorithm and the Null Basis recursion yields the same (A_n, B_n) pair, if the two sequences satisfy the following relationship:

$$u_j = \frac{\beta_{\omega_{j-1}, u_{j-1}}(\omega_j^*)^* \dots \beta_{\omega_1, u_1}(\omega_j^*)^* v_j}{\| \beta_{\omega_{j-1}, u_{j-1}}(\omega_j^*)^* \dots \beta_{\omega_1, u_1}(\omega_j^*)^* v_j \|_2} \quad (326)$$

Proof: See 6.6 Appendix

■

The following theorem states the relationship between Null basis and the linear space it spans.

Theorem 6.5.2 Let $\{b_1, \dots, b_n\}$ be a set of Null Basis pairs and $\{f_1, \dots, f_n\}$ be the corresponding vector valued functions, the recursive updating formulae in Theorem 6.5.1 yields (A_n, B_n) pair whose Krylov row space is spanned by $\{f_1, \dots, f_n\}$, where the n th row corresponding to orthonormal basis Φ_n .

Proof: See 6.6 Appendix

■

Corollary 6.5.1 Let $\{b_1, \dots, b_n\}$ be a set of Null Basis pairs, the recursive updating formulae in Theorem 6.5.1 maps any ordering of the set into pairs belonging to the same equivalent class in $E\Omega$. The linear Krylov row subspace of this equivalent class is given by

$$\text{span}(f_1, \dots, f_n) \quad (327)$$

Proof: See 6.6 Appendix

■

6.6 Appendix

Proof of Theorem 6.4.1:

1) For $k = 1$, let

$$\Phi_1(z) = \frac{\sqrt{1 - |\omega_1|^2}}{1 - \omega_1 z} v_1 = f_1 \quad (328)$$

$$x_1 = v_1 \quad (329)$$

2) For $k = 2$,

$$\Phi_2(z) = (\text{span}\{f_1, f_2\} \cap \text{span}\{f_1\}^\perp) \quad (330)$$

we then have:

$$\Phi_2(z) = K_2(f_2 - \langle f_2, f_1 \rangle f_1) \quad (331)$$

where $\|K_2\|$ is the normalizing factor, and we have

$$\frac{\Phi_2(z)}{K_2} = (f_2 - \langle f_2, f_1 \rangle f_1) \quad (332)$$

$$= \frac{\sqrt{1 - |\omega_2|^2}}{1 - \omega_2 z} v_2 - \frac{\sqrt{1 - |\omega_1|^2} \sqrt{1 - |\omega_2|^2} v_1^* v_2}{1 - \omega_1^* \omega_2} \frac{\sqrt{1 - |\omega_1|^2}}{1 - \omega_1 z} v_1 \quad (333)$$

$$= \frac{\sqrt{1 - |\omega_2|^2}}{1 - \omega_2 z} \left(v_2 - \frac{(1 - \omega_2 z)(1 - |\omega_1|^2) v_1 v_1^*}{(1 - \omega_1^* \omega_2)(1 - \omega_1 z)} v_2 \right) \quad (334)$$

$$= \frac{\sqrt{1 - |\omega_2|^2}}{1 - \omega_2 z} \left(I + \left(\frac{(\omega_1^* - z)(\omega_1 - \omega_2)}{(1 - \omega_2 \omega_1^*)(1 - \omega_1 z)} - 1 \right) v_1 v_1^* \right) v_2 \quad (335)$$

$$\begin{aligned} &= \frac{\sqrt{1 - |\omega_2|^2}}{1 - \omega_2 z} \left(I + \left(\frac{(\omega_1^* - z)}{(1 - \omega_1 z)} - 1 \right) v_1 v_1^* + \left(\frac{\omega_1 - \omega_2}{(1 - \omega_1^* \omega_2)} - 1 \right) v_1 v_1^* \right. \\ &\quad \left. + \left(\frac{(\omega_1^* - z)}{(1 - \omega_1 z)} - 1 \right) \left(\frac{\omega_1 - \omega_2}{(1 - \omega_1^* \omega_2)} - 1 \right) v_1 v_1^* v_1 v_1^* \right) v_2 \end{aligned} \quad (336)$$

$$= \frac{\sqrt{1-|\omega_2|^2}}{1-\omega_2 z} \left(I + \left(\frac{\omega_1^* - z}{1-\omega_1 z} - 1 \right) v_1 v_1^* \right) \left(I + \left(\frac{\omega_1 - \omega_2}{1-\omega_1^* \omega_2} - 1 \right) v_1 v_1^* \right) v_2 \quad (337)$$

$$= \frac{\sqrt{1-|\omega_2|^2}}{1-\omega_2 z} \beta_{\omega_1, x_1}(z) \beta_{\omega_1, x_1}(\omega_2^*)^* v_2 \quad (338)$$

straight forward computation yields that $K_2 = \| \beta_{\omega_1, x_1}(\omega_2^*)^* v_2 \|_2$ and thus

$$\Phi_2(z) = \frac{\sqrt{1-|\omega_2|^2}}{1-\omega_2 z} \beta_{\omega_1, x_1}(z) x_2 \quad (339)$$

$$x_2 = \frac{\beta_{\omega_1, x_1}(\omega_2^*)^* v_2}{K_2} \quad (340)$$

3) For $k = n$, assume the theorem is true for $k = n - 1$

$$\Phi_n(z) = \left(\text{span} \{f_1, f_2, \dots, f_n\} \perp \text{span} \{f_1, f_2, \dots, f_{n-1}\} \right) \quad (341)$$

$$= \left(\text{span} \{f_1, \dots, f_{n-2}, f_{n-1}, f_n\} \perp \text{span} \{f_1, \dots, f_{n-2}\} \right) \perp \left(\text{span} \{f_1, \dots, f_{n-2}, f_{n-1}\} \perp \text{span} \{f_1, \dots, f_{n-2}\} \right) \quad (342)$$

denote

$$\Psi_n(z) = \left(\text{span} \{f_1, f_2, \dots, f_{n-2}, f_n\} \perp \text{span} \{f_1, f_2, \dots, f_{n-2}\} \right) \quad (343)$$

$$= \frac{\sqrt{1-|\omega_n|^2}}{1-\omega_n z} \prod_{j=1}^{n-2} \beta_{\omega_j, x_j}(z) y_n \quad (344)$$

$$y_n = \frac{\beta_{\omega_{n-2}, x_{n-2}}(\omega_n^*)^* \dots \beta_{\omega_1, x_1}(\omega_n^*)^* v_n}{\| \beta_{\omega_{n-2}, x_{n-2}}(\omega_n^*)^* \dots \beta_{\omega_1, x_1}(\omega_n^*)^* v_n \|_2} \quad (345)$$

and we also have

$$\Phi_{n-1}(z) = \left(\text{span} \{f_1, f_2, \dots, f_{n-2}, f_{n-1}\} \perp \text{span} \{f_1, f_2, \dots, f_{n-2}\} \right) \quad (346)$$

$$= \frac{\sqrt{1 - |\omega_{n-1}|^2}}{1 - \omega_{n-1} z} \prod_{j=1}^{n-2} \beta_{\omega_j, x_j}(z) x_{n-1} \quad (347)$$

$$x_{n-1} = \frac{\beta_{\omega_{n-2}, x_{n-2}}(\omega_{n-1}^*)^* \dots \beta_{\omega_1, x_1}(\omega_{n-1}^*)^* v_{n-1}}{\| \beta_{\omega_{n-2}, x_{n-2}}(\omega_{n-1}^*)^* \dots \beta_{\omega_1, x_1}(\omega_{n-1}^*)^* v_{n-1} \|_2} \quad (348)$$

$$\equiv \beta_{\omega_{j-1}, x_{j-1}}(\omega_{n-1}^*)^* \dots \beta_{\omega_1, x_1}(\omega_{n-1}^*)^* v_{n-1} \quad (349)$$

And,

$$\langle \Psi_n(z), \Phi_{n-1}(z) \rangle \quad (350)$$

$$= \langle \frac{\sqrt{1 - |\omega_n|^2}}{1 - \omega_n z} \prod_{j=1}^{n-2} \beta_{\omega_j, x_j}(z) y_n, \frac{\sqrt{1 - |\omega_{n-1}|^2}}{1 - \omega_{n-1} z} \prod_{j=1}^{n-2} \beta_{\omega_j, x_j}(z) x_{n-1} \rangle \quad (351)$$

$$= \langle \frac{\sqrt{1 - |\omega_n|^2}}{1 - \omega_n z} y_n, \frac{\sqrt{1 - |\omega_{n-1}|^2}}{1 - \omega_{n-1} z} x_{n-1} \rangle \quad (352)$$

Thus

$$\Phi_n(z) = K_n(\Psi_n(z) - \langle \Psi_n(z), \Phi_{n-1}(z) \rangle \Phi_{n-1}(z)) \quad (353)$$

$$\begin{aligned} & \frac{\Phi_n(z)}{K_n} \\ &= \prod_{j=1}^{n-2} \beta_{\omega_j, x_j}(z) \left(\frac{\sqrt{1 - |\omega_n|^2}}{1 - \omega_n z} y_n - \langle \frac{\sqrt{1 - |\omega_n|^2}}{1 - \omega_n z} y_n, \frac{\sqrt{1 - |\omega_{n-1}|^2}}{1 - \omega_{n-1} z} x_{n-1} \rangle \frac{\sqrt{1 - |\omega_{n-1}|^2}}{1 - \omega_{n-1} z} x_{n-1} \right) \end{aligned} \quad (354)$$

This is in exactly the same structure as in the case for $k = 2$, and it readily follows

$$\frac{\Phi_n(z)}{K_n} \quad (355)$$

$$= \frac{\sqrt{1 - |\omega_n|^2}}{1 - \omega_n z} \prod_{j=1}^{n-2} \beta_{\omega_j, x_j}(z) \left(I + \left(\frac{\omega_{n-1}^* - z}{1 - \omega_{n-1} z} - 1 \right) x_{n-1} x_{n-1}^* \right) \left(I + \left(\frac{\omega_{n-1} - \omega_n}{1 - \omega_{n-1}^* \omega_n} - 1 \right) x_{n-1} x_{n-1}^* \right) y_n \quad (356)$$

$$\Phi_n(z) = \frac{1}{\| \beta_{\omega_{n-2}, x_{n-2}}(\omega_j^*)^* \dots \beta_{\omega_1, x_1}(\omega_j^*)^* v_n \|_2} \frac{\sqrt{1 - |\omega_n|^2}}{1 - \omega_n z} \prod_{j=1}^{n-1} \beta_{\omega_j, x_j}(z) \left(\prod_{j=1}^{n-1} \beta_{\omega_j, x_j}(\omega_n^*) \right)^* v_n \quad (357)$$

Normalization yields:

$$\Phi_n(z) = \frac{\sqrt{1 - |\omega_n|^2}}{1 - \omega_n z} \prod_{j=1}^{n-1} \beta_{\omega_j, x_j}(z) x_n \quad (358)$$

$$x_n = \frac{\beta_{\omega_{n-1}, x_{n-1}}(\omega_n^*)^* \dots \beta_{\omega_1, x_1}(\omega_n^*)^* v_n}{\| \beta_{\omega_{n-1}, x_{n-1}}(\omega_n^*)^* \dots \beta_{\omega_1, x_1}(\omega_n^*)^* v_n \|_2} \quad (359)$$

■

Proof of Lemma 6.5.1: Let (A_1, B_1) and (A_2, B_2) be any two pairs in the equivalent class, they satisfy for some full rank $n \times n$ matrix T that

$$A_1 = T A_2 T^{-1} \quad (360)$$

$$B_1 = T B_2 \quad (361)$$

Denote

$$K_1 = \text{rKrylov}(B_1, A_1 B_1, A_1^2 B_1, \dots) \quad (362)$$

$$= \text{rKrylov}(T B_2, T A_2 B_2, T A_2^2 B_2, \dots) \quad (363)$$

$$= \text{rKrylov}(T(B_2, A_2 B_2, A_2^2 B_2, \dots)) \quad (364)$$

$$K_2 = \text{rKrylov}(B_2, A_2 B_2, A_2^2 B_2, \dots) \quad (365)$$

It is then obvious that these two Krylov row subspaces are equivalent.

■

Proof of Theorem 6.5.1:

First we prove the triangular input balanced property of (A_k, B_k) by induction. We denote the recursive updating formulae as :

$$L_1 = [] , A_1 = \omega_1 , B_1 = y_1^* \quad (366)$$

$$L_k = [J_{k-1} L_{k-1} \quad t_{k-1} y_{k-1}] \quad (367)$$

$$t_k = \sqrt{1 - |\omega_k|^2} \quad (368)$$

$$y_k = \frac{\prod_{i=k-1}^1 \beta_{\omega_i, \frac{y_i}{\|y_i\|}} (\omega_k^*)^* v_k}{\| \prod_{i=k-1}^1 \beta_{\omega_i, \frac{y_i}{\|y_i\|}} (\omega_k^*)^* v_k \|_2} \quad (369)$$

$$J_k = I - (1 + \omega_k^*) y_k y_k^* \quad (370)$$

$$B_k = \begin{bmatrix} B_{k-1} \\ y_k^* \prod_{i=n-1}^1 J_i \end{bmatrix} \quad (371)$$

$$A_k = \begin{bmatrix} A_{k-1} & 0 \\ y_k^* L_k & \omega_k \end{bmatrix} , B_k = \begin{bmatrix} B_{k-1} \\ y_k^* \prod_{i=n-1}^1 J_i \end{bmatrix} \quad (372)$$

$$\Lambda_k = A_k A_k^* , \Delta_k = B_k B_k^* \quad (373)$$

For the case when $k = 2$

$$A_2 A_2^* + B_2 B_2^* \quad (374)$$

$$= \begin{pmatrix} \omega_1 & 0 \\ y_2^* y_1 & \omega_2 \end{pmatrix} \begin{pmatrix} \omega_1 & y_1^* y_2 \\ 0 & \omega_2 \end{pmatrix} + \begin{pmatrix} y_1^* \\ y_2^* - (y_2^* y_1) \frac{1+\omega_1^*}{1-|\omega_1|^2} y_1^* \end{pmatrix} \begin{pmatrix} y_1 & y_2 - (y_1^* y_2) \frac{1+\omega_1}{1-|\omega_1|^2} y_1 \end{pmatrix} \quad (375)$$

$$= \begin{pmatrix} |\omega_1|^2 & \omega_1 y_1^* y_2 \\ \omega_1^* y_2^* y_1 & |\omega_2|^2 + |y_1^* y_2|^2 \end{pmatrix} + \begin{pmatrix} 1 - |\omega_1|^2 & (y_1^* y_2) (1 - (1 + \omega_1)) \\ (y_2^* y_1) (1 - (1 + \omega_1^*)) & y_2^* y_2 - |y_1^* y_2|^2 \end{pmatrix} \quad (376)$$

$$= I_2 \quad (377)$$

Assume the equality holds for $k-1$, then we have

$$A_{k-1} A_{k-1}^* + B_{k-1} B_{k-1}^* = I_{k-1} \quad (378)$$

$$y_{k-1}^* L_{k-1} L_{k-1}^* y_{k-1} + y_{k-1}^* \left(\prod_{i=n-2}^1 J_i \right) \left(\prod_{i=n-2}^1 J_i \right)^* y_{k-1} = 1 - |\omega_{k-1}|^2 \quad (379)$$

$$(380)$$

$$y_{k-1}^* L_{k-1} A_{k-2}^* + y_{k-1}^* \left(\prod_{i=n-2}^1 J_i \right) B_{k-2}^* = 0 \quad (381)$$

For $n = k$

$$A_k A_k^* + B_k B_k^* \quad (382)$$

$$= \begin{bmatrix} A_{k-1} & 0 \\ y_k^* L_k & \omega_k \end{bmatrix} \begin{bmatrix} A_{k-1}^* & L_k^* y_k \\ 0 & \omega_k^* \end{bmatrix} + \begin{bmatrix} B_{k-1} \\ y_k^* \prod_{i=k-1}^1 J_i \end{bmatrix} \begin{bmatrix} B_{k-1}^* & (\prod_{i=k-1}^1 J_i)^* y_k \end{bmatrix} \quad (383)$$

$$= \begin{bmatrix} A_{k-1} A_{k-1}^* & A_{k-1} L_k^* y_k \\ y_k^* L_k A_{k-1}^* & |\omega_k|^2 + y_k^* L_k L_k^* y_k \end{bmatrix} + \begin{bmatrix} B_{k-1} B_{k-1}^* & B_{k-1} (\prod_{i=k-1}^1 J_i)^* y_k \\ y_k^* \prod_{i=k-1}^1 J_i B_{k-1}^* & y_k^* \prod_{i=k-1}^1 J_i (\prod_{i=k-1}^1 J_i)^* y_k \end{bmatrix} \quad (384)$$

$$(385)$$

Notice that

$$B_k(k, :) B_k(k, :)^* + A_k(k, 1) * A_k(k, 1)^* \quad (386)$$

$$= y_k^* J_{k-1} J_{k-2} \dots J_1 (y_k^* J_{k-1} J_{k-2} J_{n-2} \dots J_1)^* + y_k^* J_{k-1} J_{k-2} \dots J_2 y_1 (y_k^* J_{n-1} J_{n-2} \dots J_2 y_1)^* \quad (387)$$

$$= y_k^* J_{k-1} J_{k-2} \dots J_2 (J_1 J_1^* + y_1 y_1^*) (y_k^* J_{k-1} J_{k-2} \dots J_2)^* \quad (388)$$

$$= y_k^* J_{k-1} J_{k-2} \dots J_2 (y_k^* J_{k-1} J_{k-2} \dots J_2)^* \quad (389)$$

and

$$B_k(k, :) B_k(k, :)^* + A_k(k, 1) * A_k(k, 1)^* + A_k(k, 2) * A_k(k, 2)^* \quad (390)$$

$$= y_k^* J_{k-1} J_{k-2} \dots J_2 (y_k^* J_{k-1} J_{k-2} \dots J_2)^* + y_k^* J_{k-1} J_{k-2} \dots J_3 y_2 (y_k^* J_{k-1} J_{k-2} \dots J_2 y_2)^* \quad (391)$$

$$= y_k^* J_{k-1} J_{k-2} \dots J_3 (y_k^* J_{k-1} J_{n-2} \dots J_3)^* \quad (392)$$

and thus we have

$$\left| \omega_k \right|^2 + y_k^* L_k L_k^* y_k + y_k^* \left(\prod_{i=k-1}^1 J_i \right) \left(\prod_{i=k-1}^1 J_i \right)^* y_k \quad (393)$$

$$= \left| \omega_k \right|^2 + B_k(k, :) B_k(k, :)^* + \sum_{j=1}^{k-1} A_k(j, 1) * A_k(j, 1)^* \quad (394)$$

$$= \left| \omega_k \right|^2 + y_k y_k^* = 1 \quad (395)$$

and we have

$$\Lambda_k(k, j) + \Delta_k(k, j) \quad (396)$$

$$= A(k, :) * A(j, :)^* + B(k, :) * B(j, :)^* \quad (397)$$

Notice that

$$A(k, 1) * A(j, 1)^* + B(k, :) * B(j, :)^* \quad (398)$$

$$= y_k^* J_{k-1} \dots J_2 u_1 u_1^* J_2^* \dots J_j y_j + y_k^* J_{k-1} \dots J_1 J_1^* \dots J_j^* y_j \quad (399)$$

$$= y_k^* J_{k-1} \dots J_2 J_2^* \dots J_j y_j \quad (400)$$

and

$$A(k, 2) * A(j, 2)^* + A(k, 1) * A(j, 1)^* + B(k, :) * B(j, :)^* \quad (401)$$

$$= y_k^* J_{k-1} \dots J_3 u_2 u_2^* J_3^* \dots J_{j-1} y_j + y_k^* J_{k-1} \dots J_2 J_2^* \dots J_{j-1} y_j \quad (402)$$

$$= y_k^* J_{k-1} \dots J_3 J_3^* \dots J_{j-1} y_j \quad (403)$$

and thus we have

$$\Lambda_k(k, j) + \Delta_k(k, j) \quad (404)$$

$$= \sum_{l=1}^j A(k, l) * A(j, l)^* + (k, :) * B(j, :)^* \quad (405)$$

$$= y_k^* J_{k-1} \dots J_{j+1} y_j \omega_j^* + y_k^* J_{k-1} \dots J_{j+1} J_j y_j \quad (406)$$

$$= 0 \quad (407)$$

Now since

$$A_{k-1} A_{k-1}^* + B_{k-1} B_{k-1}^* = I_{k-1} \quad (408)$$

we have proved that

$$A_k A_k^* + B_k B_k^* = I_k \quad (409)$$

and

$$(I - z A_k)^{-1} B \left((I - z A_k)^{-1} B \right)^* \quad (410)$$

$$= \sum_{j=0}^{\infty} A^j B B^* A^{j*} = I_k \quad (411)$$

■

Proof of Lemma 6.5.2:

The proof is by direct substitution, which yields

$$A_k(j, k) = \begin{cases} \omega_j & j = k \\ t_1 t_2 u_j^* \prod_{i=j-1}^{k+1} J_i u_k & j > k \\ 0 & j < k \end{cases} \quad (412)$$

$$B_k(:, k) = t_k u_k^* \prod_{i=k-1}^1 J_i \quad (413)$$

■

Proof of Theorem 6.5.2:

We prove that $\{f_1, \dots, f_n\}$ is in the span of the Krylov space of (A_n, B_n) by induction, or equivalently, $\{f_1, \dots, f_n\}$ is in the space spanned by $(I - z A_k)^{-1} B_k$. First recall that in the TIB Tangential Schur Algorithm, the transfer function end up being a updating Blaschke Potapov Factor, whose step wise interpolation condition are determined by the null directions..

First in the Null Basis frame work, for $n = 1$, this is obviously true for

$$f_1 = \frac{v_1}{1 - \omega_1^* z} \quad (414)$$

$$A = \omega_l, \quad B = \sqrt{1 - |\omega_l|^2} v_l^* \quad (415)$$

that the basis for $(I - zA)^{-1}B$ is the interpolation basis for the degree $n = l$ Blaschke Potapov factor.

Suppose that the theorem holds for $k = n - 1$, then for $k = n$,

$$A_n = \begin{bmatrix} A_{n-1} & 0 \\ y_n^* L_n & \omega_n \end{bmatrix} \quad B_k = \begin{bmatrix} B_{n-1} \\ y_n^* \prod_{i=n-1}^l J_i \end{bmatrix} \quad (416)$$

Notice that the first $(n - 1)$ rows of the Krylov space remain unchanged from that spanned by (A_{n-1}, B_{n-1}) , so that $\{f_i\}_{i=1, \dots, n-1}$ is in the space spanned by (A_n, B_n) .

From lemma 6.5.2 we know that the equivalent null vector representation is satisfied when

$$u_j = \frac{\beta_{\omega_{j-1}, x_{j-1}}(\omega_j^*)^* \dots \beta_{\omega_1, x_1}(\omega_j^*)^* v_j}{\| \beta_{\omega_{j-1}, x_{j-1}}(\omega_j^*)^* \dots \beta_{\omega_1, x_1}(\omega_j^*)^* v_j \|_2} \quad (417)$$

and

$$G_{n-1}(z) = \beta_1(z) \beta_2(z) \dots \beta_{n-1}(z) \quad (418)$$

$$G_n(z) = \beta_1(z) \beta_2(z) \dots \beta_n(z) \quad (419)$$

where

$$\beta_j = \beta_{\omega_j, u_j}(z) \quad (420)$$

so that

$$\langle \beta_n(z), \frac{u_n}{1 - \omega_n^* z} \rangle = 0 \quad (421)$$

But we also have

$$\langle G_n(z), \frac{v_n}{1 - \omega_n^* z} \rangle \quad (422)$$

$$= v_n^* \beta_1(\omega_n^*) \beta_2(\omega_n^*) \dots \beta_{n-1}(\omega_n^*) \beta_n(\omega_n^*) \quad (423)$$

$$= \| \beta_{\omega_{j-1}, x_{j-1}}(\omega_j^*)^* \dots \beta_{\omega_1, x_1}(\omega_j^*)^* v_j \|_2 u_j^* \beta_n(\omega_n^*) \quad (424)$$

$$= K \langle \beta_n(z), \frac{u_n}{1 - \omega_n^* z} \rangle = 0 \quad (425)$$

where K is some constant. It is then obvious that

$$f_n \in \mathcal{H}(G_n) = \text{span}(A_n, B_n) \quad (426)$$

Since we already know that

$$\text{span}\{f_1, \dots, f_{n-1}\} = \mathcal{H}(G_{n-1}) = \text{span}(A_{n-1}, B_{n-1}) \quad (427)$$

and that the Krylov row space of (A_n, B_n) is orthonormal, we then have

$$K_n \equiv (B \ A \ B \ A^2 \ B \ \dots) \quad (428)$$

$$\{f_1, \dots, f_{n-1}, f_n\} \perp \{f_1, \dots, f_{n-1}, f\} \quad (429)$$

$$= K_n(n, \cdot) \quad (430)$$

So the theorem also holds for $k = n$.

■

Proof of Corollary 6.5.1: Since the equivalent class in $E\Omega$ is characterized by its own Krylov row space, this is direct from Theorem 6.5.2

■

7. Dimensionality Reduction on Frequency Domain

Model reduction has a long history in the systems and control literature. Denote the system of order n as G_n , the goal of model order reduction methods is to optimize

$$\min_{G_r} \|G_n - G_r\| \quad (431)$$

where $r < n$. For a more detailed discussion on the error measure of approximation, please see Chapter 3.

The motivation behind model reduction stems from a few issues on working with large scale systems. The most important reason here is: the state space model will be used as a parametric set up for system identification. The scale/order of the system will directly determine how many parameters are needed for estimation. A large dimension of the parameter space can introduce large variance and over fitting problems. Large scale systems also have defects in other considerations including computational speed, ill-conditioning.

In this Chapter, we will present some of our main results on model reduction. In 7.1 we briefly review some of the state-of-the art model reduction methods. In 7.2 we introduce the partial block Hankel SVD method that improves computation efficiency in reduction method. In 7.3 we state the main result, Null Basis reduction algorithm. In 6.7, Numerical examples compares the performance of balanced truncation, tangential Schur algorithm and Null Basis reduction.

7.1 Model Reduction Overview

Approximation methods can be categorized into three species:

- 1) Gramian methods
- 2) Krylov-based methods
- 3) Hankel Norm approximation

1. Gramian Methods[37].

For linear problems, as considered in systems theory and control of ordinary differential or partial differential equations, balanced truncation[38] and related methods are the methods of choice since they have some desirable properties: they preserve the stability of the system[37] and provide a global computable error bound between the transfer function of the original system and that of the reduced-order system[39].

The basic idea of BT relies on balancing the two system Gramians, the controllability Gramian P and the observability Gramian Q , defined as follows (see, e.g. [40]):

$$A P A^* + B B^* = P \quad (432)$$

$$A^* Q A + C^* C = Q \quad (433)$$

The goal of balanced truncation involves finding the invertible transformation $T \in \mathbb{C}^{n \times n}$, such that

$$\hat{P} = T P T^* = \hat{Q} = T^{-*} Q T^{-1} = \Sigma \quad (434)$$

where Σ is diagonal matrix of the Hankel singular values of the system. Then projecting the system onto the eigenspace that corresponds to the largest singular values of the Hankel matrix can keep the important dynamics of the system: the most reachable and observable states. Global error between G_n and G_r is then given by

$$\|G_n - G_r\|_{H_\infty} \leq 2(\sigma_{r+1} + \dots + \sigma_n) \quad (435)$$

where $\sigma_{r+1}, \dots, \sigma_n$ are the neglected Hankel singular values.

There exists a few variations of algorithms in applying Balanced Truncation. The original balanced truncation [49] computes a balancing transformation matrix T by spectral decompositions of P and of the projected Gramian. An efficient and numerically more robust implementation of BT, the square-root (SR) method[41], is based on Cholesky factorizations of the Gramians. and the SVD are computed by a singular value decomposition of the product of the Cholesky factors with

$$P = Z_P Z_P^* \quad (436)$$

$$Q = Z_Q Z_Q^* \quad (437)$$

and the transformation matrices V and W are computed by SVD method

$$Z_P^* Z_Q = \begin{pmatrix} U_1 & U_2 \end{pmatrix} \begin{pmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{pmatrix} \begin{pmatrix} V_1^* \\ V_2^* \end{pmatrix} \quad (438)$$

$$W^* = \Sigma_1^{-\frac{1}{2}} V_1^* Z_Q^T \quad (439)$$

$$V^* = \Sigma_1^{-\frac{1}{2}} U_1^* Z_P^T \quad (440)$$

For highly unbalanced systems, the transformation matrix T will be ill conditioned. A.P. Mullhaupt[42] showed that the similarity T that improves the sensitivity of a system by a factor ρ must have condition number greater than or equal to ρ . This motivates the development of Balancing Free Methods. The Schur method computes a reduced-order system by projecting onto the dominant left and right eigenspace of P, Q without computing balancing transformations. The reduced system is no longer balanced but still obeys the error bound of Balanced

Truncation.

A closely related class of methods which contains information about the controllability as well as the observability of the system is based on another system Gramian, the cross Gramian. The reduced order model has the same properties as in BT model reduction, i.e. the stability is preserved and a computable global error bound exists.[43]

The computation of the system Gramians (or of factors of them) as first computational step is required in all methods described above. Subsequently, the eigenvalues of the product of two positive definite matrices are required; these are the reachability and observability Gramians. The exact solution of Lyapunov Equations require dense computations. But beyond that, Mullhaupt, Andrew P., and Kurt S. Riedel. [44]proved that the condition number of the Equation increases proportionally to $\exp(n/d)$ where n is the order of the system and d is the size of the input vector. This property of discrete Lyapunov equation prevents the method from yielding consistent and stable results to high order systems.

2. Krylov Methods[45]

The next important class of MOR methods for linear systems is based on Krylov subspaces. They are often called moment-matching methods or Padé approximation methods. These methods are very efficient in many engineering applications, including circuit simulation and simulation of machine tools.

The goal in moment-matching model reduction is the construction of a reduced-order system where some moments r of the associated transfer function G_r match some moments of the original transfer function G_n . The matrices V and W for model order reduction can be computed from the vectors which are associated with the moments.

$$\text{range}(V) = \text{span}[AB, A^2B, \dots, AB] \quad (441)$$

$$\text{range}(W) = \text{span}[C^*, A^*C^*, A^{2*}C^*, A^{r*}C^*] \quad (442)$$

Using a set of k distinct expansion points, the reduced order system matches the first two moments at each expansion point[46]. For the case of one expansion point, it can be seen that the columns of V, W span Krylov subspaces which can easily be computed by Arnoldi or Lanczos methods[47].

In general, methods based on Krylov subspaces do not preserve important properties of the original system as stability and passivity. Besides, there is not a convincing error estimation for the reduced model. Other drawbacks includes that Lanczos algorithm breaks down when $\det(G_n) = 0$. And that the reduced system tends to approximate the high frequency poles of the original system, which introduces potential bias and undesired prior to the system.

3. Hankel Norm Approximation

Hankel norm approximation is a direct generalization of the Schmidt-Mirsky, Eckart Young result regarding integral operators resulting from LTI systems. See [48] for details of this method.

The procedures of Hankel norm approximation are as follows

1. Given original transfer function G_n , construct Γ_{G_n} whose symbol is given by G_n .
2. Find Γ_r such that $\|\Gamma_{G_n} - \hat{\Gamma}_r\|_2$ is controlled by σ_{r+1} .
3. Project $\hat{\Gamma}_r$ on to its stable part Γ_r .
4. The lower order approximant G_r we find is the symbol of Γ_r .

Now this solution preserves the stability of the original system. Global error bound also holds for this method.

Two major drawbacks of this method includes dense computation in solving for (A, B, C, D) , where factorization and inverse of ill-conditioned matrices are involved. Moreover, truncation of the system involves forming the Hankel/Block Hankel matrix in the first place and performing SVD to it, which leads to $O(n^3)$ computations and $O(n^2)$ storage.

7.2 Fast Partial Block Hankel SVD^[36]

An important part of our model reduction method involves solving the partial SVD of a block Hankel matrix multiplied by its transpose. In this case we can exploit matrix structure and use FFT reduce to compute.

Given a Hankel matrix

$$H = \begin{pmatrix} h_1 & h_2 & \dots & h_n \\ h_2 & & & h_{n+1} \\ \dots & & & \dots \\ h_n & h_{n+1} & \dots & h_{2n-1} \end{pmatrix} \quad (443)$$

The fast multiplication of H by a vector can be obtained via Fast Fourier transformation in $O(n \log n)$ operations. The same method can be extended to Block Hankel case, where matrix entries are replaced by $k \times l$ block submatrices. In this case, the Hankel Multiplication is given by:

$$H = \begin{pmatrix} H_1 & H_2 & \dots & H_n \\ H_2 & & & H_{n+1} \\ \dots & & & \dots \\ H_n & H_{n+1} & \dots & H_{2n-1} \end{pmatrix} \quad (444)$$

$$y(i : k : \text{end}) = \sum_{j=1}^n H(i : k : \text{end}, j : l : \text{end}) x(j : l : \text{end}) \quad (445)$$

We can utilize the fact that, despite H being unsymmetric, HH^* and H^*H are symmetric. We can exploit this fact and apply Lanczos algorithm together with Block Hankel FFT multiplication to compute the partial decomposition with n largest eigenvalues where

$$H \simeq \tilde{H} = \tilde{U} \tilde{\Sigma} \tilde{V} \quad (446)$$

$$HH^* = \tilde{U} \tilde{\Sigma}^2 \tilde{U}^* \quad (447)$$

$$H^*H = \tilde{V} \tilde{\Sigma}^2 \tilde{V}^* \quad (448)$$

7.3 Null Basis Reduction

The realization problem via decomposition of block Hankel matrices was first proposed by [Ho-Kalman 1966] [49]. Kung's SVD realization procedure was subsequently shown to be equivalent to doing balanced truncation on an exact state space realization of the finite impulse response. In [50], [Kung 1978] applied singular-value decomposition shown to be optimal in rank reducing case. Kung's SVD realization procedure was subsequently shown to be equivalent to doing balanced truncation on an exact state space realization of the finite impulse response[51]. See [52] for a more detailed review and discussion.

We propose a new system reduction method called Null Basis reduction. It is closely related to tangential Schur algorithm in [Xiao Yu 2015][36], in the framework of Triangular Input Balanced(TIB) system. The tangential Schur algorithm suffers from a few drawbacks. It is not guaranteed to converge, It's limited to work with lossless functions only. It is variant and very sensitive to permutation of basis. The Null Basis method employs a different set of parameterization system and construction method. It is numerically more stable and overcomes the aforementioned drawbacks. It is also more flexible to use in different contexts given that the parameters commute by design. We first introduce the definition of VMOA. The Ho-Kalman-Kung Realization Algorithm will be reviewed, and then we will introduce the Null Basis Reduction which is closely related. We will also discuss

some details on the computations and algorithms involved.

Let f be in L^1 and let $I(f)$ denote $|I|^{-1} \int_I f$, where I is a subarc of the unit circle T . We say f is of bounded mean oscillation and write $f \in \text{BMO}$ if

$$\|f\|_* = \sup_I (|f - I(f)|) < \infty \quad (449)$$

The space of function of vanishing mean oscillation (i.e. VMO) is the closure in BMO of the continuous functions on T . $\text{BMOA} = \text{BMO} \cap H^1$ and $\text{VMOA} = \text{VMO} \cap H^1$. In the univariate case, the Hankel Operator of a function is bounded if and only if the function is in BMOA and is compact if and only if the function is in VMOA. This is an essential requirement for the use of Hankel approximation

In the multivariate case, where the operator of the symbol is Block Hankel, compactness is implied by matrix $H_{m \times n}^\infty \cap C$ [53]. Such kind of functions and Hankel operators can be approximated in Hankel norm by the finite rank partial SVDs. Global error bound of H^∞ functions is guaranteed by AAK theorem.

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Now the functions in VMOA are approximated in Hankel norm by their truncations. Global error bound of H^∞ functions is guaranteed by AAK theorem. We provide a reduction algorithm calculates TIB pair and C which preserves H^∞ bound.

Given the transfer function of $m \times m$ Multiple-Input-Multiple-Output (MIMO) system,

$$T(z) = \sum_{k=0}^{\infty} T_k z^k \quad (451)$$

where $H(z) \in H_{m \times m}^\infty \cap C$, its associated Hankel operator is compact and given by

$$H_T = \begin{pmatrix} T_1 & T_2 & T_3 & \dots \\ T_2 & T_3 & & \dots \\ T_3 & & & \dots \\ \dots & \dots & \dots & \dots \end{pmatrix} \quad (452)$$

denote its $mn \times mn$ sized truncation

$$H_{T,n} = \begin{pmatrix} T_1 & T_2 & \dots & T_n \\ T_2 & & & T_{n+1} \\ \dots & & & \dots \\ T_n & T_{n+1} & \dots & T_{2n-1} \end{pmatrix} \quad (453)$$

Moreover, if M_{AB} and N_{CA} are the controllability and observability matrices, then

$$H_{T,mn} = N_{CA} M_{AB} \quad (454)$$

$$= \begin{pmatrix} C \\ CA \\ \dots \\ CA^{n-1} \end{pmatrix} (B \quad AB \quad \dots \quad A^{n-1}B) \quad (455)$$

The Ho-Kalman-Kung Realization Algorithm is the following:

1. Let $H_{T,n}$ have the singular value decomposition

$$H_{T,n} = [U_r \quad U_s] \begin{bmatrix} \Sigma_r & 0 \\ 0 & \Sigma_s \end{bmatrix} \begin{bmatrix} V_r^* \\ V_s^* \end{bmatrix} \quad (456)$$

2. The Rank- r estimate of $H_{T,n}$ is then $H_r = U_r \Sigma_r V_r^*$

3. The reduced state space description is estimated as

$$A_r = \Sigma_r^{-\frac{1}{2}} U_r^* H_{T,n} (m+1 : m(r+1), m+1 : m(r+1)) V_r \Sigma_r^{-\frac{1}{2}} \quad (457)$$

$$B_r = \left(\Sigma_r^{\frac{1}{2}} V_r^* \right) (:, 1 : m) \quad (458)$$

$$C_r = \left(U_r \Sigma_r^{\frac{1}{2}} \right) (1 : m, :) \quad (459)$$

$$D_r = T_0 \quad (460)$$

This algorithm achieves the rank-r estimate H_r where

$$H_r = \arg \min_{\text{rank}(H) = r} \|H_{T,n} - H\| \quad (461)$$

However it is worth noticing that the truncated matrix H_r is in general not Hankel. We proceed to introduce the Null Basis reduction algorithm.

1. Perform partial rank r SVD of the block Hankel matrix associated with the $k \times k$ MIMO system:

$$H_{T,n} = [U_r \ U_s] \begin{bmatrix} \Sigma_r & 0 \\ 0 & \Sigma_s \end{bmatrix} \begin{bmatrix} V_r^* \\ V_s^* \end{bmatrix} \quad (462)$$

2. The Rank-r estimate of $H_{T,n}$ is then $H_r = U_r \Sigma_r V_r^*$ where

$$\begin{pmatrix} \tilde{B} & \tilde{A} \tilde{B} & \tilde{A}^2 \tilde{B} & \dots \end{pmatrix} = V_r^* \quad (463)$$

$$\tilde{C} = U_r \Sigma_r T_r = (U_r \Sigma_r) (1 : k, :) \quad (464)$$

3. Recover A, B

$$\tilde{A} = \underline{V_r \Sigma_r V_r^*} = \underline{Q \Lambda Q^{-1}} \quad (465)$$

$$\tilde{B} = V_r T_r = V_r^* (:, 1 : k) \quad (466)$$

where S is a block shift matrix

$$S_r = \begin{pmatrix} 0_{k \times k} & 0_{k \times k} & 0_{k \times k} & \dots & 0_{k \times k} \\ I_{k \times k} & 0_{k \times k} & 0_{k \times k} & \dots & 0_{k \times k} \\ 0_{k \times k} & I_{k \times k} & 0_{k \times k} & \dots & 0_{k \times k} \\ \dots & \dots & \dots & \dots & 0_{k \times k} \\ 0_{k \times k} & 0_{k \times k} & \dots & I_{k \times k} & 0_{k \times k} \end{pmatrix} \quad (467)$$

$$= \begin{pmatrix} 0 & 0 & 0 & \dots & 0 \\ 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & \dots & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \otimes I_{k \times k} \quad (468)$$

$$T_r = \begin{pmatrix} 1 \\ 0 \\ 0 \\ \dots \\ 0 \end{pmatrix} \otimes I_{k \times k} \quad (469)$$

4. Let $\omega_i = \Lambda_{ii}$, $v_i = (T\tilde{B})(i, :)$, then for $k = 1, \dots, r$, recursively update the system until we get degree r system A_r, B_r

$$L_k = [J_{k-1} \ L_{k-1} \quad y_{k-1}] \quad (470)$$

$$y_k = \frac{\sqrt{1 - |\omega_k|^2} \prod_{i=k-1}^1 \beta_{\omega_i, \frac{y_i}{\|y_i\|}} (\omega_k^*)^* v_k}{\| \prod_{i=k-1}^1 \beta_{\omega_i, \frac{y_i}{\|y_i\|}} (\omega_k^*)^* v_k \|_2} F \quad (471)$$

$$J_k = I - \frac{(1 + \omega_k^*)}{1 - |\omega_k|^2} y_k y_k^* \quad (472)$$

$$A_k = \begin{bmatrix} A_{k-1} & 0 \\ y_k^* L_k & \omega_k \end{bmatrix}, \quad B_k = \begin{bmatrix} B_{k-1} \\ y_k^* \prod_{i=n-1}^1 J_i \end{bmatrix} \quad (473)$$

5. Denote the K_r as the Krylov matrix spanned by A_r, B_r , $\tilde{K}$ as the Krylov matrix spanned by $\tilde{A}, \tilde{B}$.

$$\tilde{K} = Q K_r \quad (474)$$

and we have

$$\tilde{C} \tilde{K} = \tilde{C} Q K_r \quad (475)$$

Let

$$C_r = \tilde{C} Q \quad (476)$$

and we have

$$\tilde{C} \tilde{K} = C_r K_r \quad (477)$$

This is equivalently

$$C_r = (T_1 \ T_2 \ \dots) (B_r \ A_r B_r \ A_r^2 B_r)^* \quad (478)$$

■

Remarks:

1) Partial Block Hankel SVD

The Partial Block Hankel SVD is calculated with Lanczos algorithm and FFT Hankel multiplication. Notice that

$$H = U \Sigma V^* \quad (479)$$

$$H H^* = U |\Sigma|^2 U^* \quad (480)$$

$$H^* H = V |\Sigma|^2 V^* \quad (481)$$

The multiplication in (14) and (15) can be calculated in $O(n \log n)$. The results are Hermitian, so the unitary matrices as well as the squared singular values of H can be calculated with Lanczos method.

$$H_{T,n} = N_{CA} M_{AB} \quad (482)$$

$$\begin{pmatrix} C \\ C A \\ \dots \\ C A^{n-1} \end{pmatrix} \begin{pmatrix} B & A B & \dots & A^{n-1} B \end{pmatrix} \quad (483)$$

2) Calculation of (A, B) for the state space system

Matrix A is calculated by an inner product of truncated of partial isometries and thus numerically stable. B is from a direct truncation of the Krylov matrix. So this step is numerically stable.

3) Parameterization with pole and Null Basis

The parameterization and reconstruction of system with pole and Null Basis is a recursive process. Unlike direct solve of Lyapunov Equations which is extremely ill conditioned, our process is based on matrix multiplications and matrix concatenations. This is also the major difference between the Null Basis method and the tangential Schur reduction algorithm in [Xiao Yu 2015] [54]. There are several differences in performance by design:

I. In (15)-(16), the way we obtain A and B does not guarantee that they are input balanced, i.e. $AA^* + BB^* = I$. This is partly due to the fact that we are working with truncated Krylov matrices, and the difference can be huge in some cases. Numerical perturbations can also cause the absence of input balance. This violates the tangential Schur algorithm assumption of input balance and the error is reflected in approximation result when the Controllability Gramian is not equal to identity.

II. This assumption of input balance has also limited tangential Schur algorithm to be applied to balancing free systems where the Null Basis parameterization suffice to work.

III. The Tangential Schur Algorithm is a greedy algorithm, attaining local optimal, not global ones. The space spanned by the subset of the parameter set is not a subspace of the whole set space. The (lack of) property

deprives the algorithm from having monotonicity in reduction order. But this is what one would expect from reduction: the more dimension we have, the less approximation error there should be from the original system

IV The parameter of an LTI system in terms of Tangential Schur Algorithm is a sequence, not a set, i.e. it is order sensitive. Permutations of the parameters yields a different system. The Null Basis method is invariant under permutations. The dimension of parameter space is largely reduced by a factor of $n!$ which brings more flexibility to incorporate priors into the parameter space.

4) Calculation of C in the state space system

In this part we show that the Null Basis reduction method achieves the same bound as does the Kung realization method. We will show, once truncated, the system is preserved in each step of the Null Basis reduction method

I. In Step 1 Null Basis reduction and Kung realization is exactly the same.

II. In Step 2

In the Kung realization the Controllability Gramian and C is given by

$$\begin{pmatrix} B & AB & A^2B & \dots \end{pmatrix} = \Sigma_r^{\frac{1}{2}} V_r^* \quad (484)$$

$$C = \left(U_r \Sigma_r^{\frac{1}{2}} \right) (1 : m, :) \quad (485)$$

and the Null Basis reduction method yields

$$\begin{pmatrix} \tilde{B} & \tilde{A} \tilde{B} & \tilde{A}^2 \tilde{B} & \dots \end{pmatrix} = V_r^* \quad (486)$$

$$\tilde{C} = (U_r \Sigma_r) (1 : m, :) \quad (487)$$

It is obvious these two realizations are equivalent.

III. In Step 3 and Step 4

In these two steps, A and B are decomposed and reconstructed into TIB pairs. In both steps, the Krylov space that A and B span is preserved, and that $\tilde{K}$ and K_r span the same row space. Also since they are both partial isometries, we have

$$\tilde{K} K_r = Q \quad (488)$$

where Q is unitary matrix. And thus we have the transformation

$$\begin{cases} \tilde{C} \rightarrow \tilde{C} Q = C_r \\ \tilde{K} \rightarrow Q^{-1} \tilde{K} = K_r \end{cases} \quad (489)$$

which is essentially

$$\begin{cases} \tilde{A} \rightarrow Q^{-1} \tilde{A} Q = A_r \\ \tilde{B} \rightarrow Q^{-1} \tilde{B} = B_r \\ \tilde{C} \rightarrow \tilde{C} Q = C_r \end{cases} \quad (490)$$

again this preserves the system so the Triangular Input Balanced system from Null Basis reduction is equivalent to the Kalman-Ho-Kung result, the H^∞ bound is preserved.

5) As a result the algorithm works differently than the tangential Schur algorithm. The algorithm works with systems that are not required to be input balanced, so that when we work with a truncated system, the algorithm is guaranteed to converge. The space spanned by subset is a subspace of the whole set space. The H_2 norm of approximation is guaranteed to be monotonic. The recursive updating formulae is a mapping from an equivalent set to a null basis set, invariant to permutations and scaling to null basis.

7.4 Numerical Examples

We present three numerical examples which compares the approximation error of a larger system with a lower degree system. A comparison is made between 1) Tangential Schur Algorithm, which uses null vectors and poles to parameterize rational linear systems, 2) Balanced Truncation Method (Matlab 2017a), which is a state of the art reduction method, and our algorithm, i.e. Null Basis Reduction. The code of our is implemented in Matlab which takes impulse response as input and computes Null Basis parameterization of the TIB system.

Example 7.4.1 ($1/f^\alpha$ noise) with impulse response

$$F(z) = \sum_{j=1}^{\infty} \begin{pmatrix} j^{-1.1} & j^{-1.2} & j^{-1.3} \\ j^{-1.4} & j^{-1.5} & j^{-1.6} \\ j^{-1.7} & j^{-1.8} & j^{-1.9} \end{pmatrix} \quad (491)$$

We pick 3000 truncation of the infinitely length impulse response and apply three different reduction methods, i.e. Tangential Schur, Null Basis Reduction and Balanced Truncation (Matlab 2017a Version, Square Root Method). We reduce the system to various degrees and compare the relative H_2 difference of the first 3000 coefficients. The Tangential Schur loses monotonicity in approximation error when the number of poles increases. Null basis reduction achieves extremely close accuracy to Balanced Truncation with a Triangular Input Balanced realization.

Result:

# of poles	Tangential Schur	Null Basis	Balanced Truncation
5	$1.825 e - 02$	$1.784 e - 02$	$1.844 e - 02$
15	$2.698 e - 03$	$2.420 e - 03$	$2.399 e - 03$
25	$1.817 e - 03$	$1.168 e - 03$	$1.147 e - 03$
35	$2.772 e - 03$	$8.207 e - 04$	$8.210 e - 04$
45	$5.018 e - 03$	$6.277 e - 04$	$6.294 e - 04$
55	$6.386 e - 03$	$5.200 e - 04$	$5.063 e - 04$
65	$1.010 e - 03$	$4.374 e - 04$	$4.408 e - 04$

(492)

Table. Relative H_2 Norm of the first 3000 Coefficients

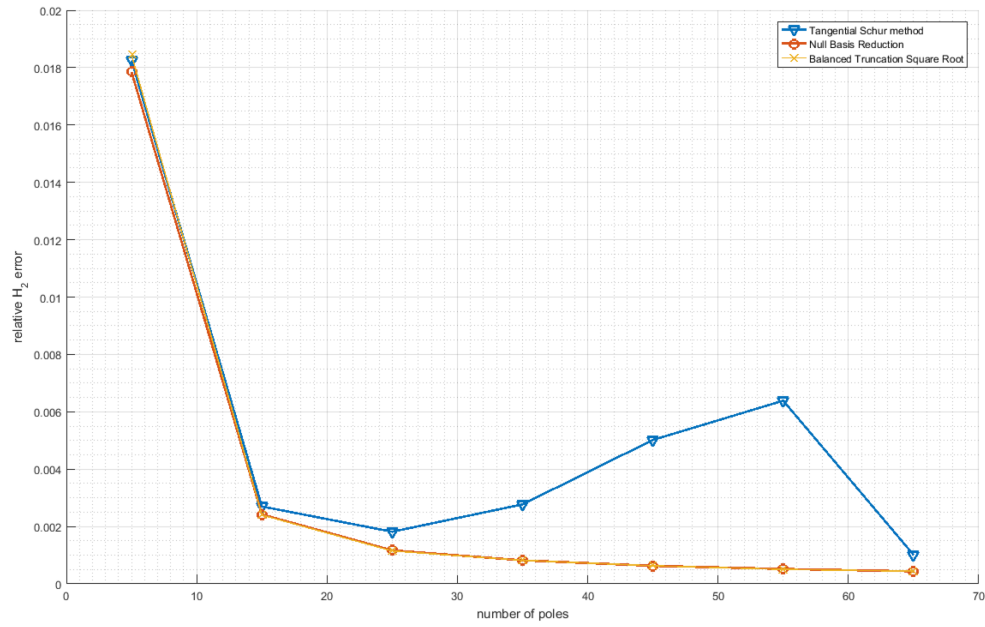


Figure. Relative H_2 Error

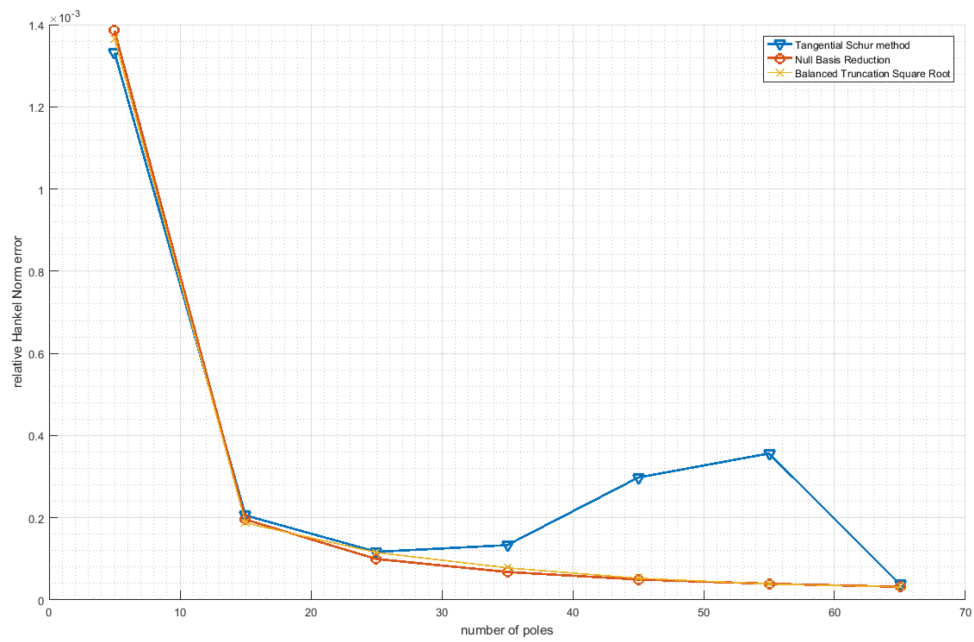


Figure. Relative Hankel Norm Error

Example 7.4.2 (Rational Finite Dimension System) with a 10×10 impulse response $F(z)$, generated with 65 random rational basis and 3000 coefficients truncation.

$$F(z) = \sum_{j=1}^{65} \frac{A_j}{1 - \omega z} \quad (493)$$

where A_j for each j is a random 10×10 matrix, whose entries follows uniform $[0,1]$ distribution.

We pick 1000 truncation of the infinitely long impulse response and apply three different reduction methods, i.e. Tangential Schur, Null Basis Reduction and Balanced Truncation (Matlab 2017a Version, Square Root Method). We reduce the system to various degrees and compare the relative H_2 difference of the first 3000 coefficients. The Balanced Truncation Method under performs due to truncation error as well as condition number of matrix A , when it attempts to numerically balance the system. The Tangential Schur algorithm under performs Null Basis method. It captures the original system when the pole number equals the degree.

# of poles	Tangential Schur	Null Basis	Balanced Truncation
5	$9.143 e - 01$	$9.143 e - 01$	$9.233 e - 01$
15	$7.295 e - 01$	$7.288 e - 01$	$8.015 e - 01$
25	$5.275 e - 01$	$5.081 e - 01$	$5.801 e - 01$
35	$3.935 e - 01$	$3.064 e - 01$	$3.589 e - 01$
45	$3.911 e - 01$	$1.487 e - 01$	$1.737 e - 01$
55	$2.158 e - 01$	$4.525 e - 02$	$5.591 e - 02$
65	$2.062 e - 08$	$2.099 e - 09$	$2.829 e - 04$

(494)

Table. Relative H_2 Norm of the first 1000 Coefficients

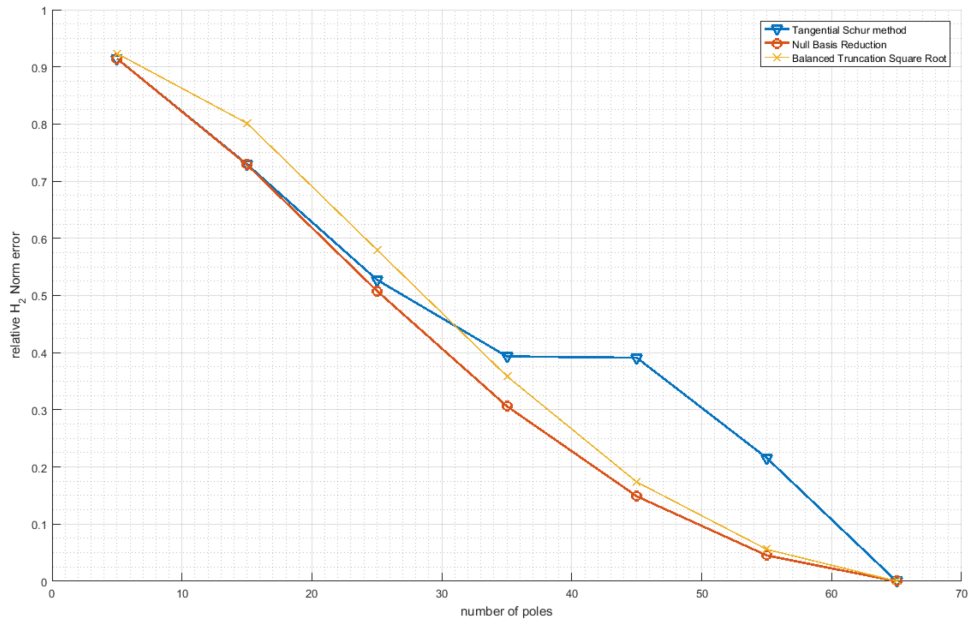


Figure. Relative H_2 Error

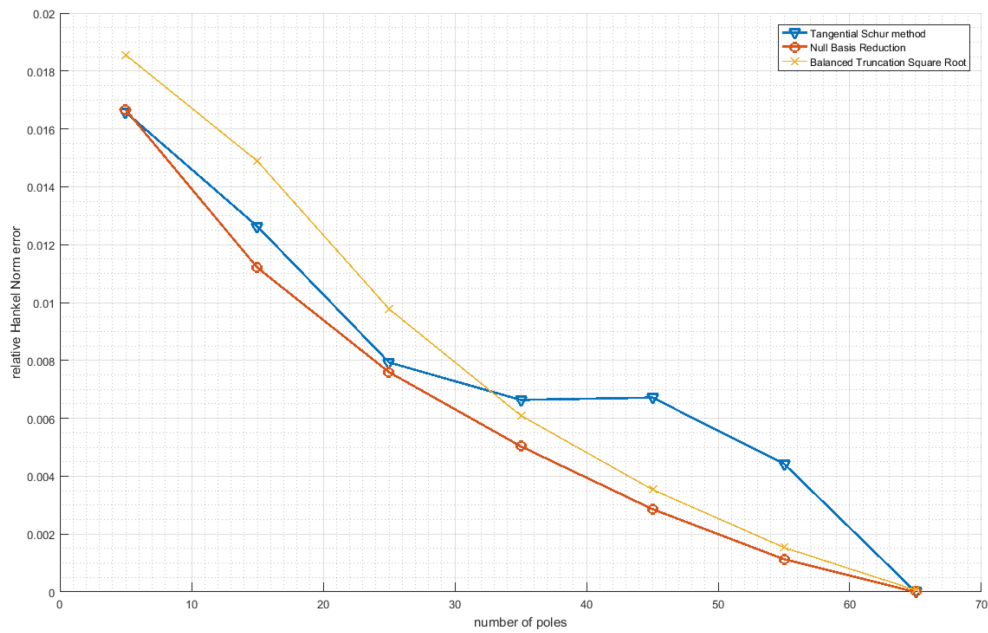


Figure. Relative Hankel Norm Error

Example 7.4.3 (Rational Finite Dimension System) with a 2×2 dimension system with 500 degrees, where Balanced Truncation fails to converge due to large condition number of A .

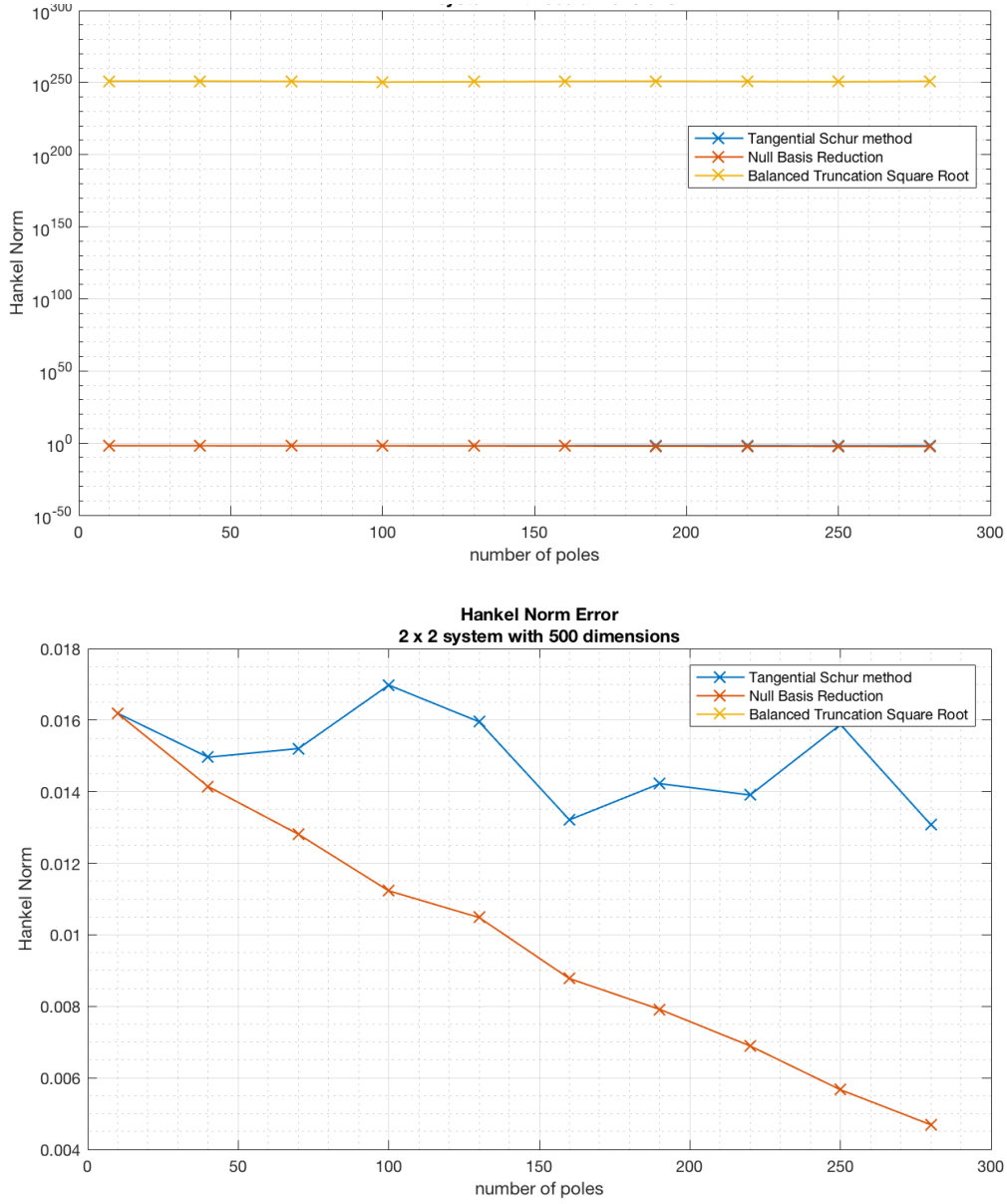


Figure. Relative Hankel Norm Error

This is an example of where the Balanced Truncation method fails due to large condition number of A . The tangential Schur method is also not converging. In the tangential Schur reduction method, the decomposed unitary matrix corresponding to (A, B) is truncated to get the reduced system. However when the reduction order is not close to the real order of the system, the truncated part will not be unitary nor approximately unitary. This would cause a problem for tangential Schur method since their parameterization would require (A, B) to be input

balanced already. And whenever it is not, the parameterization would be inaccurate and not reflecting the real system. We can try to transform A and B to preserve unitarity, but that would again suffer from the same problem as does the balanced truncation method, i.e. A being ill conditioned. This is one of the major reasons why null basis reduction performs better in such scenario: it does not require unitarity and the approximation error is monotonic and inversely proportional to the reduction degree.

8. Dimensionality Reduction on Statistical Manifold

In Chapter 3 we have derived Information Distance as a geodesic between two models in the asymptotic case . However, in the scenario where there are large but finitely many samples, The approach to making statistical inference is fundamentally different. For any deterministic parameter in a statistical model, Cramer-Rao lower bound points out that there will always be a variance in estimation, given finite sample size. The condition can deteriorate in the presence of insufficient amount of data or low signal to noise ratio with respect to the model, which are common scenarios in the applications of statistical models. This is another way of saying ‘all models are wrong’. As a matter of fact, in many occasions we can neither prove or falsify the correctness of a model, both of which would require far more evidence. What we can do is to have a model which is ‘close’ to the real one, given available samples. The ‘closeness’ is defined by hypothesis testing, i.e. their distinguishability. If we cannot tell between two models from the samples they generate, then we say they are close.

In hypothesis testing, the distinguishability is measured by making two types of mistakes. Incorrect rejection of a true hypothesis, i.e. type I error, and incorrect retaining a false hypothesis, i.e. type II error. Let’s suppose that based on the two different models in the hypothesis, the probability measures are P and Q . We reject P when event A occurs. The probability of making two types of errors is given by

$$P(A) + Q(A^c) \tag{495}$$

$$= P(A) + 1 - Q(A) \tag{496}$$

$$\geq 1 - \sup_A (Q(A) - P(A)) \tag{497}$$

$$\geq 1 - \sup_A |Q(A) - P(A)| \tag{498}$$

where last term is the total variation distance between P and Q . The probability of making mistakes is bounded from below by $1 - \text{TVD}$. This means the more we control the total variation distance, the larger probability of making mistakes is in distinguishing the real and the estimation model, and the closer these two models are. This can be achieved without even knowing what the true model is. In this Chapter, we will derive finite sample realizations of asymptotic theories. The main object of research here is information distance. The intention of making this choice stems from some of the following advantages.

- 1) $\text{TVD}(P, Q) < I(P, Q)$

In Chapter 3 we have shown that Information Distance bounds Total Variation Distance from above. So the choice of information distance is consistent with our approach to estimate models that are hard to distinguish by hypothesis testing.

- 2) Invariant to change of parameterization

A very important property of information distance is: it is invariant to re-parameterization. This is immediate from the its definition, definite integrals are invariant with respect to changes of variables.

3) Computability

Under the parameterization of cepstrum, the information distance between a rational function and white noise is defined by the sum of squares of the cepstrum coefficients. This is great simplification than direct computation of total variation. In this chapter, different methods of cepstrum computation will be illustrated for both scalar and matrix valued transfer function.

4) $\log f(z) \in H^2(z)$

Compared to some classical distance measures, information distance expanded the space where we can apply approximation theory. A classical example would be a random walk with transfer function

$$f(z) = \frac{1}{1-z} \quad (499)$$

Apparently, the H^2 norm will not be convergent as we have more and more samples. However this process do have finite information length:

$$\| \log \frac{1}{1-z} \|_{H^2}^2 \quad (500)$$

$$= \lim_{m \rightarrow \infty} \sum_{k=1}^m \frac{1}{k^2} = \frac{\pi^2}{6} \quad (501)$$

which means we can find a model, which is close to random walk with respect to hypothesis testing, while the real model and the approximant can be infinitely far away in the sense of H^2 norm. In fact, the real model itself is not convergent

5) Pole and Zeros of a Rational Function

In conventional rational approximation, the basis of the system is characterized by the poles of the transfer function. In the context of approximating cepstrum, poles of both the system and its inverse will be captured.

In order to learn the behavior of logarithm and other functions of the transfer function, the best approach under some scenarios, is to work with the truncated Toeplitz operator. Structured linear algebra enables us to achieve evaluation of the Laurent coefficients with speed and efficiency.

8.1 Functions of (Block) Toeplitz Matrices

Given a function $f(z)$ and

$$f(z) = \sum_{k=-\infty}^{\infty} a_k z^k \quad (502)$$

be the Laurent expansion of the arbitrary $m \times m$ function, then the (block) Toeplitz matrix $T(f)$ where

$$T(f) = (a_{i-j}) \quad (503)$$

are called Laurent matrices. The Laurent matrix generates a bounded operator on l^2 if and only if function $f \in L^\infty$. Specifically in our case, we are interested in causal transfer function, i.e. all the fourier coefficients with negative indices vanishes. Then for $f \in H^\infty$, the multiplication operator

$$T(f) : H^2 \rightarrow H^2, \quad g \rightarrow fg \quad (504)$$

is obviously bounded. In order to solve equations and calculate coefficients numerically, we work with finite truncation of the multiplication operator,

$$T_n(f) = \begin{cases} (a_{i-j}) & n \geq i \geq j \\ 0 & i < j \leq n \end{cases} \quad (505)$$

In order to combine computational application with theories, we now rephrase some of the properties of functions in terms of matrices. The proceeding work in next few sections will focus on truncated part of infinite dimension matrices. Given a stable and minimum phase $m \times m$ Matrix Valued transfer function (SISO will be a 1×1 special case of this generalization)

$$f(z) = \sum_{i=0}^{\infty} f_i z^i \quad (506)$$

We denote its corresponding block Toeplitz matrix as

$$T_f = \begin{pmatrix} f_0 & 0 & 0 & 0 & 0 \\ f_1 & f_0 & 0 & 0 & 0 \\ f_2 & f_1 & f_0 & 0 & 0 \\ f_3 & f_2 & f_1 & f_0 & 0 \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix} \quad (507)$$

and its $mn \times mn$ truncation as

$$T_{f,n} = \begin{pmatrix} f_0 & 0 & 0 & 0 & 0 \\ f_1 & f_0 & 0 & 0 & 0 \\ f_2 & f_1 & f_0 & 0 & 0 \\ \dots & \dots & \dots & \dots & 0 \\ f_{n-1} & f_{n-2} & \dots & \dots & f_0 \end{pmatrix} \quad (508)$$

The input-output mapping relationship is given by

$$Y(t) = T_f X(t) \quad (509)$$

$$\begin{pmatrix} y_t \\ y_{t-1} \\ y_{t-2} \\ y_{t-3} \\ \dots \end{pmatrix} = \begin{pmatrix} f_0 & 0 & 0 & 0 & 0 \\ f_1 & f_0 & 0 & 0 & 0 \\ f_2 & f_1 & f_0 & 0 & 0 \\ f_3 & f_2 & f_1 & f_0 & 0 \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix} \begin{pmatrix} x_t \\ x_{t-1} \\ x_{t-2} \\ x_{t-3} \\ \dots \end{pmatrix} \quad (510)$$

where $X(t) \sim N(0, I_{mn})$. Now let's assume f_0 is not positive definite, and admits singular value decomposition

$$f_0 = U \Sigma V \quad (511)$$

Since $f(z)$ is minimum phase, diagonal elements of S are all positive. Let

$$f_0' = U \Sigma U^* = f_0 V^* U^* \quad (512)$$

$$f_k' = f_k V^* U^* \quad (513)$$

$$x_t' = U V x_t \quad (514)$$

and we can rewrite the linear system as

$$\begin{pmatrix} y_t \\ y_{t-1} \\ y_{t-2} \\ \dots \\ \dots \end{pmatrix} = \begin{pmatrix} f_0' & 0 & 0 & 0 & 0 \\ f_1' & f_0' & 0 & 0 & 0 \\ f_2' & f_1' & f_0' & 0 & 0 \\ f_3' & f_2' & f_1' & f_0' & 0 \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix} (V^* U^* \otimes I_n) (U V \otimes I_n) \begin{pmatrix} x_t \\ x_{t-1} \\ x_{t-2} \\ \dots \\ \dots \end{pmatrix} \quad (515)$$

$$\begin{pmatrix} y_t \\ y_{t-1} \\ y_{t-2} \\ y_{t-3} \\ \dots \end{pmatrix} = \begin{pmatrix} f_0' & 0 & 0 & 0 & 0 \\ f_1' & f_0' & 0 & 0 & 0 \\ f_2' & f_1' & f_0' & 0 & 0 \\ f_3' & f_2' & f_1' & f_0' & 0 \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix} \begin{pmatrix} x_t' \\ x_{t-1}' \\ x_{t-2}' \\ \dots \\ \dots \end{pmatrix} \quad (516)$$

Since $(U V \otimes I_n)$ is a unitary transformation, we have $X(t)' \sim N(0, I_{mn})$. In the innovation representation, without loss of generality, we can always assume f_0 is positive definite. Then the principal logarithm of T_f is given by

$$T_a = \begin{pmatrix} a_0 & 0 & 0 & 0 & 0 \\ a_1 & a_0 & 0 & 0 & 0 \\ a_2 & a_1 & a_0 & 0 & 0 \\ a_3 & a_2 & a_1 & a_0 & 0 \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix} \quad (517)$$

The eigenvalues of block triangular block Toeplitz matrix are given by its diagonal blocks. Since the diagonal blocks f_0 are all positive definite, with positive eigenvalues, matrix T_a is a real matrix. Then it's straight forward that

$$a(z) = \log f(z) = \sum_{i=0}^{\infty} a_i z^i \quad (518)$$

$$T_f = e^{T_a} = \sum_{n=0}^{\infty} \frac{T_a^n}{n!} \quad (519)$$

Now that both T_f and T_a are (block) Triangular, a truncation of upper left corner of the matrices on both sides of the equation won't change the equality. And we have

$$T_{f,n} = e^{T_{a,n}} = \sum_{n=0}^{\infty} \frac{T_{a,n}^n}{n!} \quad (520)$$

Having finite information length means that

$$\infty > \lim_{m \rightarrow \infty} \frac{1}{m} I^2 = \sum_{i=1}^{\infty} a_n^2 \geq \lim_{m \rightarrow \infty} \frac{1}{m} \|T_a\|_F \quad (521)$$

We will introduce the functions, operations and algorithms with which we can actually perform numerical computation of several truncated matrix functions in the section.

8.1.1 Multiplication

The multiplication of two functions can be equivalently expressed as the multiplication of a Toeplitz operator and a vector of the corresponding Fourier coefficients. Numerically for finite triangular (block) Toeplitz matrices, the multiplication can be efficiently done by embedding the Toeplitz matrix into a circulant one. This technique is well developed and applies to regular and block Toeplitz matrices alike.

Now given $n \times n$ triangular (block) Toeplitz matrix T_n and $n \times 1$ vector x , let

$$C_{2n} = \begin{pmatrix} T_n & S_n \\ S_n & T_n \end{pmatrix} \quad (522)$$

such that C_{2n} is circulant, then we have

$$C_{2n} = F_{2n}^* \text{diag}(F_{2n} C(:, 1)) F_{2n} \quad (523)$$

where F_{2n} is the Fourier matrix. The product then becomes the lower half of

$$C_{2n} \begin{pmatrix} 0 \\ x \end{pmatrix} = \begin{pmatrix} S_n x \\ T_n x \end{pmatrix} \quad (524)$$

which can be formed in $O(n \log n)$ time. For more details and other classes of structured matrices, we refer to [55].

8.1.2 Inverse

The exact inverse methods for triangular Toeplitz matrices with forward substitution requires about $n(n+1)$ arithmetic operations. A faster divide and conquer method which requires $O(n \log n)$ is proposed by [56]. Without loss of generality assume the size of the matrix is an integer power of 2, (we can always append zero if not). The idea is to recursively partition the larger matrix into 2×2 equally sized pieces until the size of the matrix is appropriate for direct solve. Now given triangular Toeplitz system

$$T_{2n} = \begin{pmatrix} t_n & 0 \\ s_n & t_n \end{pmatrix} \quad (525)$$

$$T_{2n} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \quad (526)$$

$$\Rightarrow \begin{cases} t_n x_1 = y_1 \\ t_n x_2 = y_2 - s_n x_1 \end{cases} \quad (527)$$

where the product $s_n x_1$ can be evaluated by FFT. The method can be generalized to the block Toeplitz case by substituting the scalars with blocks.

8.1.3 Square Root

Matrix square root is one of the most commonly occurring matrix functions. It's also one of the key steps in the computation of matrix logarithms in some methods. We will exclusively be concerned with the principal square root, $A^{\frac{1}{2}}$. For $A \in \mathbb{C}^{n \times n}$ with no eigenvalues on $\mathbb{R}^{-}$, $A^{\frac{1}{2}}$ is the unique square root X of A whose spectrum lies in the open right half-plane. One of the state of the art method to compute matrix square root is proposed by Cheng, Higham, Kenney, and Laub [57].

Algorithm 8.1.1:

0. Given Triangular Toeplitz A , we compute $A^{\frac{1}{2}}$
 1. $\mu_k = |\det(M_k)|^{-1/(2n)}$
 2. $M_{k+1} = \frac{1}{2} \left(I + \frac{1}{2} (\mu_k^2 M_{k+1} + \mu_k^{-2} M_k^{-1}) \right)$, $M_0 = A$
 3. $X_{k+1} = \frac{1}{2} \mu_k X_k (I + \mu_k^{-2} M_k^{-1})$, $X_0 = A$
 4. $Y_{k+1} = \frac{1}{2} \mu_k Y_k (I + \mu_k^{-2} M_k^{-1})$, $Y_0 = I$
- $A^{\frac{1}{2}} = \lim_{k \rightarrow \infty} X_k$ or Y_k

■

The iteration is applied to M_k and either X_k or Y_k . To remedy in case of ill conditioned A , preconditioning are some times necessary, such as using appropriate fractional differencing matrix as preconditioner. Notice that all the Toeplitz inverse and multiplication can exploit structure, and this can be easily generalized to the block case. A different approach is to use divide and conquer

Algorithm: 8.1.2

0. Given Triangular Toeplitz A , we compute $A^{\frac{1}{2}}$
1. $A_1 = \left(A \left(1 : \frac{n}{2}, 1 : \frac{n}{2} \right) \right)^{\frac{1}{2}}$
2. $A_2 = \frac{1}{2} A_1^{-1} A \left(\frac{n}{2} + 1 : n, 1 : \frac{n}{2} \right)$

$$A^{\frac{1}{2}} = \begin{pmatrix} A_1 & 0 \\ A_2 & A_1 \end{pmatrix}$$

■

Notice that in this algorithm, Step 1 is again a square root problem and can be divide further until the size of the matrix is small enough. Step 2 is exploiting the property that lower(upper) triangular Toeplitz matrices multiplication commute. This is only true for the Toeplitz case, not the block Toeplitz case. In the block case we would need to solve a Sylvester equation, which is far more complicated. Both the inverse and multiplication can exploit structure.

Let us now suppose we have a fast way to compute the matrix logarithm and exponential, the computation of which also don't involve matrix square root, then we can evaluate the matrix pth root as fast. The specific algorithm will be introduced in the next section, and the following square root algorithm would be based on that.

Algorithm: 8.1.3

0. Given Triangular Toeplitz A , we compute $A^{\frac{1}{2}}$

1. $B = \frac{1}{2} \log m(A)$

2. $A^{\frac{1}{2}} = e^B$

■

This method is generally true for all matrices as long as its logarithm exists. In terms of its performance, it completely depends on the algorithm of logarithm and exponential of structured matrix.

8.1.4 Logarithm

The matrix exponential is defined for $A \in C^{n \times n}$ by

$$e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots \quad (528)$$

A logarithm of $A \in C^{n \times n}$ is any matrix X such that $e^X = A$. For A with no eigenvalues on $\mathbb{R}^-$ and we exclusively work with the principal logarithm, i.e. whose spectrum lies in the strip $\{z : -\pi < \text{Im}(z) < \pi\}$. We next introduce some algorithms to compute the logarithm of triangular (block) Toeplitz matrices.

The first method is inverse scaling and squaring. This is a most widely used method proposed by Kenney and Laub[58]. The method computes $A^{\frac{1}{2^s}}$ for an integer s large enough so that $A^{\frac{1}{2^s}}$ is close enough to the identity, and

$\log(A^{\frac{1}{2^s}})$ can then be well approximated. Then we scale the logarithm back to remove the 2^s th root. The speed and accuracy of inverse scaling and squaring algorithm was greatly improved in [Mohy, Higham 2012][52], by the use of backward error instead of forward error, the steps taken to avoid cancellation in Padé approximant as well as the triangularization by Schur decomposition. For $n \times n$ matrices, the improved algorithm takes kn^3 flops where $k \geq 30$.

Algorithm:8.1.4[59]

0. Given Triangular Toeplitz A , we compute $\log(A)$

1. $A \leftarrow A^{\frac{1}{2}}$

2. $Z_0 = A - I$

3. for $i = 1 : s - 2$

$A \leftarrow A^{\frac{1}{2}}$

$P = P(I + A)$

4. Solve $XP = Z_0$ for $X = A^{\frac{1}{2^s}} - I$

5. Evaluate $U = r_m(X)$, where r_m is a expansion approximation method

6. $\log(A) = 2^s U$

■

The computation of square root and multiplication can exploit Toeplitz structure. This method can be easily generalized to the block Toeplitz case. Another method is approximating logarithm with arithmetic-geometric mean. Given two complex matrices with positive real parts X and Y ,

$$A_k = \frac{1}{2}(A_k + G_k), \quad A_0 = X \quad (529)$$

$$G_k = \sqrt{A_k G_k}, \quad G_0 = Y \quad (530)$$

It was shown in [60] that A_k and G_k have a common limit, we denote it by $AGM(X, Y)$ and the matrix $C_k = A_k - G_k$ converges quadratically to the zero matrix. In [61], stability as well as convergence behavior are discussed. The algorithm is as the follow

Algorithm:8.1.5

0. Given Triangular Toeplitz A , we compute $\log(A)$

1. Compute $AGM(A, I)$ with by iteratively evaluation of

$A_k = \frac{1}{2}(A_k + G_k), \quad A_0 = X$

$G_k = \sqrt{A_k G_k}, \quad G_0 = Y$

2. $\log(A) = \pi(2 AGM(I, 2^{2-m} A^{-1}))^{-1} - m \ln(2)$

m is chosen s.t. $x2^m > 2^{\frac{p}{2}}$, and approximation precision is 2^{-p}

■

The computation of square root and multiplication can exploit Toeplitz structure. This method can be easily generalized to the block Toeplitz case. Specifically for the scalar case, there is a different thread of logarithm calculation proposed by A.P.Mullhaupt[62]. The function logarithm has the following equality

$$\frac{\partial \log f(z)}{\partial z} = \frac{1}{f(z)} \frac{\partial f(z)}{\partial z} \quad (531)$$

and equivalently

$$\begin{pmatrix} f_0 & 0 & 0 & 0 & 0 \\ f_1 & f_0 & 0 & 0 & 0 \\ f_2 & f_1 & f_0 & 0 & 0 \\ f_3 & f_2 & f_1 & f_0 & 0 \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix} \begin{pmatrix} a_1 \\ 2 a_2 \\ 3 a_3 \\ 4 a_4 \\ \dots \end{pmatrix} = \begin{pmatrix} f_1 \\ 2 f_2 \\ 3 f_3 \\ 4 f_4 \\ \dots \end{pmatrix} \quad (532)$$

Now given $\{f_i\}_{i=0,\dots,k}$, solving the coefficients $\{a_i\}_{i=1,\dots,k}$ is equivalent as solving the linear system and $a_0 = \log f_0$, which has complexity is $O(n^2)$.

Algorithm : 8.1.6

0. Given lower Triangular Toeplitz $A_{ij} = f_{i-j}$, we compute $\log(A)_{ij} = a_{i-j}$

1. $a_0 = \log f_0$

2. $\begin{pmatrix} a_1 \\ a_2 \\ \dots \\ a_n \end{pmatrix} = D^{-1} A^{-1} D \begin{pmatrix} f_1 \\ f_2 \\ \dots \\ f_n \end{pmatrix}$ where $D = \text{diag}(1, 2, \dots, n)$

$\log A$ is Toeplitz with first column given by $(a_0, a_1, \dots, a_n)$

■

This method turns to be extremely concise and requires no iterations. However, it does not work in the block Toeplitz case since for general matrices

$$d(e^A) \quad (533)$$

$$= d\left(I + A + \frac{1}{2}A^2 + \frac{1}{6}A^3 + \dots\right) \quad (534)$$

$$= \sum_{i=1}^{\infty} \sum_{j=0}^{i-1} A^j (dA) A^{i-j-1} \quad (535)$$

which is generally not the same as $e^A (dA)$ unless A commutes with its derivative. A more detailed description of matrix valued functions that commutes with their derivatives is still an open problem, see [2012 Holt, Mehrmann, Schneider][63].

8.2 ϵ - (block) Circulant Matrix

A different approach for evaluation of triangular Toeplitz Matrix Functions is based on a special class of Circulant matrices called ϵ -Circulant matrices. The idea of using ϵ - Circulant as approximation to triangular Toeplitz matrices is first used in [64], see also [Bini, Latouche, Meini 2005][65]. Bini elaborated this topic for the Toeplitz case systematically in [66]. The generalization to the block Toeplitz and block ϵ -Circulant case was proposed in [Bini, Dendievel, Latouche, Meini 2016][67] to calculate Block Toeplitz matrix exponentials. In this section we review and introduce the notions on ϵ -Circulant matrices. We also generalize the method to block case and compute matrix logarithms. A niche method which combines ϵ -Circulant and extrapolation is proposed, achieving better efficiency compared to the original approximation.

8.2.1 Circulant Matrix, Block Circulant Matrix and ϵ -Circulant Matrix

An $n \times n$ circulant matrix is a special kind of Toeplitz matrix which takes the form

$$C(a)_{i,j} = \begin{cases} a_{i-j} & i \geq j \\ a_{n-j+i} & j > i \end{cases} \quad (536)$$

where the first column is given by

$$a = \begin{pmatrix} a_0 \\ \dots \\ a_{n-1} \end{pmatrix} \quad (537)$$

It can be diagonalized by Fourier transform[68], i.e.

$$C = F_n^{-1} \text{diag}(F_n a) F_n \quad (538)$$

$$F_n C F_n^{-1} = \text{diag}(F_n a) \quad (539)$$

where

$$\omega = e^{-2\pi i/n} \quad (540)$$

$$F_n = \begin{pmatrix} 1 & 1 & \dots & 1 \\ 1 & \omega & \dots & \omega^{n-1} \\ \dots & \dots & \dots & \dots \\ 1 & \omega^{n-1} & \dots & \omega^{(n-1)(n-1)} \end{pmatrix} \quad (541)$$

$$F_n^{-1} = \frac{1}{n} F_n^* \quad (542)$$

The result can also be generalized to $n \times n$ Block Circulant matrix with $k \times k$ blocks,

$$C(A)_{i,j} = \begin{cases} A_{i-j} & i \geq j \\ A_{n-j+i} & j > i \end{cases} \quad (543)$$

$$C(A) = F_{n,k}^{-1} \text{blkdiag}(F_{n,k} A) F_{n,k} \quad (544)$$

where

$$A = \begin{pmatrix} A_0 \\ \dots \\ A_{n-1} \end{pmatrix} \quad (545)$$

$$F_{n,k} = F_n \otimes I_k \quad (546)$$

$$F_{n,k}^{-1} = \frac{1}{n} F_{n,k}^* \quad (547)$$

A block ϵ -Circulant matrix with block size $k \times k$ takes the form

$$U = \begin{pmatrix} U_0 \\ \dots \\ U_{n-1} \end{pmatrix} \quad (548)$$

$$C_\epsilon(U)_{i,j} = \begin{cases} \epsilon U_{i-j} & i > j \\ U_{n-j+i} & j \geq i \end{cases} \quad (549)$$

$$C_\epsilon(U) = \begin{pmatrix} U_0 & U_n & U_{n-1} & U_{n-2} & \dots \\ \epsilon U_1 & U_0 & U_n & U_{n-1} & \dots \\ \epsilon U_2 & \epsilon U_1 & U_0 & U_n & \dots \\ \epsilon U_3 & \epsilon U_2 & \epsilon U_1 & U_0 & \dots \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix} \quad (550)$$

which is related to block circulant matrix in the following way. Let

$$d_\epsilon = \text{diag}(1, \theta^1, \theta^2, \dots, \theta^{n-1}) \quad (551)$$

$$\theta = \epsilon^{\frac{1}{n}} \quad (552)$$

$$\mathcal{D}_\epsilon = d_\epsilon \otimes I_k \quad (553)$$

and we have

$$\mathcal{D}_\epsilon C_\epsilon(U) \mathcal{D}_\epsilon^{-1} = C(\mathcal{D}_\epsilon U) \quad (554)$$

Then $C_\epsilon(U)$ can be block diagonalized as

$$F_{n,k} \mathcal{D}_\epsilon C_\epsilon(U) \mathcal{D}_\epsilon^{-1} F_{n,k} = \text{blkdiag}(F_{n,k} U) \quad (555)$$

Notice that the computation of $\mathcal{D}_\epsilon C_\epsilon(U) \mathcal{D}_\epsilon^{-1}$ is $O(nk^2)$. The computational complexity of diagonalization of the structured matrix is greatly reduced. The following algorithm calculates the matrix logarithm of

$$U(z) = \sum_{k=0}^{n-1} U_k z^k \quad (556)$$

Algorithm 8.2.1: Given the first k columns of $nk \times nk$ block ϵ -Circulant matrix with block size k , this algorithm block diagonalize the matrix

$$1. U = \begin{pmatrix} U_0 \\ U_{n-l} \\ \dots \\ U_l \end{pmatrix}$$

$$2. Z = \mathcal{D}_\epsilon U$$

$$3. V = \mathcal{F}^* Z = \begin{pmatrix} V_0 \\ V_{n-l} \\ \dots \\ V_l \end{pmatrix}$$

$$F_{n,k} \mathcal{D}_\epsilon C_\epsilon(U) \mathcal{D}_\epsilon^{-1} F_{n,k} = \text{blkdiag}(V)$$

■

8.2.2 Logarithm of Triangular Block Toeplitz Matrices

With Algorithm 8.2.1 where we diagonalize the block ϵ -Circulant matrix with FFT, the computation of logarithm is then straight forward

Algorithm: 8.2.2 Logarithm of a block ϵ -Circulant matrix

0. Given a complex number ϵ , and block vector $U = (U_i)_{i=0, \dots, n-l}$

$$1. Z = \mathcal{D}_\epsilon U$$

$$2. V = \mathcal{F}^* Z = \begin{pmatrix} V_0 \\ V_{n-l} \\ \dots \\ V_l \end{pmatrix}$$

$$3. Y = \begin{pmatrix} \log V_0 \\ \log V_{n-l} \\ \dots \\ \log V_l \end{pmatrix}$$

$$4. A = \frac{1}{n} \mathcal{D}_\epsilon^{-l} \mathcal{F}^* Y = \begin{pmatrix} A_0 \\ A_{n-l} \\ \dots \\ A_l \end{pmatrix}$$

and we have

$$\log \begin{pmatrix} U_0 & U_1 & \dots & U_{n-1} \\ \epsilon U_{n-1} & U_0 & \dots & U_{n-2} \\ \dots & \dots & \dots & \dots \\ \epsilon U_1 & \epsilon U_2 & \dots & U_0 \end{pmatrix} = \begin{pmatrix} A_0 & A_1 & \dots & A_{n-1} \\ \epsilon A_{n-1} & A_0 & \dots & A_{n-2} \\ \dots & \dots & \dots & \dots \\ \epsilon A_1 & \epsilon A_2 & \dots & A_0 \end{pmatrix} \quad (557)$$

Cost: The cost is $2nk^2 + 2nk^2 \log(nk) + c_1 nk^3 \sim O(nk^2 \log(nk))$

■

The Block ϵ -Circulant can be decomposed into a sum of two parts, i.e.

$$C_\epsilon(U) = T(U) + \epsilon L(U) \quad (558)$$

where

$$L(U) = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ U_1 & 0 & 0 & 0 & 0 \\ U_2 & U_1 & 0 & 0 & 0 \\ U_3 & U_2 & U_1 & 0 & \dots \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix} \quad (559)$$

$$T(U) = \begin{pmatrix} U_0 & U_{n-1} & U_{n-2} & U_{n-3} & \dots \\ 0 & U_0 & U_{n-1} & U_{n-2} & \dots \\ 0 & 0 & U_0 & U_{n-1} & \dots \\ 0 & 0 & 0 & U_0 & \dots \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix} \quad (560)$$

So that

$$\lim_{\epsilon \rightarrow 0} C_\epsilon(U) = T(U) \quad (561)$$

This means that when ϵ is small enough, $T(U)$ is approximated by $C_\epsilon(U)$. We now show that this approximation is inherited by matrix logarithms. First we introduce the definition of Fréchet derivative.

Definition: Given a matrix function $G: T \in \mathbb{R}^{n \times n} \rightarrow G(T) \in \mathbb{R}^{n \times n}$, a linear mapping $G'(T)Z: Z \in \mathbb{R}^{n \times n} \rightarrow G'(T)Z \in \mathbb{R}^{n \times n}$ is the Fréchet derivative of G at T if for any $Z \in \mathbb{R}^{n \times n}$ we have

$$\lim_{\epsilon \rightarrow 0} \left\| \frac{G(T + \epsilon Z) - G(T)}{\epsilon} - G'(T)Z \right\|_2 = 0 \quad (562)$$

the norm of the Fréchet derivative is given by $\|G'(T)\| = \max_{\|Z\|=1} \|G'(T)Z\|$. If G has a Fréchet derivative, we say that G is differentiable.

Now assume $\rho(A) < 1$, let

$$G(T) := \log(T) = \log(I - A) = - \sum_{k=1}^{\infty} \frac{A^k}{k} \quad (563)$$

$$\log(T + \epsilon L) = \log(T) + (\log(T))' \epsilon L + E(\epsilon L) \quad (564)$$

where $\|E(L)\| \leq O(\|\epsilon L\|^2)$.

$$G'(T) = \lim_{\epsilon \rightarrow 0} \frac{G(T + \epsilon L) - G(T)}{\epsilon} \quad (565)$$

$$G'(T) : L \rightarrow \sum_{n=1}^{\infty} \frac{1}{n} \sum_{k=0}^{n-1} A^k L A^{n-1-k}, \quad A = I - T \quad (566)$$

Without loss of generality, assume $\|A\|_2 < 1$, $\|L\|_2 < 1$. (we can always take the square root of T if this is not true, using the inverse squaring and scaling method).

$$\|G(T)\|_2 = \max_{\|L\|=1} \left\| \sum_{n=1}^{\infty} \frac{1}{n} \sum_{k=0}^{n-1} A^k L A^{n-1-k} \right\| \quad (567)$$

$$\leq \sum_{n=1}^{\infty} \frac{1}{n} \sum_{k=0}^{n-1} \|A\|^{n-1} \quad (568)$$

$$= \frac{1}{1 - \|A\|_2} \quad (569)$$

$$= \frac{1}{1 - \|I - T\|_2} \quad (570)$$

So we have

$$\|\log(T + \epsilon L) - \log(T)\| = \|(\log(T))' \epsilon L + E(\epsilon L)\| \quad (571)$$

$$\leq \|(\log(T))'\| \|\epsilon L\| + \|E(\epsilon L)\| \quad (572)$$

$$\leq |\epsilon| \frac{1}{1 - \|I - T\|_2} \|L\|_2 + O(|\epsilon L|^2) \quad (573)$$

Thus for circulant matrix, $C_{\epsilon}(U)$ can be decomposed into $T(U)$ plus a perturbation $\epsilon L(U)$, we then have, for ϵ

small enough

$$\| \log(C_\epsilon(U)) - \log(T(U)) \|_2 \leq |\epsilon| \frac{1}{1 - \|I - T\|_2} \|L\|_2 + O(\|\epsilon L\|_2^2) \quad (574)$$

Theoretically, the approximation can arbitrarily close when we pick an ϵ small enough. However, in numerical computations, the approximation error originates from two sources. Besides the approximation error mentioned above when ϵ is not exactly 0, there is the round off error when ϵ is indeed small where taking the inverse of it can be dangerous. The approximation error grows as $O(|\epsilon|)$ while the roundoff error is $O(\mu |\epsilon|^{-1})$, where μ is the machine precision. This leads to an overall error which is $O(\mu^{\frac{1}{2}})$. Using a pure imaginary ϵ and taking only the real part can further reduce the error. In [69], when they calculate the matrix exponential, they use k different calculations with different ϵ and average out some error. One good choice is to take ϵ to be the unit roots [70]. Without loss of generality, assume that $C_\epsilon(U) - I$ and $T(U) - I$ are contractions. Let the approximation be

$$\log(C_\epsilon(U)) = \log(T(U) + \epsilon L(U)) \quad (575)$$

$$= \log(I + ((T(U) - I) + \epsilon L(U))) \quad (576)$$

$$= \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} ((T(U) - I) + \epsilon L(U))^k \quad (577)$$

And the approximation error is given by

$$\log(C_\epsilon(U)) - \log(T(U)) \quad (578)$$

$$= \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} ((T(U) - I) + \epsilon L(U))^k - (T(U) - I)^k \quad (579)$$

Now let $\epsilon_j = i^{\frac{1}{m}} \omega_m^j \epsilon$, $j = 0, \dots, m-1$, where ω_m is the m^{th} root of unity, ϵ is some scalar $|\epsilon| < 1$. ϵ_j is pure imaginary. Then we have

$$\sum_{j=0}^{m-1} \epsilon_j^n = 0 \text{ for } n < m \quad (580)$$

Using this property we can evaluate $\log(C_\epsilon(U))$ on all of ϵ_j and average out the leading error terms

$$\frac{1}{m} \left(\sum_{j=0}^{m-1} \log(C_{\epsilon_j}(U)) \right) - \log(T(U)) = O(\epsilon^{2m}) \quad (581)$$

$$(582)$$

given that both $L(U)$ and $T(U) - I$ are contractions, specifically $L(U)$ is nilpotent. The factor of 2 in the power comes from the fact that ϵ_j is pure imaginary and our computation is for real result.

Algorithm: 8.2.3 Logarithm of a triangular matrix by means of ϵ -Circulant matrix

0. Given a complex number ϵ , and block vector $U = (U_i)_{i=0, \dots, n-1}$

1. set $\epsilon_j = i^{\frac{1}{m}} \omega_m^j \epsilon$

2. Compute the first block column Y_j of $\log(C_{\epsilon_j}(U))$ $j = 0, \dots, m-1$ using algorithm 8.2.2

3. set $A = \frac{1}{m} \sum_{j=0}^{m-1} \text{Re}(Y_j)$

■

The complexity of this method will be a linear multiple of the non-average one. Now here we propose a more efficient error cancelling technique by using Richardson Extrapolation. Richardson Extrapolation[71] significantly reduces error in the leading terms of error expansion. Suppose we wish to approximate $A(0)$, and we have a method to compute $A(\epsilon)$ that depends on a small parameter ϵ , so that

$$A(\epsilon) = A(0) + C\epsilon^n + O(\epsilon^{n+1}) \quad (583)$$

Define a new method

$$R(\epsilon, k) = \frac{k^n A(\epsilon) - A(k\epsilon)}{k^n - 1} \quad (584)$$

Then

$$R(\epsilon, k) = A^* + O(\epsilon^{n+1}) \quad (585)$$

And $R(\epsilon, k)$ is called Richardson Extrapolation. If we start with a relatively large ϵ and apply the extrapolation recursively, we can reduce both approximation and roundoff error at the same time.

Algorithm 8.2.4 Given the first k columns of $nk \times nk$ block ϵ -Circulant matrix with block size k , this algorithm computes the its logarithm with Richardson Extrapolation


```

1 Pick  $N$  as the number of iterations
2 Pick  $\epsilon$  as the initial step size.
3 Pick  $t$  as some scalar
4 For  $j = 1 : N + 1$ 
5      $L_{0,j} = \log(C_{\epsilon \rightarrow j}(U))$  using Algorithm 8.2.2
6 End
7 For  $k1 = 1 : N$ 
8     For  $k2 = 1 : (N+1-k1)$ 
9          $L_{k1,k2} = (t^{k1} L_{k1-1,k2+1} - L_{k1-1,k2}) \frac{1}{t^{k1}-1}$ 
10    End
11 End

```

■

Cost: $O((N+1) n k^2 \log(n k))$

The cost is much smaller than if we apply inverse squaring and scaling which costs $O((N+1) (nk)^3)$. Numerical examples show that it takes less levels to achieve the same or even better accuracy than averaging method.

8.3 Numerical Examples

The numerical experiments have been performed in Matlab 2016b. For square root we first generate result matrix, multiply by itself and use that to compute square root. For matrix logarithm, we use expansion coefficients of $f(z) = \frac{1}{1-z}$ where its logarithm is known.

8.3.1 Square Root

Methods:

Schur: Matlab built in, Schur method to compute matrix square root

EL : Exponential of half of matrix logarithm, Algorithm 8.1.3

DB : Scaled Denman and Beavers iteration, Algorithm 8.1.1

DC : Divide and conquer, Algorithm 8.1.2

PSDB: DB method with preconditioning

Example 8.1 $2^{10} \times 2^{10}$ Toeplitz Matrix

$$T_{ij} = \begin{cases} 1 & i \geq j \\ 0 & i < j \end{cases} \quad (586)$$

Result

	Schur	EL	DB	PSDB	DC
Time(s)	10.9201	0.0086	0.0203	0.0208	0.0053
Relative H_2 Error	4.9029×10^{-12}	5.9608×10^{-12}	1.7687×10^{-10}	5.0132×10^{-12}	4.8902×10^{-12}
Relative Inf Error	4.5475×10^{-13}	5.6088×10^{-13}	2.7397×10^{-11}	3.7748×10^{-13}	4.2766×10^{-13}

(587)

This example shows that in the SISO case, while the performance is within the comparable level, EL and DC are significantly faster than other methods, 1000 times faster than the Matlab builtin Function.

Example 8.2 Block Toeplitz Matrix with $2^{10} \times 2^{10}$ blocks, each block size is 2×2

$$a_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \quad a_2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad (588)$$

$$b_{i,j} = \begin{cases} a_1 & i = j \\ a_2 & i > j \end{cases} \quad (589)$$

Result:

	PSDB	DB	Schur
Time(s)	0.120	0.121	14.212
Relative H_2 Error	1.2295×10^{-8}	4.6500×10^{-8}	9.3671×10^{-14}
Relative Inf Error	4.5451×10^{-8}	1.2293×10^{-7}	3.7503×10^{-13}

(590)

Example 8.3 Block Toeplitz Matrix with $2^{10} \times 2^{10}$ blocks, each block size is 2×2

$$t_{n,i,j} = \prod_{k=1}^{i-j} \frac{k + 0.5}{k} \quad (591)$$

$$T_n = t_n \otimes I_2 \quad (592)$$

Result:

	PSDB	DB	Schur
Time(s)	0.097	NaN	10.926
Relative H_2 Error	3.6939×10^{-12}	NaN	3.3098×10^{44}
Relative Inf Error	7.4246×10^{-12}	NaN	4.0686×10^{45}

(593)

Notice that DB and PSDB are about 100 times faster than the Matlab builtin method. Preconditioning of the DB method also allows the algorithm to succeed when both the regular scaled product DB iterations and the Matlab builtin Schur method fails.

8.3.2 Logarithm

Example 8.4 : In the SISO case, let

$$T_m = \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 0 & 1 & 1 & \dots & 1 \\ 0 & 0 & 1 & \dots & 1 \\ 0 & 0 & 0 & \dots & \dots \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \quad (594)$$

And we know the closed form solution

$$(\log T_m)_{j,k} = \begin{cases} \frac{1}{k-j} & k > j \\ 0 & k \leq j \end{cases} \quad (595)$$

For a $2^{10} \times 2^{10}$ Toeplitz Matrix we compute its logarithm and compare with the closed form solution

Result:

	Matlab(Schur)	toeplog	agm	sri
Time(s)	5.529	0.0033	0.056	0.043
Relative H_2 Error	3.6936×10^{-15}	0	1.9141×10^{-14}	4.8282×10^{-14}
Relative Inf Error	5.7072×10^{-16}	0	1.7764×10^{-14}	5.6849×10^{-14}

(596)

	ϵ - Circulant	mean ϵ - Circulant	ϵ - Circulant Extrap
Time(s)	0.030	0.047	0.025
Relative H_2 Error	8.9880×10^{-8}	1.1984×10^{-14}	1.1051×10^{-14}
Relative Inf Error	3.5012×10^{-8}	2.6598×10^{-15}	9.7474×10^{-15}

(597)

The toeplog method is by far the best know approach to calculate the logarithm of a triangular Toeplitz matrix. It is faster and more accurate than any other methods. The use of divide and conquer as well as exploiting the matrix structure in the context of solving one linear system makes it extremely fast, while it avoids iterations or approximations so that it is very accurate. It can also be parallelized to be even faster, unlike the iterative ones. ϵ -Circulant method is losing about half of the machine precision, while the averaging method and Extrapolation method corrects that. The Matlab builtin, agm, sri, mean ϵ -Circulant and ϵ -Circulant Extrapolation are having similar accuracy. Matlab builtin is about 1000 times slower than others at this matrix size. Extrapolation method is as accurate but twice as much faster as the mean ϵ -Circulant method.

Example 8.5 Same matrix generation with a larger size, $2^{20} \times 2^{20}$ Toeplitz Matrix, we compute its logarithm and compare with the closed form solution

$$T_{ij} = \begin{cases} 1 & i \geq j \\ 0 & i < j \end{cases} \quad (598)$$

Result:

	toeplog	agm	sri
Time(s)	2.530	27.067	35.600
Relative H_2 Error	0	5.0772×10^{-13}	2.5528×10^{-13}
Relative Inf Error	0	2.1871×10^{-14}	2.3259×10^{-14}

(599)

	ϵ – Circulant	mean ϵ – Circulant	ϵ – Circulant Extrap
Time(s)	0.194	2.867	1.318
Relative H_2 Error	6.5222×10^{-6}	5.4447×10^{-13}	3.2057×10^{-13}
Relative Inf Error	3.5816×10^{-7}	8.1688×10^{-15}	9.7411×10^{-15}

(600)

Given a million by million size Toeplitz matrix the Matlab built in will not be able to solve the problem. the performance of the others are compared. ϵ -Circulant is the fastest but is not very accurate. Toeplog is still the winner on an overall basis. Besides that the ϵ -Circulant based methods are faster than the iterative methods. ϵ -Circulant Extrapolation is twice as fast as the mean ϵ -Circulant method, achieving similar accuracy, if not better.

Example 8.6 $2^{12} \times 2^{12}$ Toeplitz Matrix

$$T_{ij} = \begin{cases} 1 & i \geq j \\ 0 & i < j \end{cases} \quad (601)$$

where we treat the matrix as block Toeplitz with 4×4 blocks and $2^{10} \times 2^{10}$ blocks

Result:

	Matlab(Schur)	agm	sri
Time(s)	72.154	1.544	1.186
Relative H_2 Error	1.7952×10^{-13}	4.5445×10^{-14}	7.1543×10^{-14}
Relative Inf Error	1.1768×10^{-14}	4.2126×10^{-15}	9.3179×10^{-14}

(602)

	ϵ – Circulant	mean ϵ – Circulant	ϵ – Circulant Extrap
Time(s)	0.756	8.406	2.896
Relative H_2 Error	2.4243×10^{-8}	2.0857×10^{-14}	9.9919×10^{-15}
Relative Inf Error	1.4431×10^{-8}	5.9437×10^{-15}	3.6671×10^{-15}

(603)

In the block case, toeplog is no longer present, Matlab builtin is beaten by other implementations both on speed and accuracy. In the next example the speed and accuracy evolution with the growing size of the matrix is shown. The structure exploiting methods are growing at about the same speed in consumed time and ϵ -Circulant Extrapolation seems to beat all those who achieves similar accuracy. Also its approximation error grows slower than the others as the size of matrix gets bigger.

Example 8.7 Toeplitz Matrix

$$T_{ij} = \begin{cases} 1 & i \geq j \\ 0 & i < j \end{cases} \quad (604)$$

where we treat the matrix as block Toeplitz with 8×8 blocks. Logarithm was computed for a range of different sizes

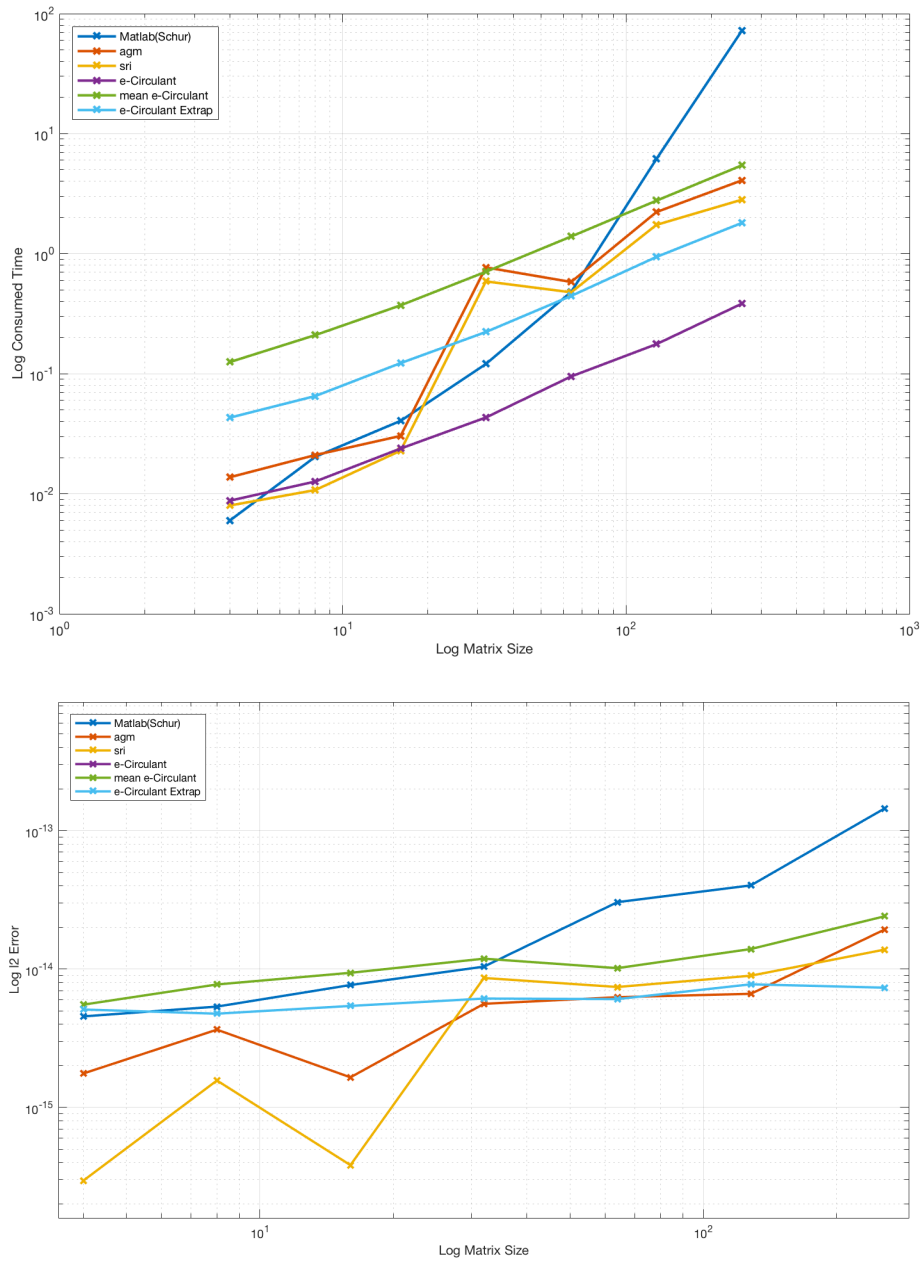


Figure: Without ϵ -Circulant method

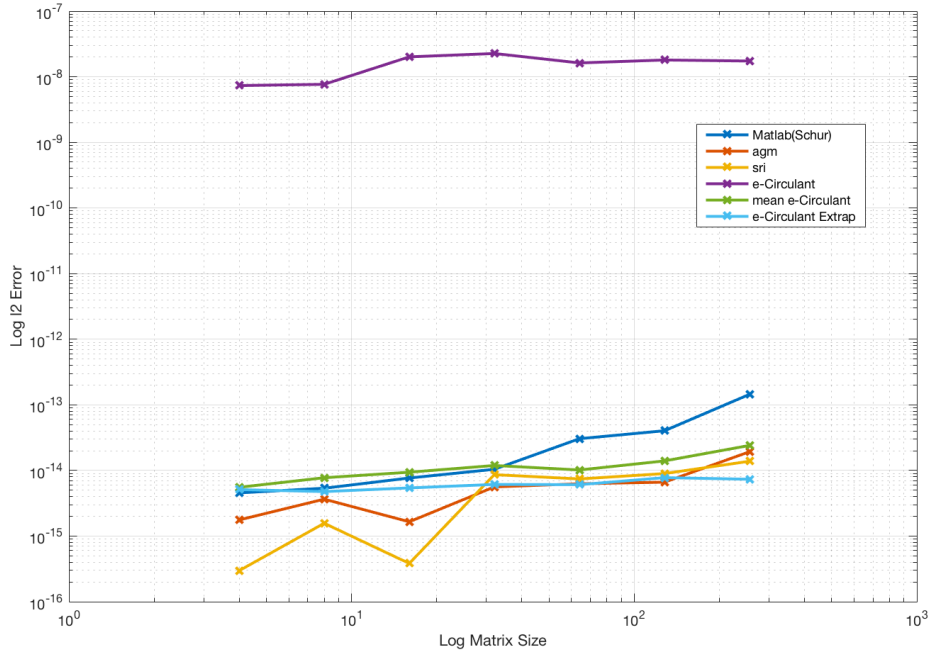


Figure: Containing ϵ -Circulant method

Even though being the fastest, the purple line(ϵ -Circulant) loses about half of the digits compared to other methods and we present it in a separate plot. At a similar approximation accuracy level, the ϵ -Circulant matrix with extrapolation consumes less time when the dimension is large enough. While when the matrix small, there are less to exploit out of matrix structure and Schur, AGM and SRI are winning methods. And at the same time when the matrix size grows bigger, the error of extrapolation method does not seem to increase while all the others have a tendency of larger numerical error.

8.4 Reduction On Cepstrum

The cepstrum is widely used and celebrated in the signal processing community. Firstly being proposed by John Tukey[72], cepstrum was later redefined by Oppenheim and Schafer[73], where the analytical formulation of the cepstrum for transfer functions was derived. The application of cepstrum includes speech analysis, communications, seismology and geophysics.

Our approach on cepstrum of transfer function stems from a different starting point, information geometry. The approach is to compare and analyze distributions by embedding them in a Riemannian manifold where the metric is well defined and distance can be measured theoretically and numerically. It has been proved by Campbell(1968) and Chentsov(1972) that Fisher information matrix is the unique invariant metric of statistical manifolds, i.e. it is invariant of change of parameters. In the case of multivariate Gaussian with common means but differing variance-covariance matrix, which in the time series analysis context turns out to be the auto covariance matrix, the information manifold is indeed a Riemannian manifold of semi-positive definite (SPD) matrices. The distance being defined as

$$\| \log(C_2 C_1^{-1}) \| = \sum_{i=1}^n \log^2(\lambda_i(C_2 C_1^{-1})) \quad (605)$$

Measuring the distance between two semi-Positive definite matrices has a large variety of applications and this very distance was proposed and used by Jensen (1981) [19] for distribution measure, Pennec, Fillard, Ayache (2006) [74-75] for computer vision, and Bhatia (2006) [76], Moakher (2006) [77] (2011) [78] for the use of Lie group theory and differential geometry. The affine invariance property of such metric is widely preferred especially over Euclidean distance of two SPD matrices. In dimension reduction problems, higher dimensional models are projected onto the lower dimensional manifold where this affine invariant distances are used to encode the similarity between the original and reduced models.

Specifically in our problem, as introduced in Chapter 3, in the SISO case the auto covariance matrix is the product of a triangular Toeplitz matrix and its conjugate transpose, the entries being fourier coefficients of the transfer function. The cepstrum comes in at this time, with which as the coordinates of Riemannian manifold, the manifold turns out to be Euclidean. This is a result discovered by A.P. Mullhaupt[79-80]. This property and specific use of information geometry in time series analysis made the computation possible to develop reduction methods and other applications based on the information distance. We can apply the model reduction method to the cepstrum of transfer function in order to get a model that is close in the statistical manifold. Notice that the information distance controls the total variation distance, so in the sense of hypothesis testing, reduction on the cepstrum should be better than reduction on the transfer function itself.

The Cepstrum Reduction is indeed the null basis reduction applied to the cepstrum. While the impulse response reduction works if the function is in VMOA, it's logarithm in $H^\infty \cap C$ is sufficient for reduction on cepstrum to

work. This is based on the result of [Hartman, 1958][81]. A classical example for the potential extension of function space by taking the logarithm is given by the transfer function of random walk. $f(z) = \frac{1}{1-z}$ is not in H_2 but its logarithm is.

Algorithm 8.4.1 :

0. Give transfer function $f(z)$ and reduction degree n

1. Use Algorithm 'Toeplitz' in 8.1.4 for SISO, and Algorithm 8.2.4 for MIMO to compute the coefficients of $a(z) = \log f(z)$

2. Use Algorithm 7.3.1 (degenerates to SISO reduction in the SISO case) to obtain reduced order model to degree n

■

However, the equivalence of H_2 distance of cepstrum and information distance is no longer valid in the MIMO case. The major reason stems from the fact that, while

$$\log f(z) g(z) = \log g(z) f(z) \quad (606)$$

is most natural in the SISO case since $g(z)$ and $f(z)$ are scalar functions, in the MIMO case the commutativity is absent in general case, and

$$\log f(z) g(z) \neq \log f(z) + \log g(z) \quad (607)$$

Let $n \times n$ matrix valued function $F(z)$ $G(z)$ denote the transfer function, $\Phi_F(z)$, $\Phi_G(z)$ denote the spectral density, with T_F , T_G , T_{Φ_F} , T_{Φ_G} be Toeplitz matrices with symbol F , G , Φ_F , Φ_G respectively. The information distance between F and G is given by

$$\lim_{m \rightarrow \infty} \frac{1}{2mn} I^2(T_{FF^*}, T_{GG^*}) \quad (608)$$

$$\lim_{m \rightarrow \infty} \frac{1}{2mn} I^2(T_G^{-1} T_F T_F^* T_G^{-*}, I) \quad (609)$$

$$\leq \frac{1}{n} \|\log(T_F T_G^{-1})\|_F^2 \quad (610)$$

$$= \frac{1}{n} \|\log(F G^{-1})\|_{H_2}^2 \quad (611)$$

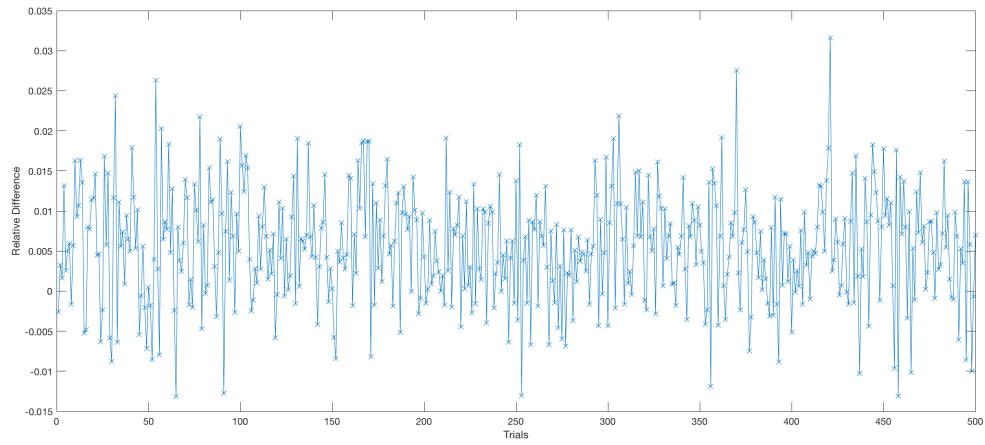
and the information distance is controlled by (596) which in the SISO case is the norm of cepstrum difference. In the MIMO case

$$\frac{1}{n} \|\log(F G^{-1})\|_{H_2} \neq \frac{1}{n} \|\log(F) - \log(G)\|_{H_2} \quad (612)$$

These two values turn out to be very similar when they commute enough. In our case, without loss of generality assume the block diagonals are identity, and the Toeplitz matrix will be close to identity given a low signal to noise ratio.

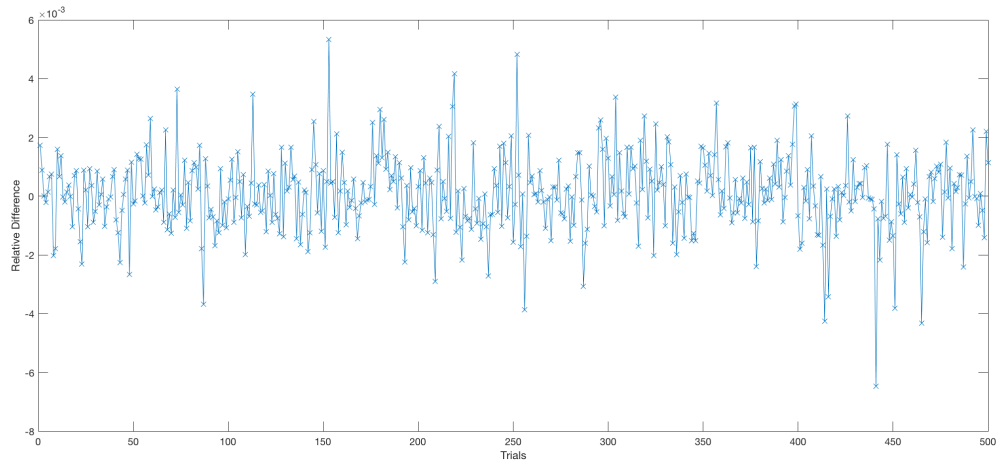
Example 8.8 Randomly generated impulse response F and G with SNR = 1, compute

$$\frac{\|\log(F G^{-1})\|_{H_2} - \|\log(F) - \log(G)\|_{H_2}}{\|\log(F G^{-1})\|_{H_2}} \quad (613)$$



Example 8.9 Randomly generate impulse response F and G with $\text{SNR} = 0.1$, compute

$$\frac{\|\log(F G^{-1})\|_{H_2} - \|\log(F) - \log(G)\|_{H_2}}{\|\log(F G^{-1})\|_{H_2}} \quad (614)$$



The relative difference within the Example 8.5.1 is within 3% and Example 8.5.2 within 0.6%. The experiments show there is no dominant relationship between the two, however the closer the matrices are to identity, the smaller the difference there is. This phenomenon can actually be explained by the Lie-Trotter formula.

Theorem 8.5.1. For arbitrary $n \times n$ real or complex matrices A and B ,

$$e^{A+B} = \lim_{m \rightarrow \infty} \left(e^{\frac{A}{m}} e^{\frac{B}{m}} \right)^m \quad (615)$$

This is an analogue of the classical exponential law and a more detailed version would be the Baker-Campbell-Hausdorff formula. Suppose that the principal logarithm exists for T_F , T_G and $T_F T_G^{-1}$. Then we have

$$e^{\log T_F - \log T_G} = \lim_{m \rightarrow \infty} \left(e^{\frac{\log T_F}{m}} e^{-\frac{\log T_G}{m}} \right)^m \quad (616)$$

Taking logarithm on both sides yields

$$\log T_F - \log T_G = \lim_{m \rightarrow \infty} m \left(T_F^{\frac{1}{m}} - T_G^{\frac{1}{m}} \right) \quad (617)$$

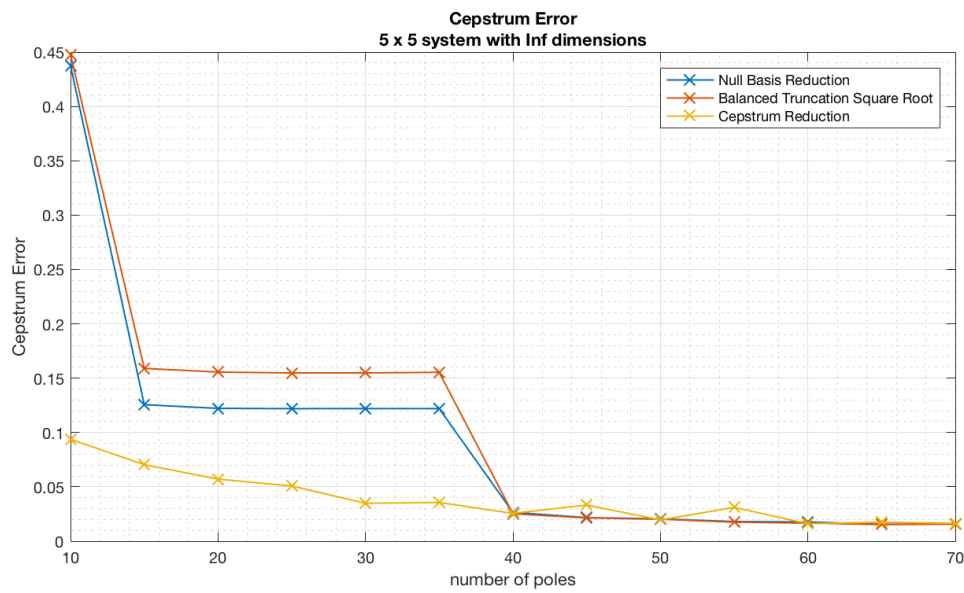
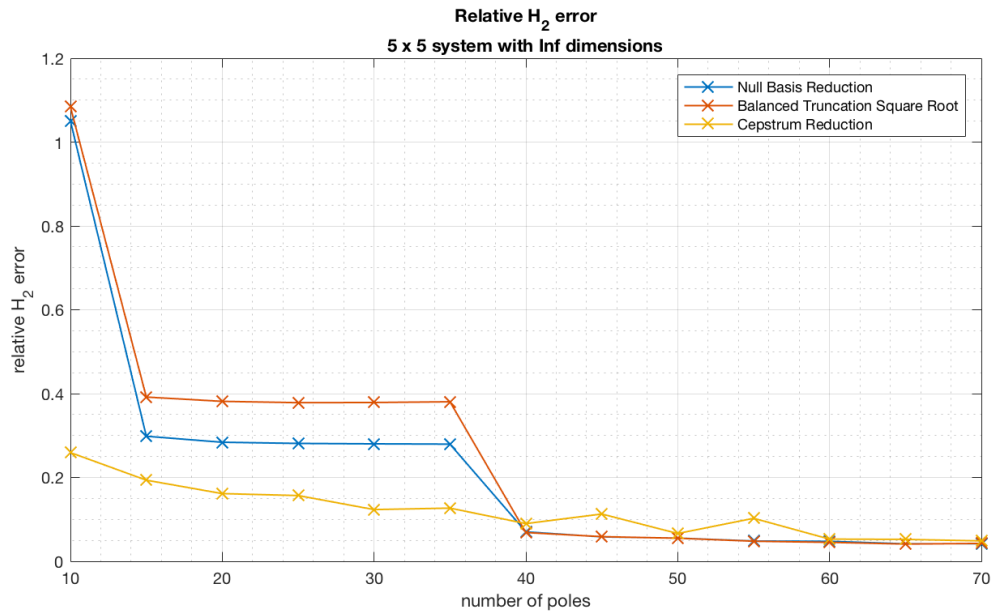
$$\lim_{m \rightarrow \infty} \log T_F^{\frac{1}{m}} - \log T_G^{\frac{1}{m}} = \lim_{m \rightarrow \infty} T_F^{\frac{1}{m}} - T_G^{\frac{1}{m}} \quad (618)$$

This equality says that when T_F and T_G are closer enough, and so that they ‘commute enough’, the value on each side get closer to each other.

Now we apply reduction methods to the cepstrum of impulse response. We pick 3000 truncation of two infinitely long impulse responses and apply three different reduction methods, i.e. Null Basis Reduction, Balanced Truncation and Cepstrum Null Basis Reduction (Matlab 2017a Version, Square Root Method). We reduce the system to various degrees and compare the relative H_2 difference of the first 3000 coefficients. The Cepstrum reduction methods are able to capture most of the system starting from a very low degree.

Example 8.10 (Infinite Dimension System) with a 3×3 impulse response $F(z)$ truncated at $j = 3000$,

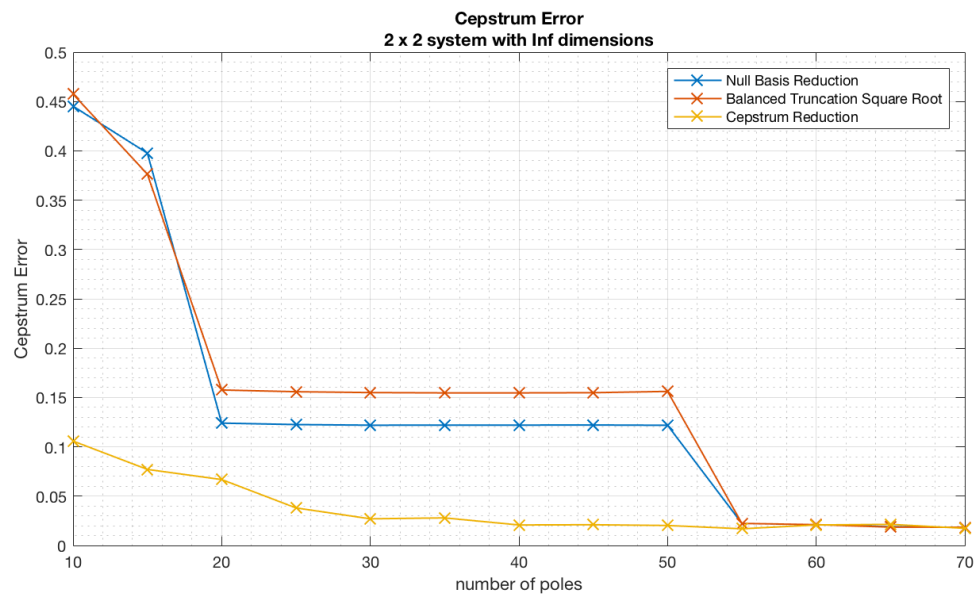
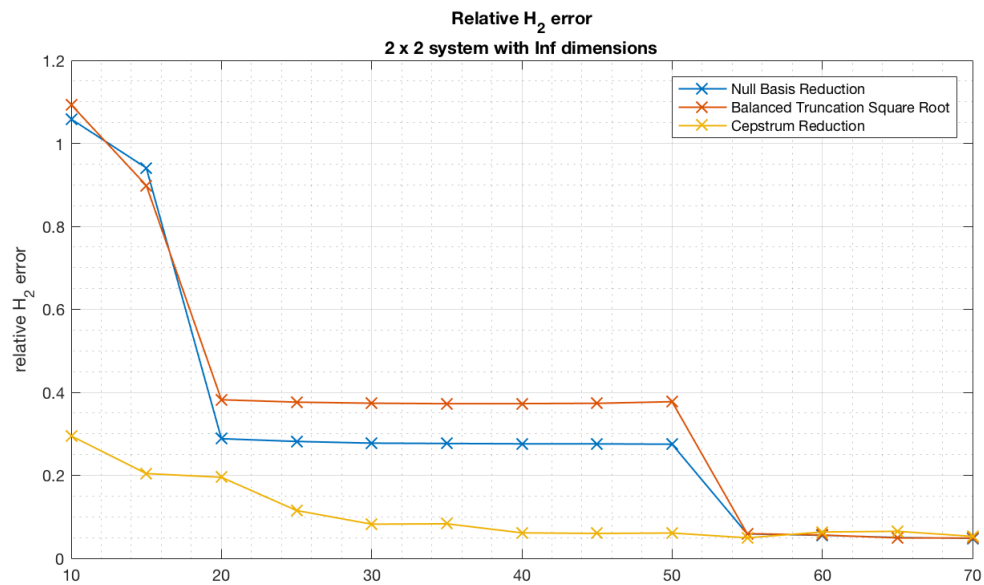
$$F(z) = \sum_{j=1}^{3000} j^{-} \begin{pmatrix} 0.9 & 0.7 \\ 0.8 & 0.6 \end{pmatrix} \quad (619)$$



...

Example 8.11 (Infinite Dimension System) with a 3×3 impulse response $F(z)$ truncated at $j = 10000$,

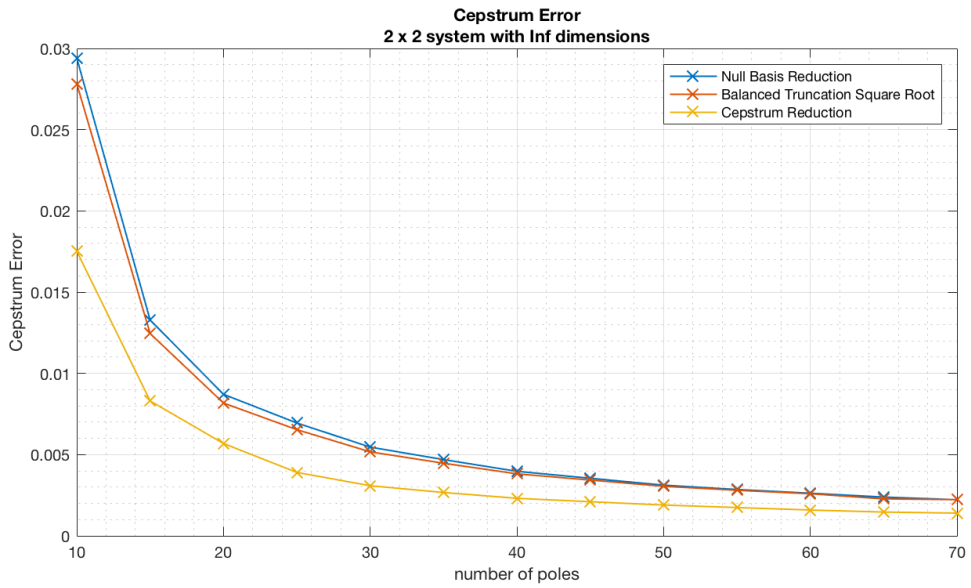
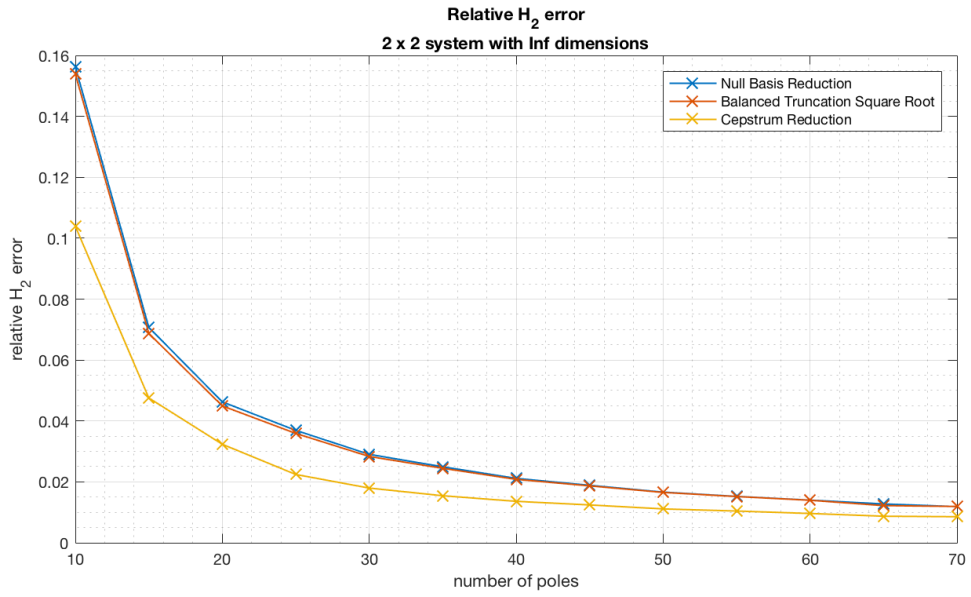
$$F(z) = \sum_{j=1}^{10000} j^{-\begin{pmatrix} 0.9 & 0.7 \\ 0.8 & 0.6 \end{pmatrix}} \quad (620)$$



Example 8.12 (Infinite Dimension System) with a 3×3 impulse response $F(z)$,

$$F(z) = A \sum_{j=1}^{3000} \frac{\begin{pmatrix} 1 & 0.1 \\ 0.1 & 1 \end{pmatrix}}{j} A^{-1} \quad (621)$$

where the elements of A follow $U(0, 1)$ distribution.



From Example 8.10 and 8.11, it's clear to see that as we truncate at larger n for the series, it takes more poles for the regular reduction methods on the impulse response to converge to the real one, while the performance of cepstrum reduction tend to be consistent. This is because the reduction on impulse response is only capturing the finite truncation part of the impulse response. Given a Hankel operator that is not compact on H_2 (symbol not in VMOA), the approximation error from truncation can blow up. Suppose we are given such impulse response up to the first 3000 observations, a regular reduction method yield approximation error within 5%, which is not going to persist given more observations. This is typically the case when the time series is not stationary as opposed to what's quite frequently assumed in the time series analysis context. The yellow curve actually shows that there is a way to capture the basis of the system in some scenarios even if it's not stationary. For an impulse response not in H_2 but whose logarithm is in H_2 , the cepstrum reduction method captures the basis of the system faster and better, which is also more consistent as the number of observations evolves. In the MIMO case, rather than stationarity of the system, which means the transfer function is in H_2 , we only require the log of spectral density to be in L_2 .

References

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